



光栅色散型光谱仪 & F-P干涉型光谱仪

虞益挺

西北工业大学



提 纲

1、引言

2、光栅色散型光谱仪

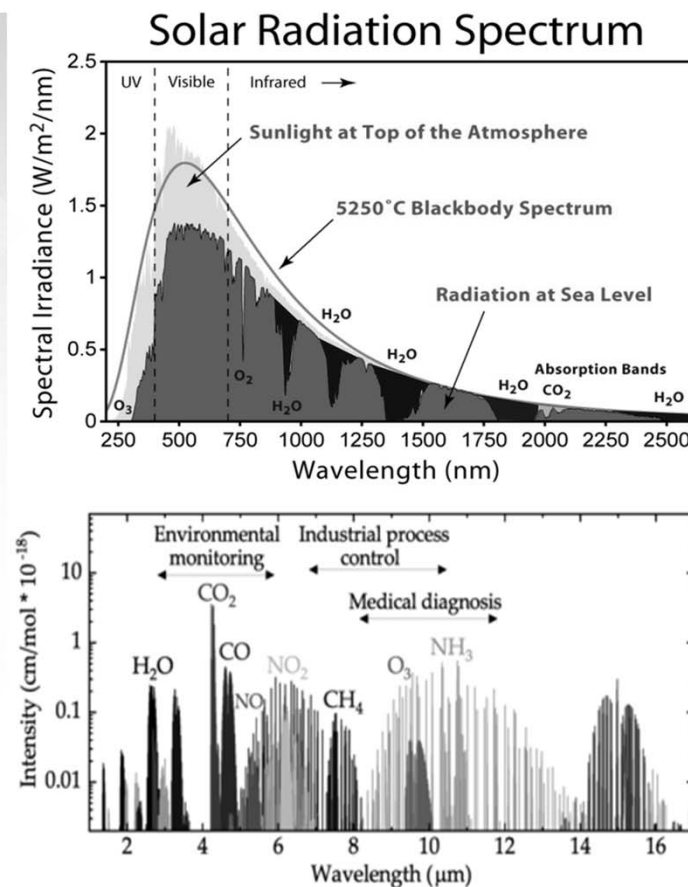
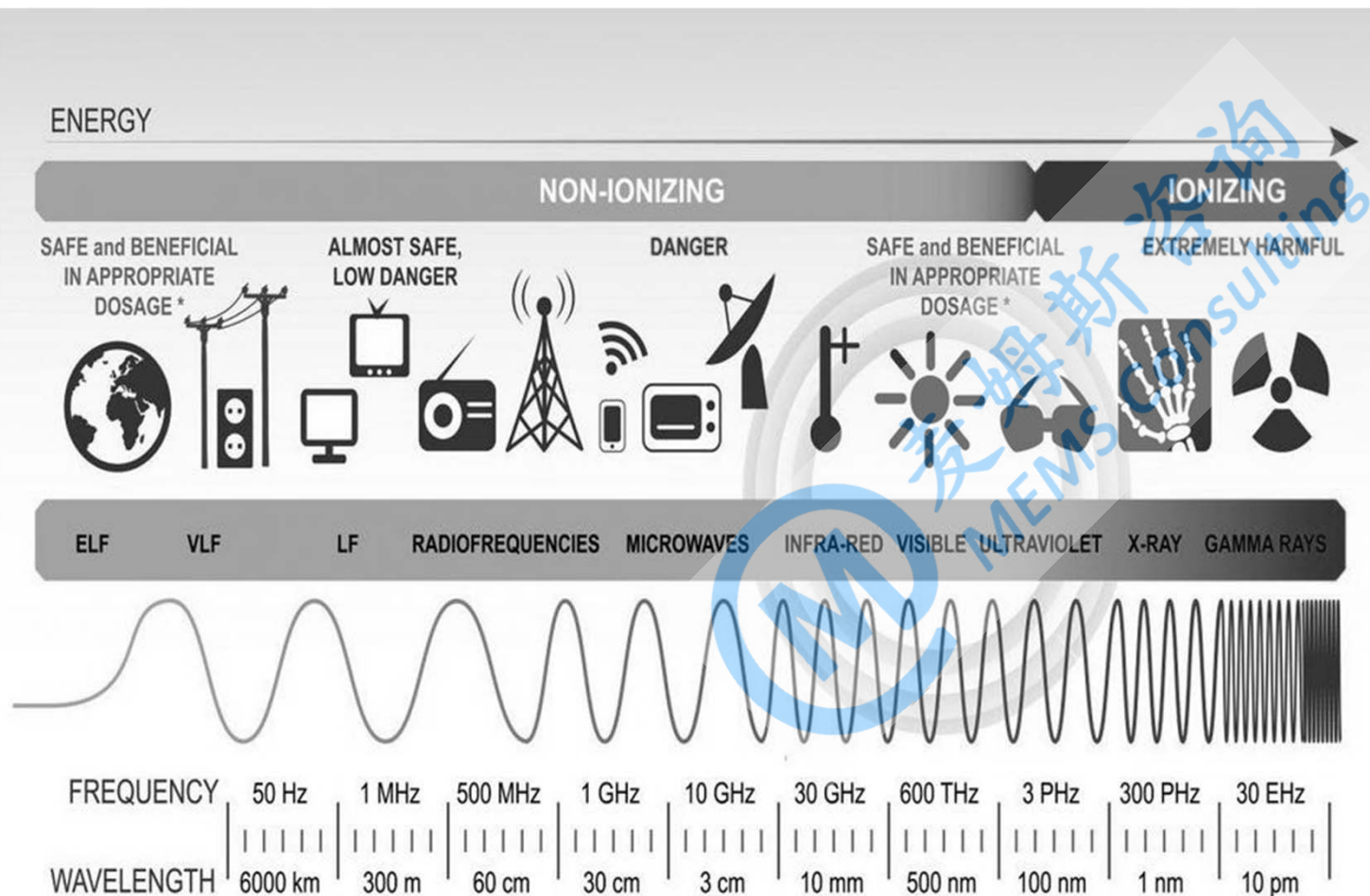
- (1) 衍射光栅（反射/透射光栅）分光原理
- (2) 硅基微纳光栅：设计与加工
- (3) 基于硅基微纳光栅的微型光谱仪及其应用

3、F-P干涉型光谱仪

- (1) F-P干涉仪分光原理
- (2) F-P全固态分光器件及其制造方法
- (3) 基于F-P全固态分光器件的微型光谱仪
- (4) F-P动态分光器件及其制造方法
- (5) 基于F-P动态分光器件的微型光谱仪
- (6) 基于F-P干涉技术的微型光谱仪典型应用

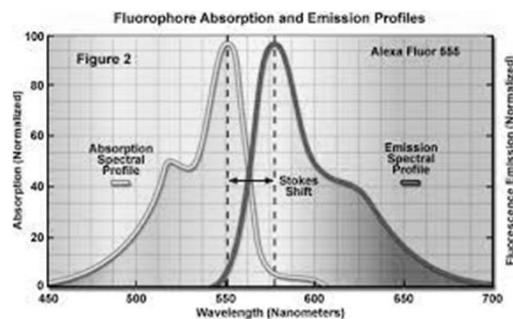
4、总结

1.引言 什么是光谱?

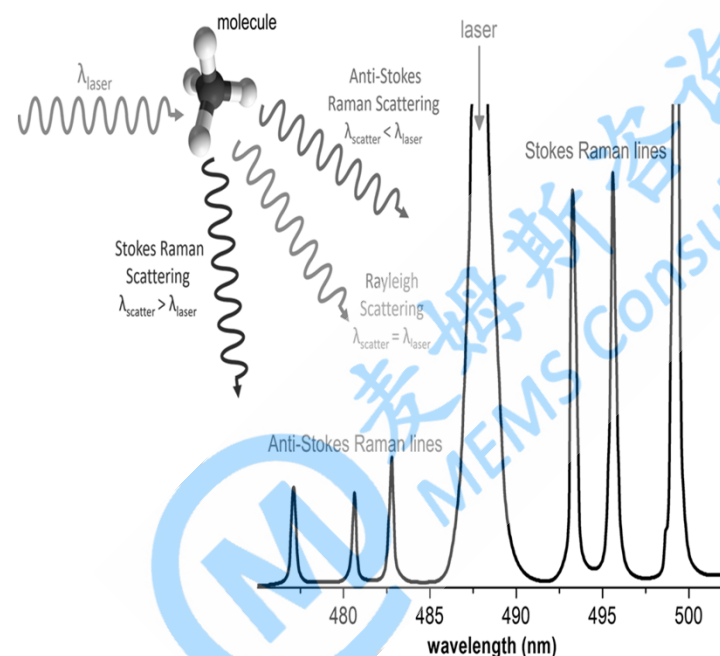


光谱是物质指纹→唯一性

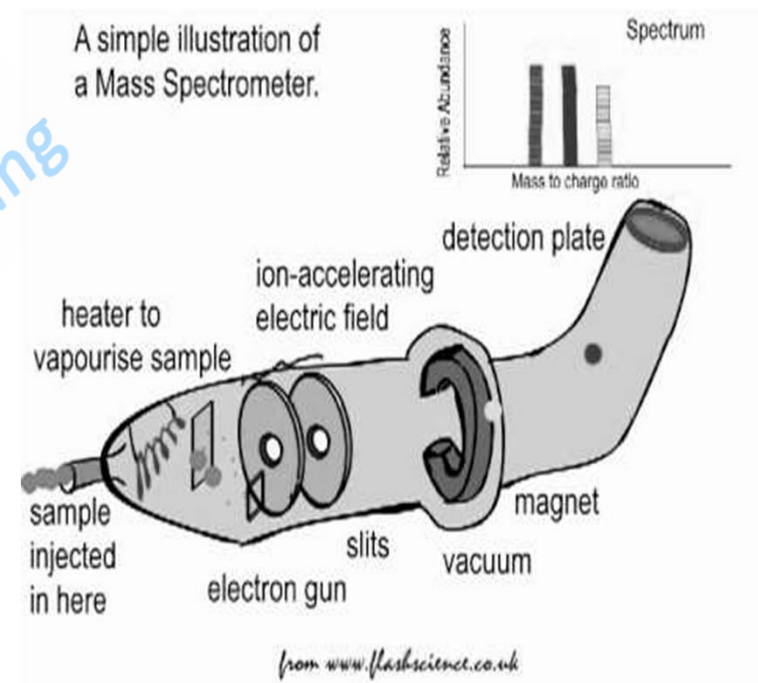
1.引言 什么是光谱?



荧光光谱



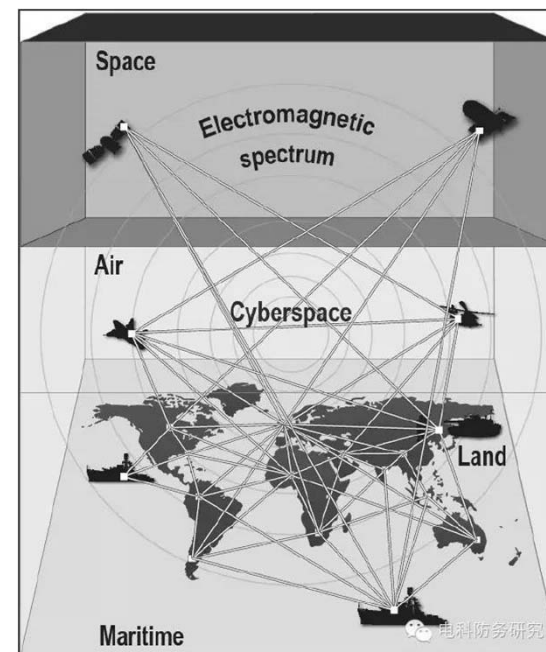
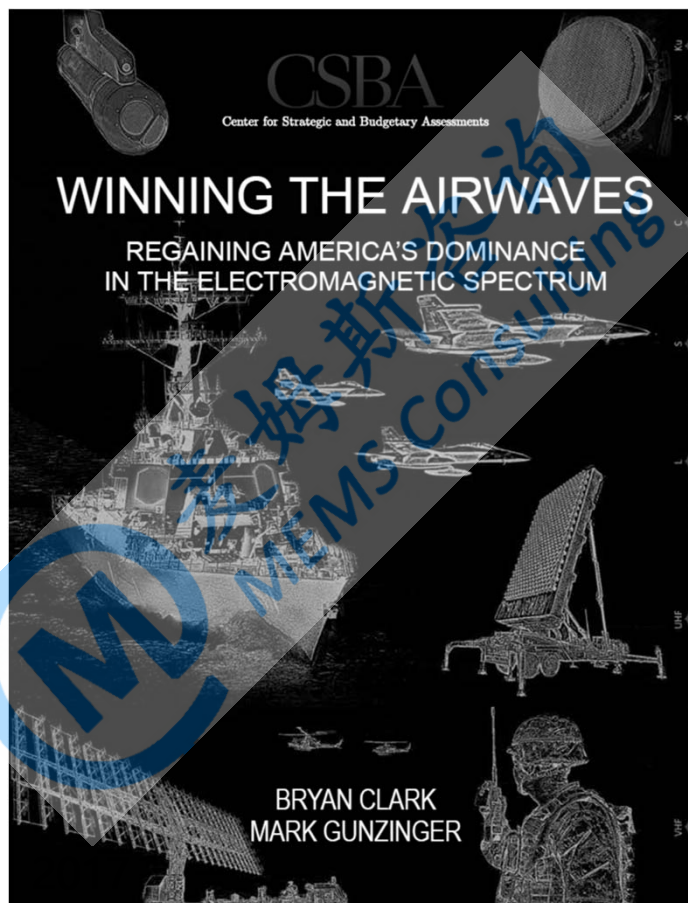
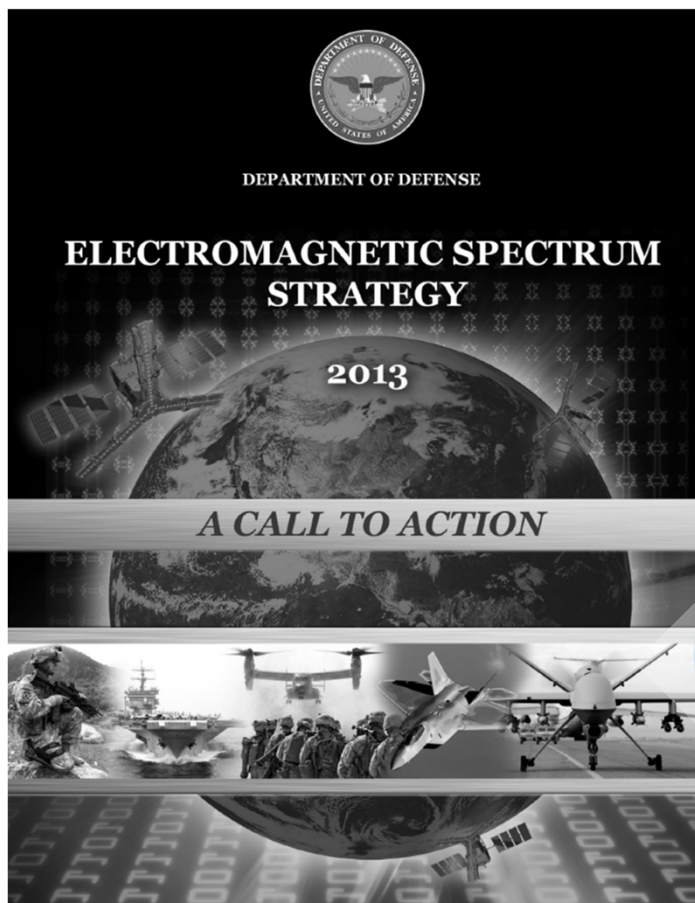
拉曼光谱



质谱

形形色色的光谱应用

1.引言 什么是光谱?

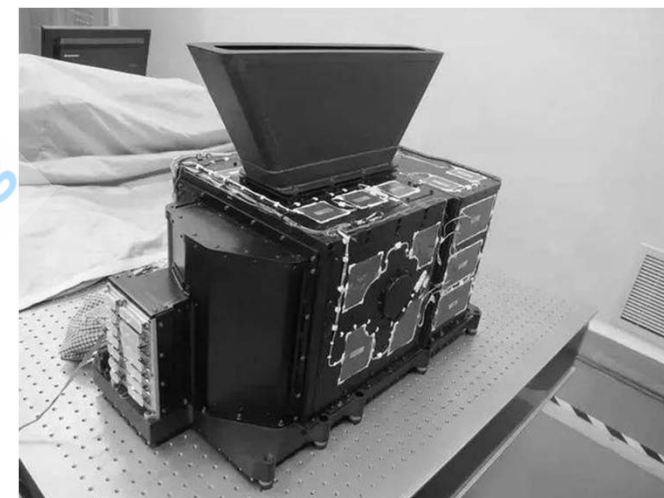


电磁频谱作战域为继
陆、海、空、天、赛博空间
之外的“第6个作战域”

美军高度重视电磁频谱域——电磁频谱战

1.引言 光谱仪及应用

Visible spectrophotometer

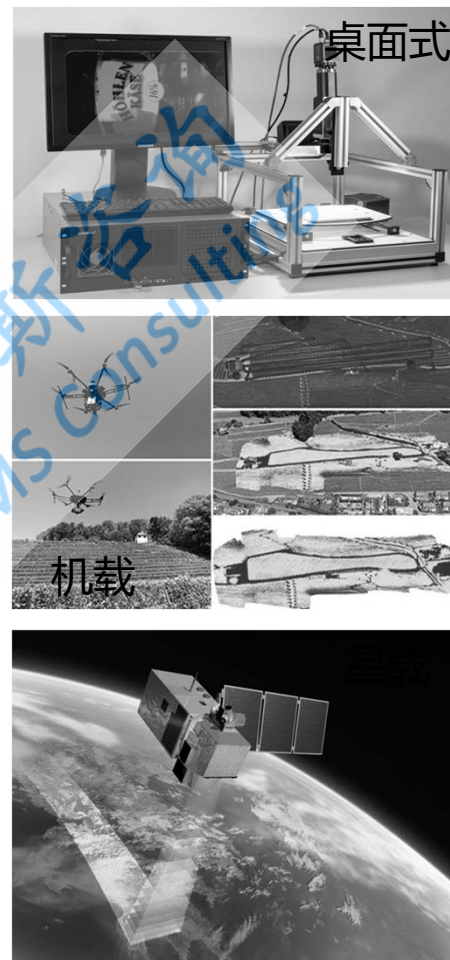
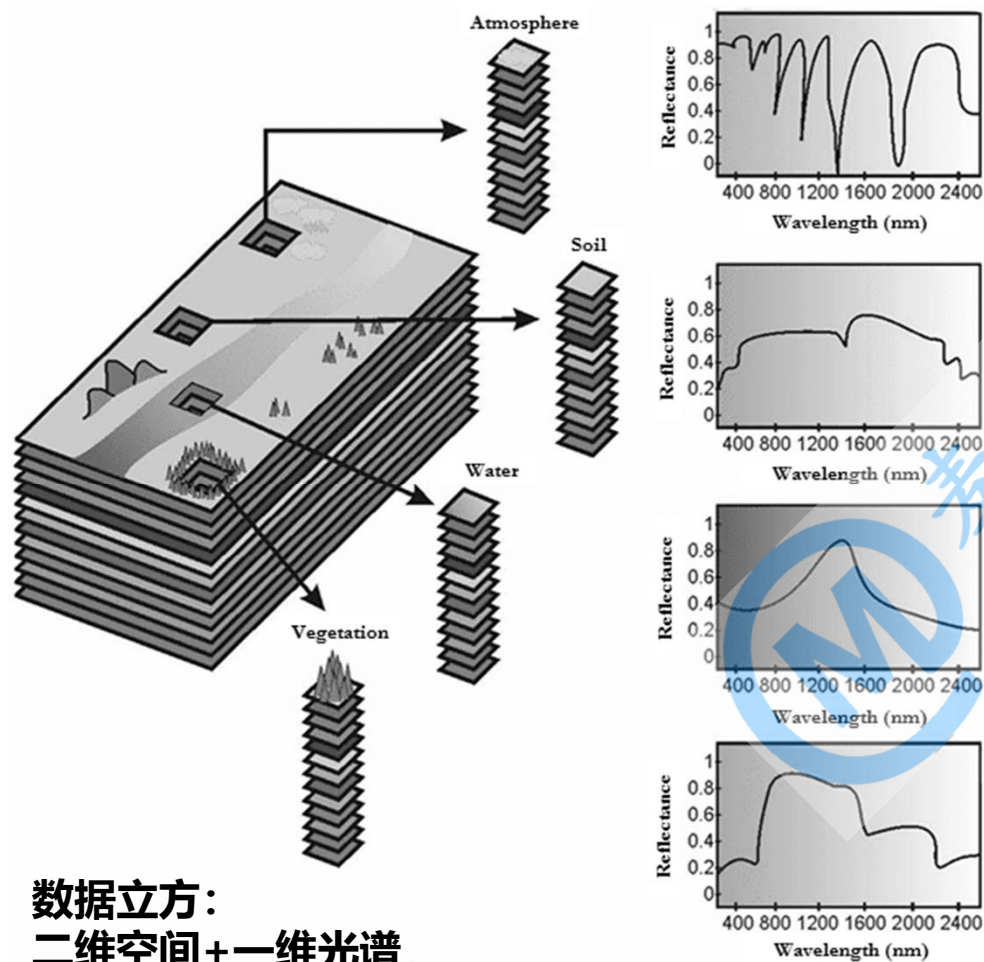


星上定标光谱仪

应用领域:

- | | |
|--------|------------|
| ✓ 医药化工 | ✓ 水质分析 |
| ✓ 食品安全 | ✓ 生命健康 |
| ✓ 气体检测 | ✓ 刑事侦查 |
| ✓ 深空探测 | ✓ 防伪鉴别 ... |

1.引言 光谱成像仪及应用



应用领域：

- ✓ 遥感遥测
- ✓ 食品安全
- ✓ 地质勘探
- ✓ 环境监测
- ✓ 智慧农业
- ✓ 石油化工
- ✓ 军事国防
- ✓ 医疗诊断
- ✓ 防伪鉴别...



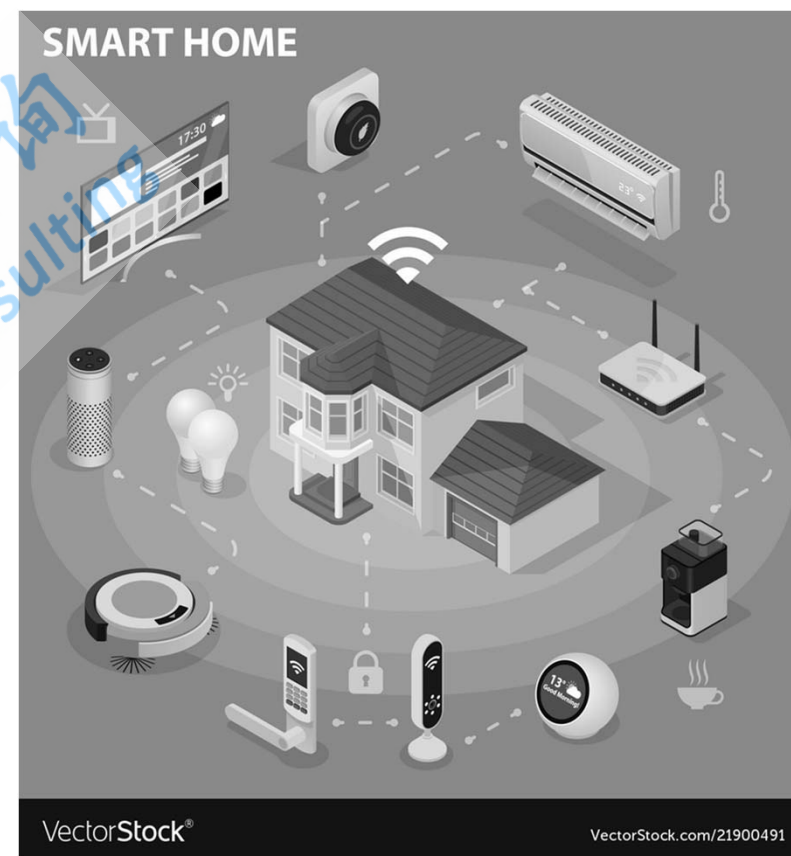
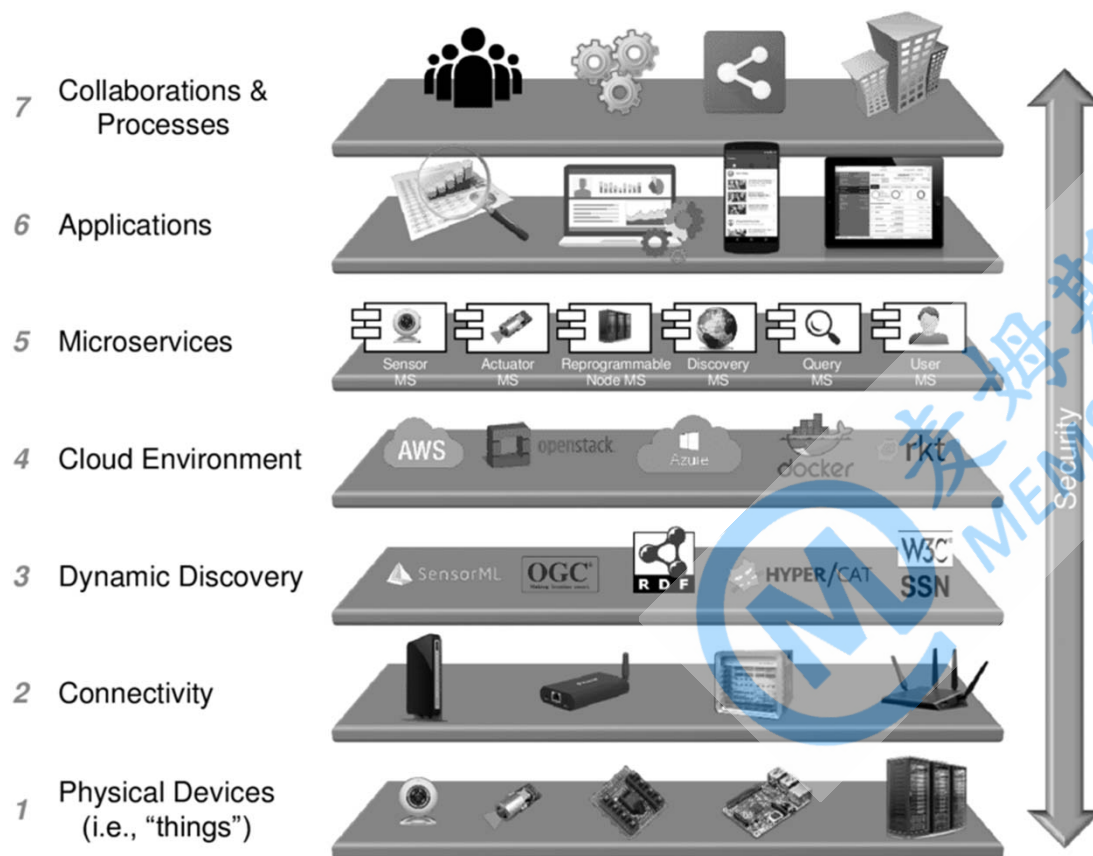
1.引言 为什么要微型化?

物联网（Internet of Things）是在互联网和通信网络的基础上，将日常用品、设施、设备、车辆和其他物品互相连通的网络。作为一个广义的概念，物联网利用传感器、通信网络、软件、控制系统等将物品与网络和其他物品进行连接和互动，实现对现实世界的数字化和自动化。**物联网改变了互联网中信息全部由人获取和创建，以及物品全部需要人类指令和操作的情况**，未来将深远地影响生产生活中的每个方面。未来，世界上物和物互联的规模将远超人和人互联的规模，这种指数型的增长主要来自物品与物品之间多种多样的连接与自主运行。



来源：艾瑞咨询研究院自主研究及绘制。

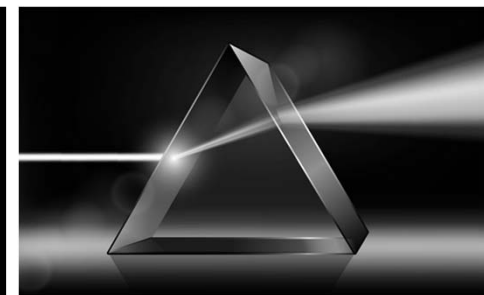
1.引言 为什么要微型化?



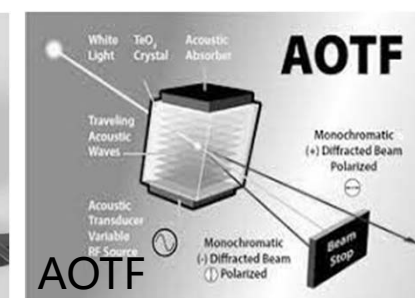
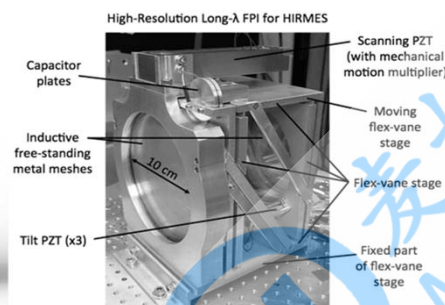
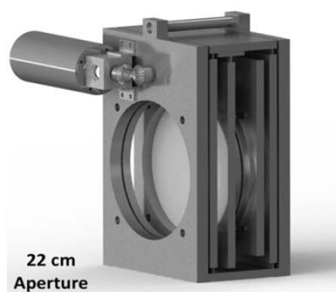
工业、医疗、交通、建筑、农业、电力、军事.....

1.引言 微型化解决方案

色散型：
光栅、棱镜



滤光型：FPI

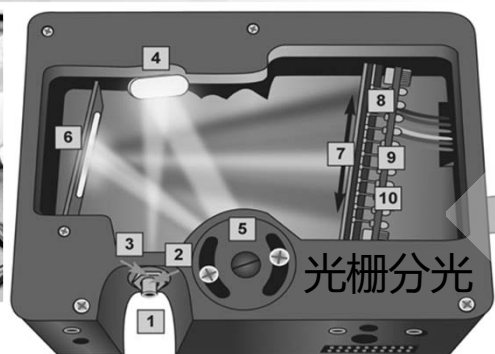


干涉型：FTIR



几种传统的光谱获取方式

1.引言 微型化解决方案



工作原理图

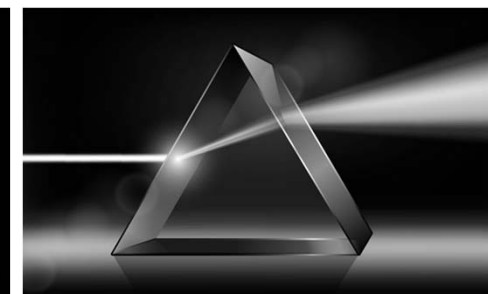
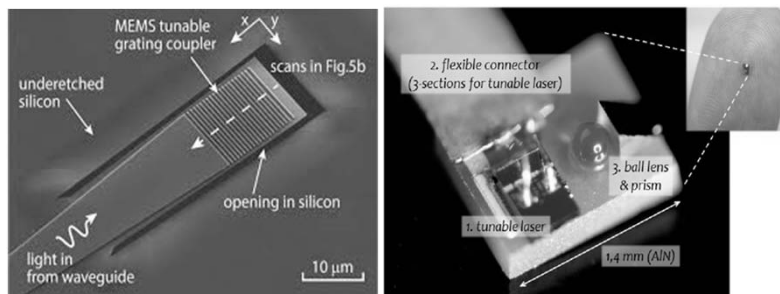
Industries		
Spectrometer		
STS	Consumer Electronics	Lighting & Illumination
Flame	Education & Academia	Manufacturing
HR	Energy, Oil & Gas	Medical & Life Sciences
Maya2000 Pro	Environmental	Pharma
QE Pro	Food & Agriculture	Safety & Security
NIRQuest	Government	Semiconductor
NanoQuest		
HDX	Laser & Photonics	
OFX		

小型化光谱仪典型代表：Ocean Optics

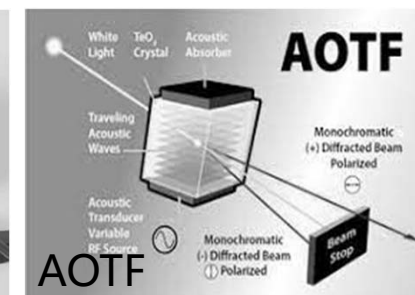
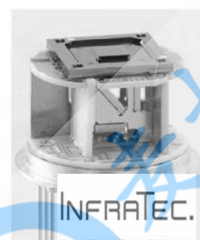
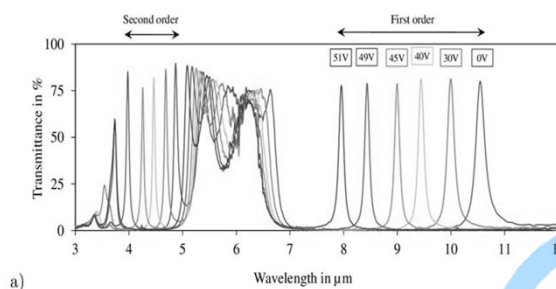


1.引言 微型化解决方案

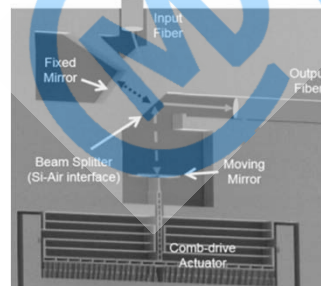
色散型：
光栅、棱镜



滤光型：FPI

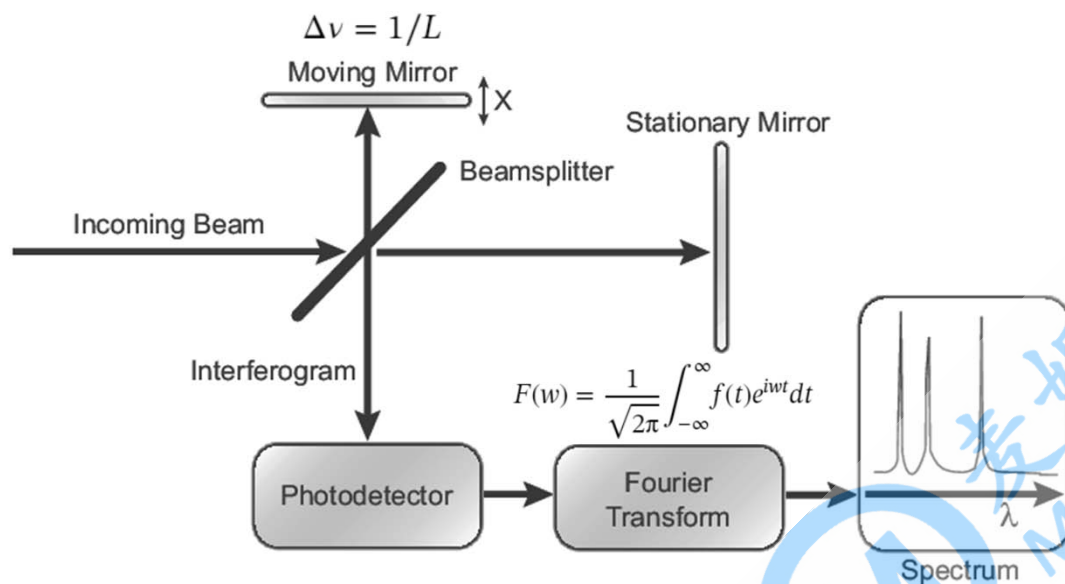


干涉型：FTIR

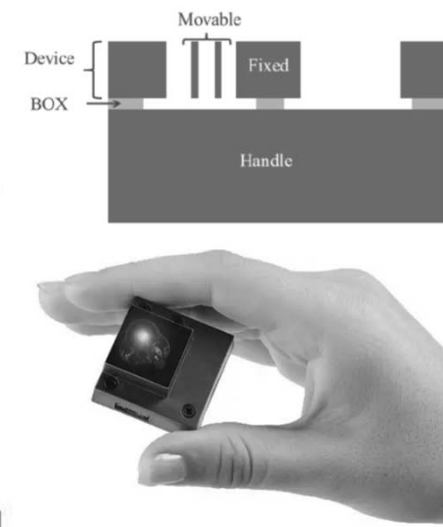
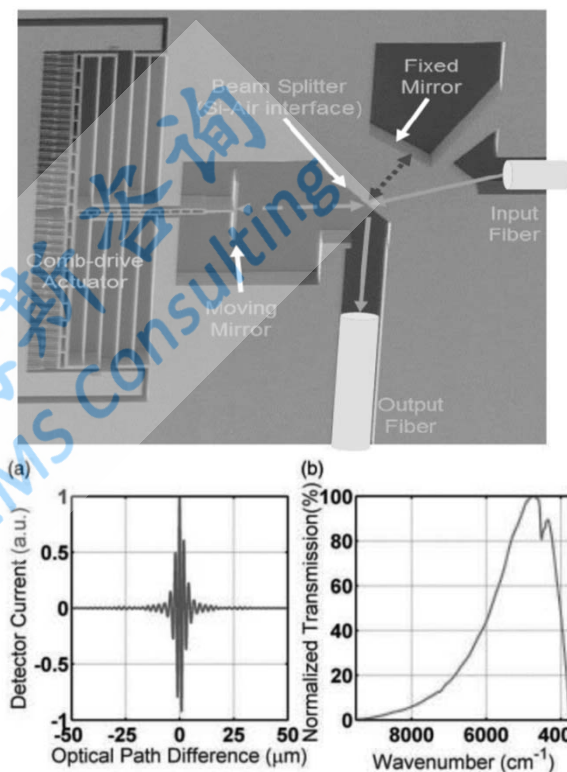


几种传统的光谱获取方式

1.引言 微型化解决方案



FTIR之迈克尔逊干涉仪



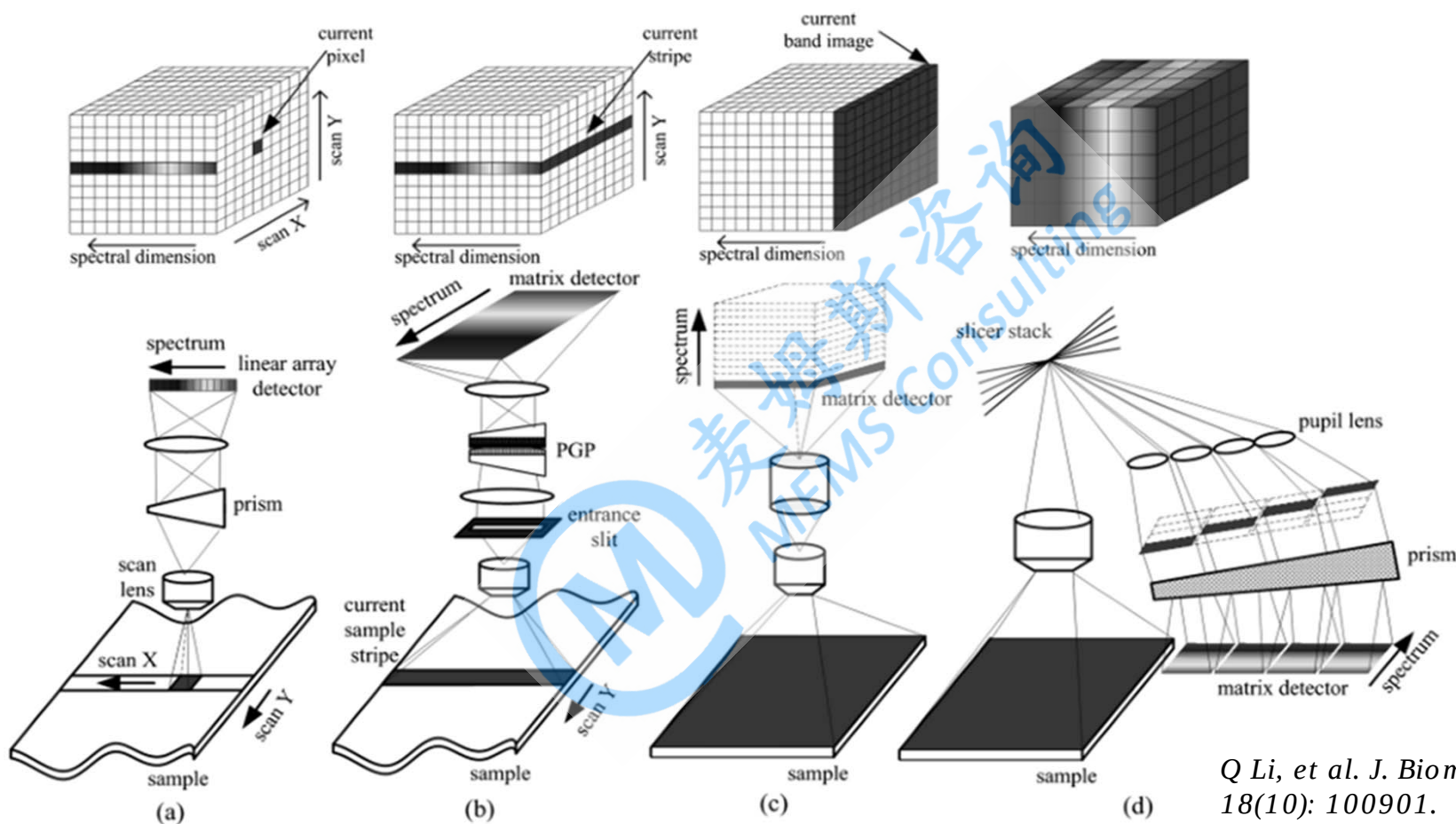
$32 \times 32 \times 22 \text{ mm}^3$
weight: 17g
 λ : 1350-2500nm
 $\Delta\lambda$: 16nm

Y M Sabry, et al. Appl. Spect., 2016.

Y M Sabry, et al. Proc. SPIE, 2017, 10116.

微型化光谱仪典型代表: **Si-Ware** (<https://www.si-ware.com/>)

1.引言 微型化解决方案



Q Li, et al. J. Biomed. Opt., 2013, 18(10): 100901.

几种常用的光谱图像获取方式

1.引言 微型化解决方案



United States Patent [19] 3,971,065
Bayer (45) July 20, 1976

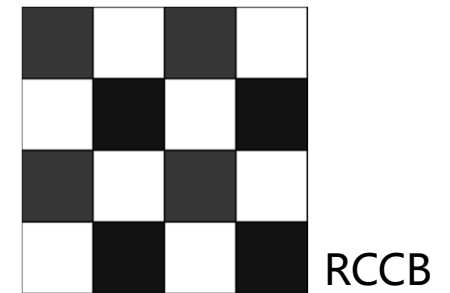
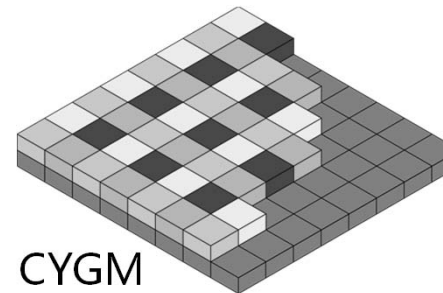
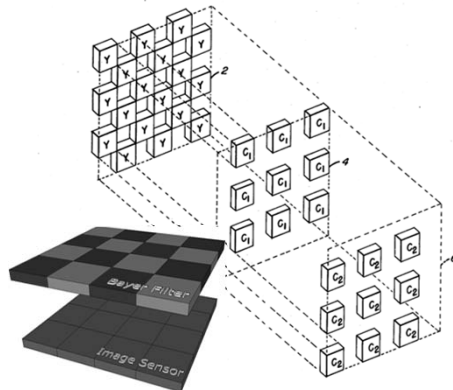
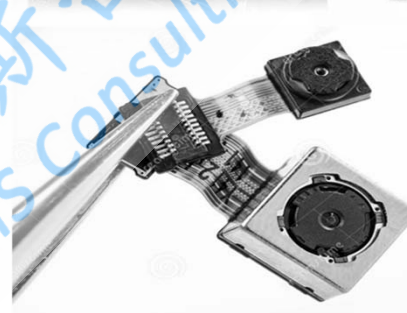
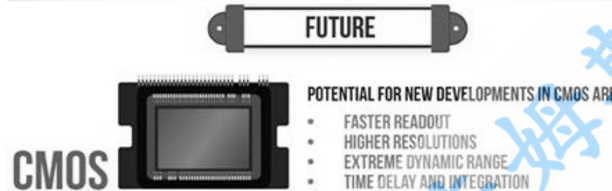
[54] COLOR IMAGING ARRAY
[75] Inventor: Bryce E. Bayer, Rochester, N.Y.
[73] Assignee: Eastman Kodak Company, Rochester, N.Y.
[22] Filed: Mar. 5, 1975
[21] Appl. No.: 555,477

[52] U.S. Cl. 358/41; 350/162 SP; 350/317; 358/44
[51] Int. Cl. H04N 9/24
[58] Field of Search 358/44, 45, 46, 47, 358/48; 350/317, 162 SP; 315/169 TV

[56] References Cited
UNITED STATES PATENTS
2,446,791 8/1948 Schroeder 358/44
2,508,267 5/1950 Kasperowicz 358/44
2,884,483 4/1959 Eberhart et al. 358/44
3,725,572 4/1973 Kurokawa et al. 358/46

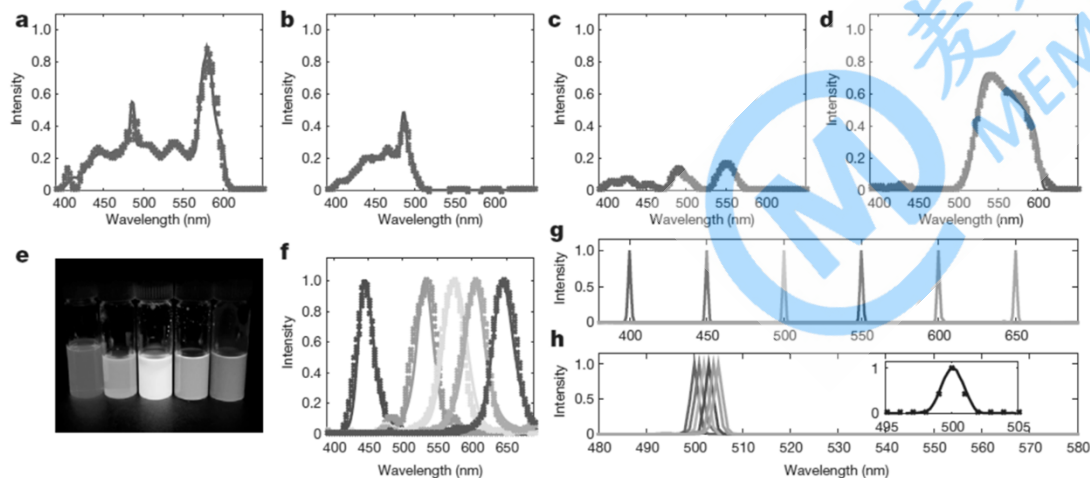
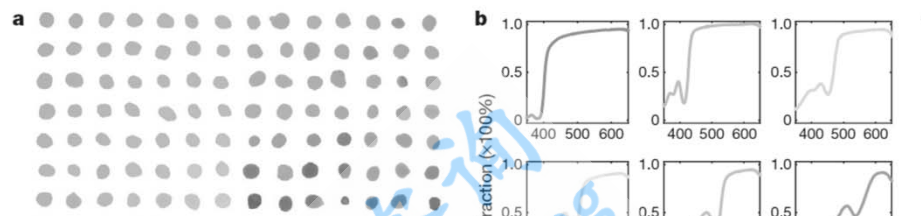
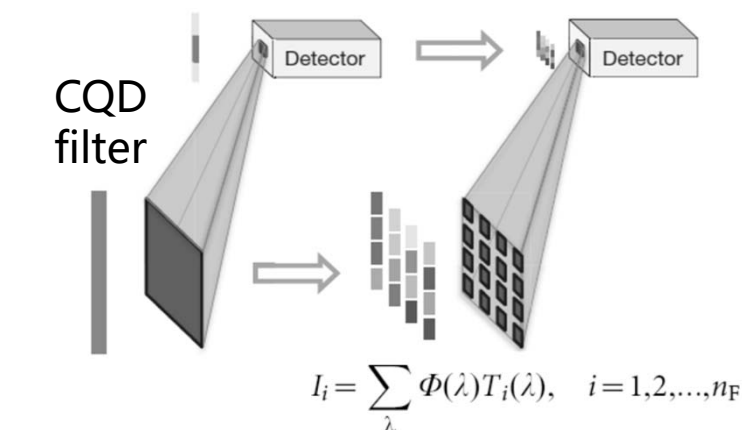
Primary Examiner—George H. Libman
Attorney, Agent, or Firm—George E. Grosser

11 Claims, 10 Drawing Figures



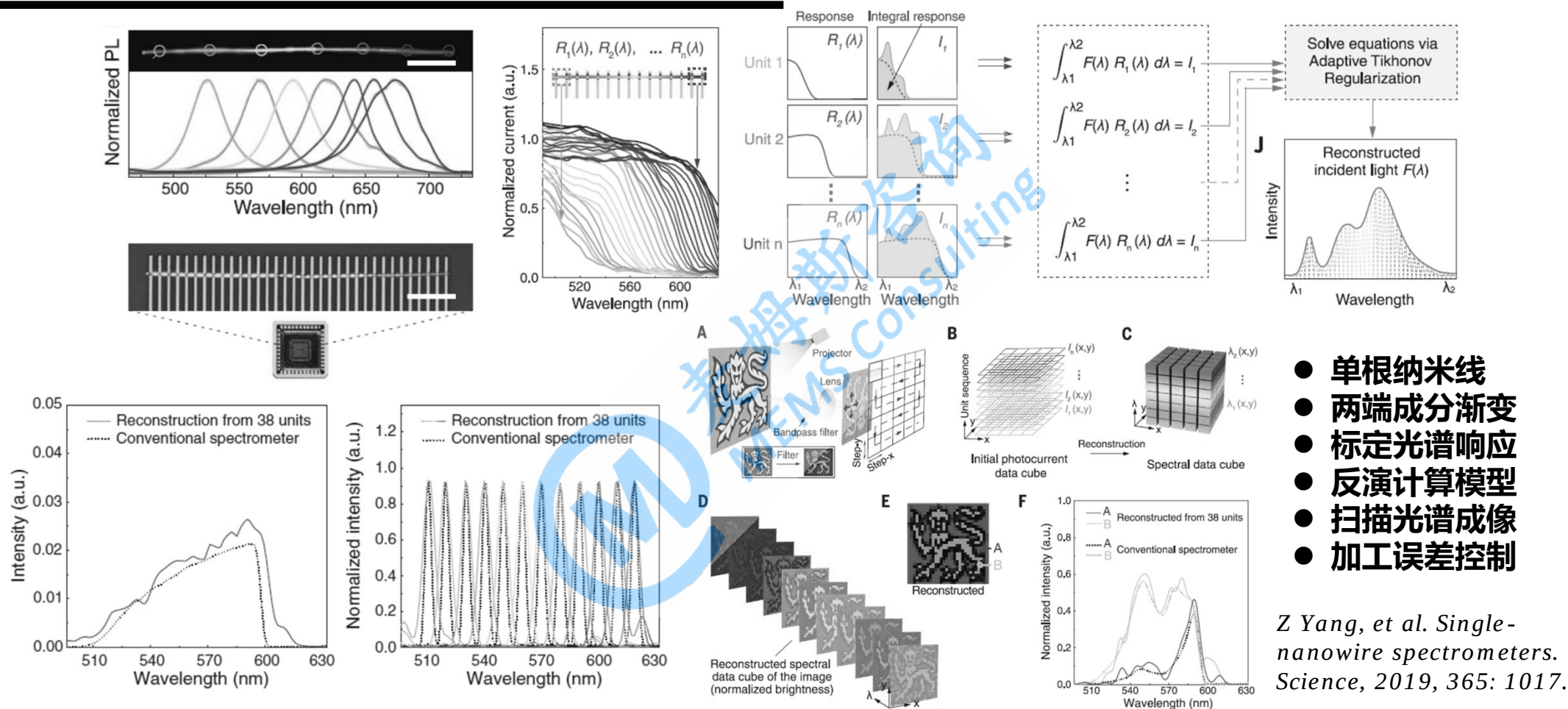
一种常被忽略的光谱图像获取方式——Bayer滤波

1.引言 微型化解决方案



新型光谱获取方式：量子点（计算重构）

1.引言 微型化解决方案

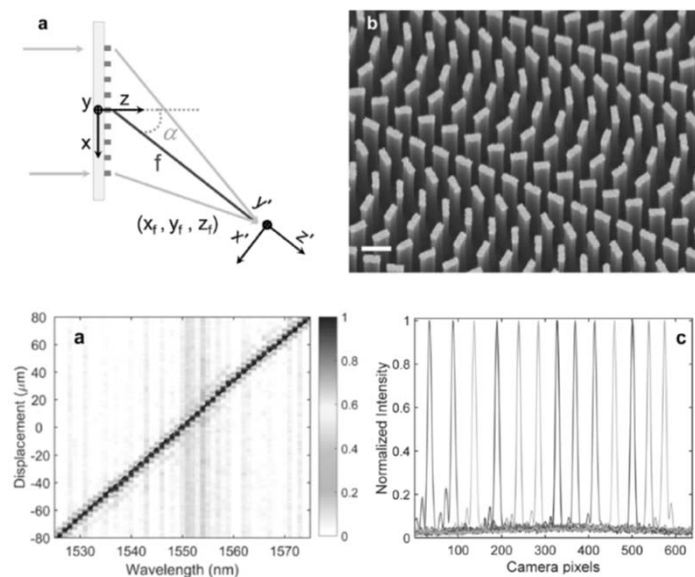


- 单根纳米线
- 两端成分渐变
- 标定光谱响应
- 反演计算模型
- 扫描光谱成像
- 加工误差控制

Z Yang, et al. Single-nanowire spectrometers. *Science*, 2019, 365: 1017.

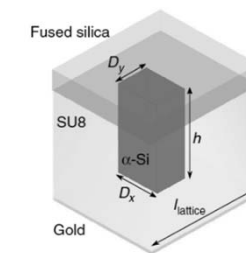
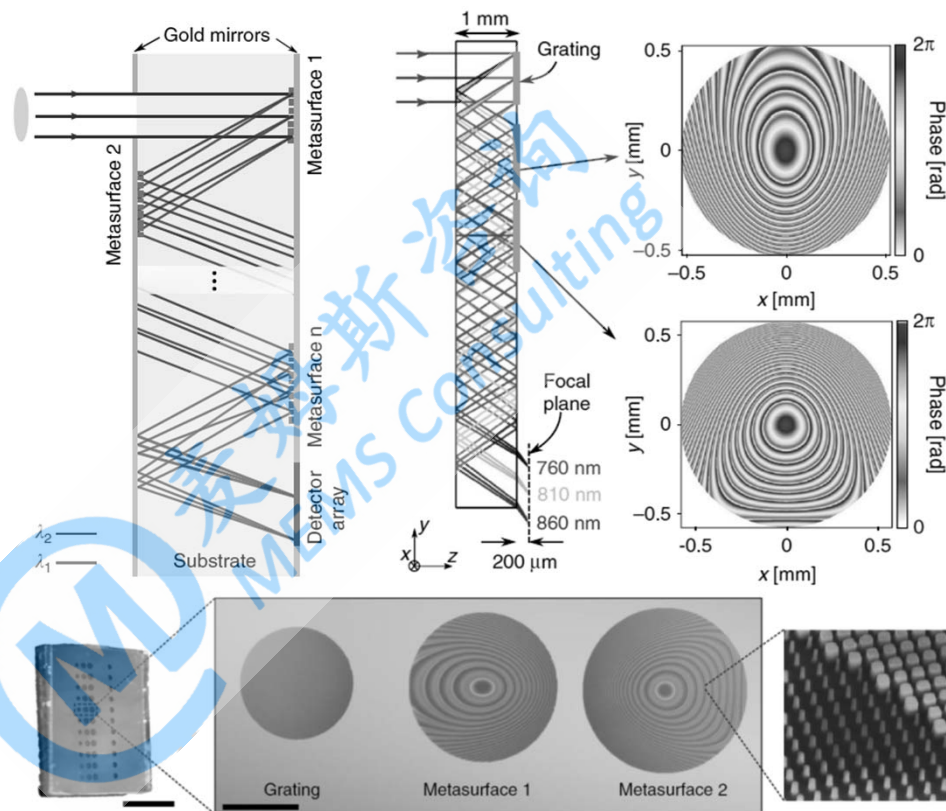
新型光谱获取方式：纳米线（计算重构）

1.引言 微型化解决方案



F Capasso et al. Nano Lett., 2016, 16(6): 3732.

硅纳米柱; 偏轴聚焦超透镜; 内在色散
波长范围1.1-1.6 μm ; 光谱分辨率0.2 μm



硅纳米柱; 光栅+超表面
聚焦反射镜+反射镜
波长范围760-860nm;
光谱分辨率1.2nm

A Faraon et al. Nat. Commun., 2018, 9: 4196.

新型光谱获取方式：超表面



提 纲

1、引言

2、光栅色散型光谱仪

- (1) 衍射光栅（反射/透射光栅）分光原理
- (2) 硅基微纳光栅：设计与加工
- (3) 基于硅基微纳光栅的微型光谱仪及其应用

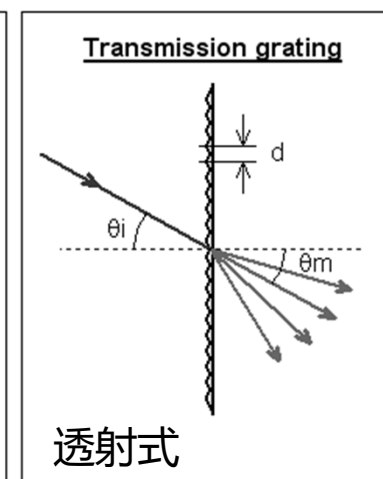
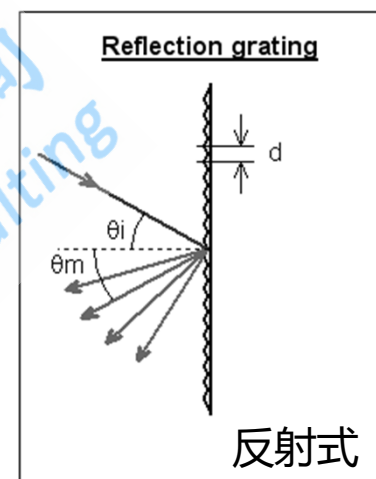
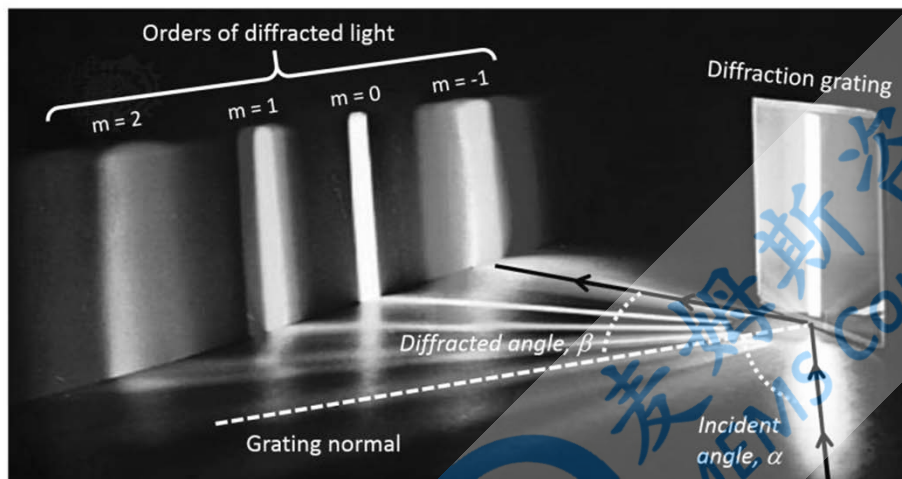
3、F-P干涉型光谱仪

- (1) F-P干涉仪分光原理
- (2) F-P全固态分光器件及其制造方法
- (3) 基于F-P全固态分光器件的微型光谱仪
- (4) F-P动态分光器件及其制造方法
- (5) 基于F-P动态分光器件的微型光谱仪
- (6) 基于F-P干涉技术的微型光谱仪典型应用

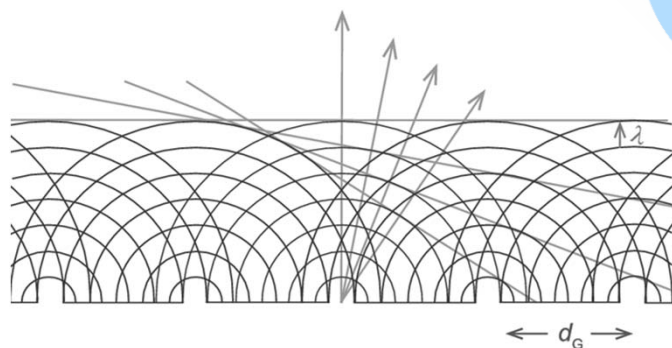
4、总结

2.光栅色散型光谱仪

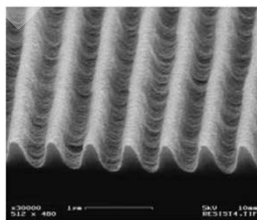
✓ 衍射光栅分光原理



光栅方程: $m\lambda = d_G(\sin\alpha + \sin\beta_m)$.



机械刻划
全息记录

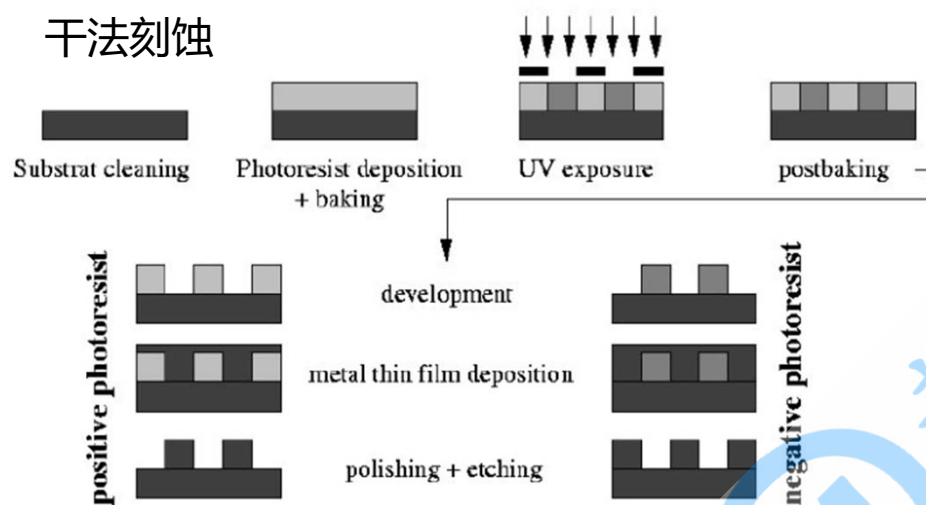


<https://www.newport.com.cn/n/diffraction-grating-physics>

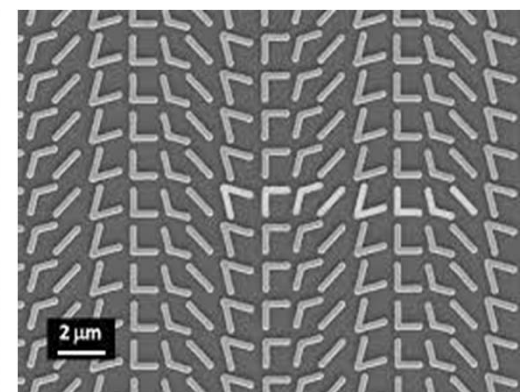
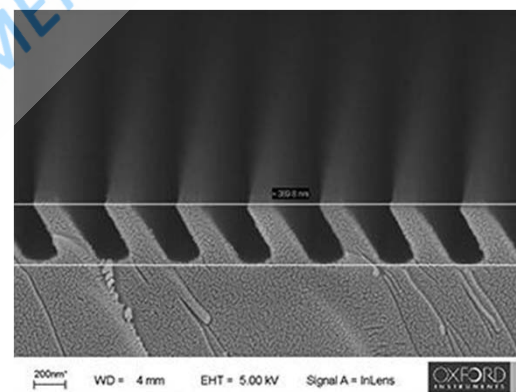
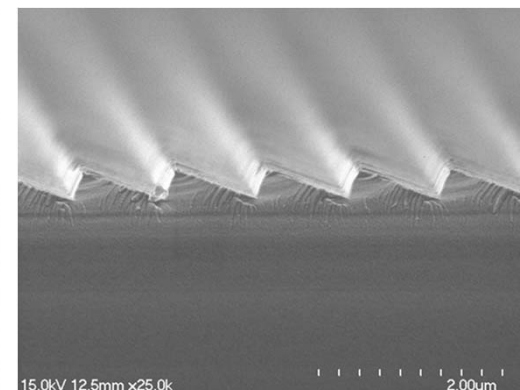
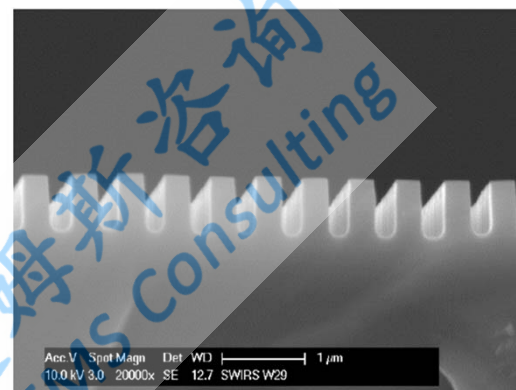
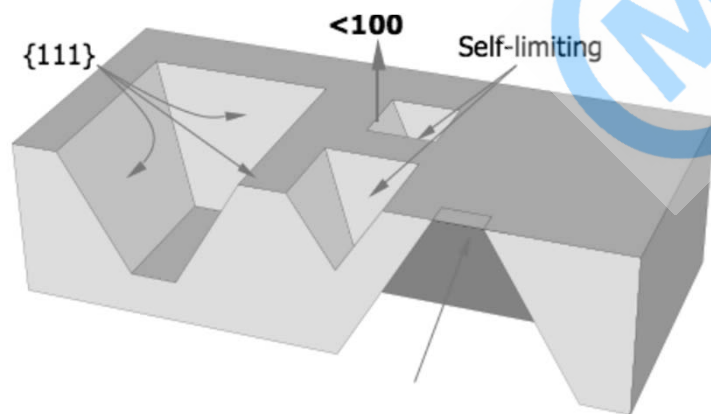
2.光栅色散型光谱仪

✓ 硅微光栅

干法刻蚀

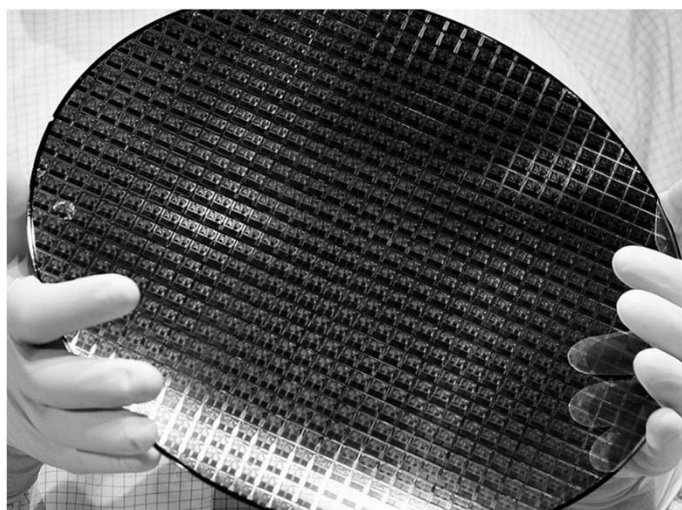


各向异性湿法腐蚀

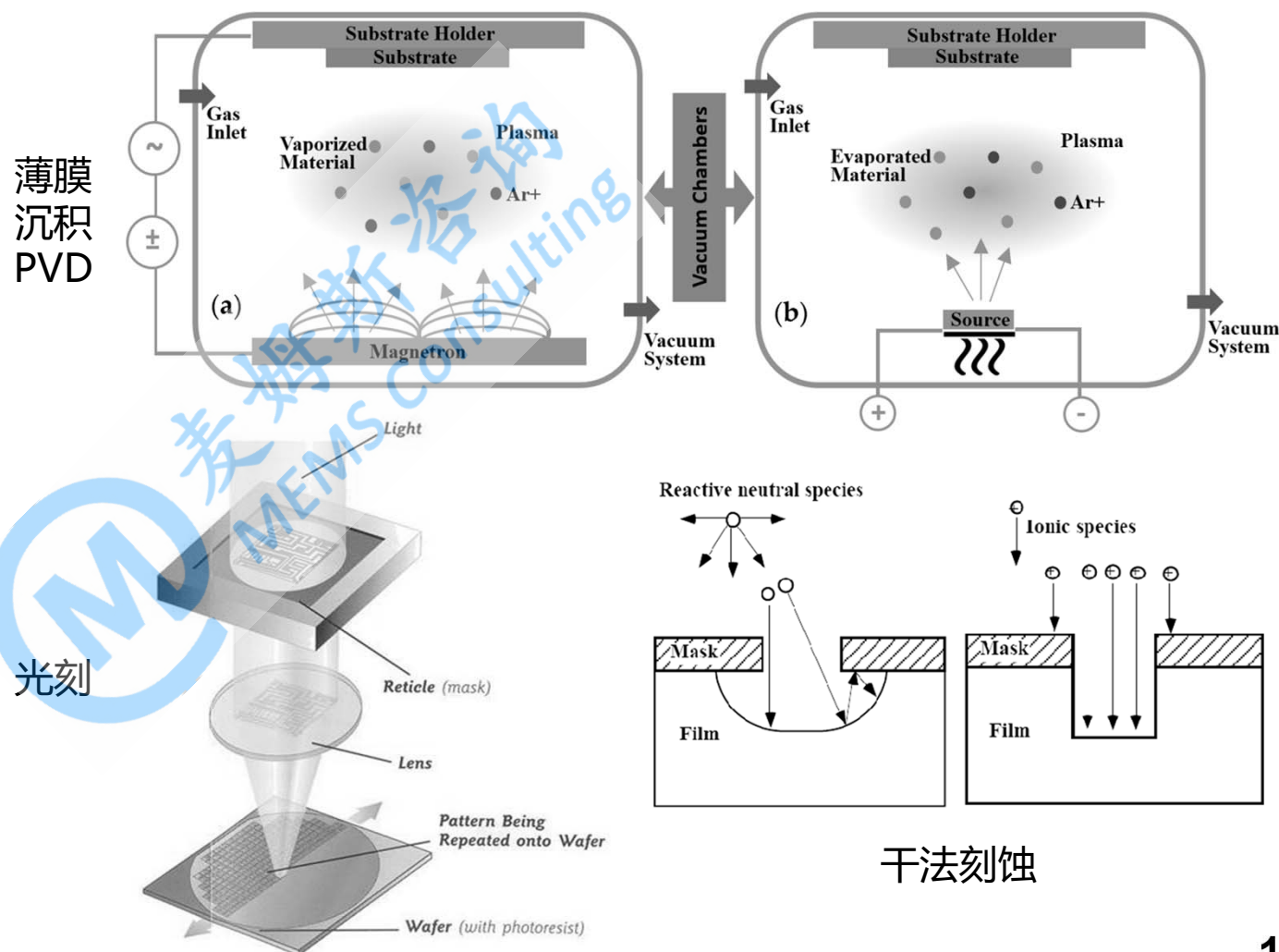


2.光栅色散型光谱仪

✓ 半导体微加工工艺

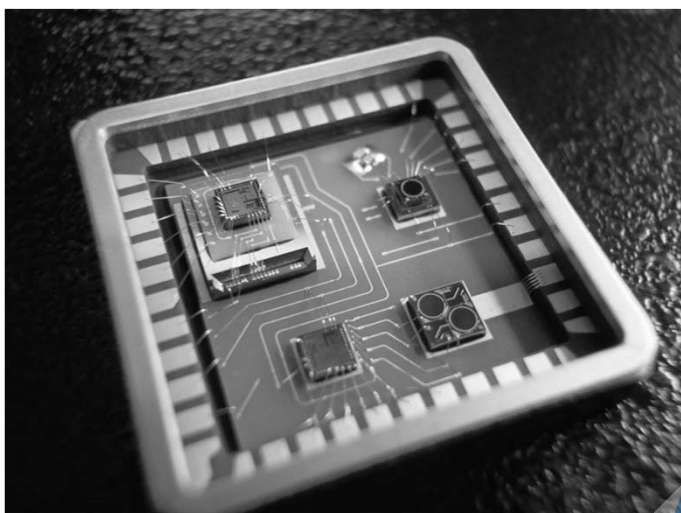


前道工艺:
薄膜沉积、光刻、刻蚀
氧化、注入、清洗、抛光、
退火等

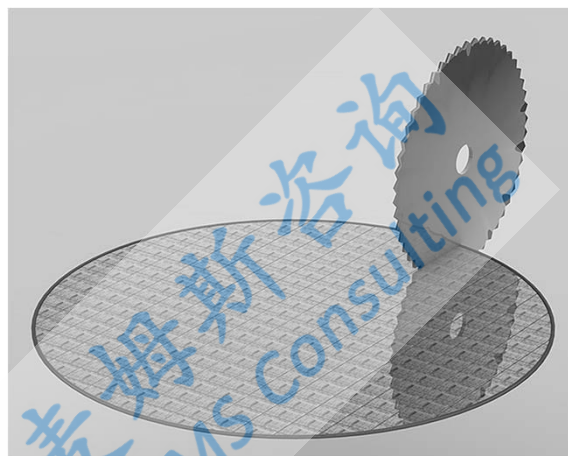


2.光栅色散型光谱仪

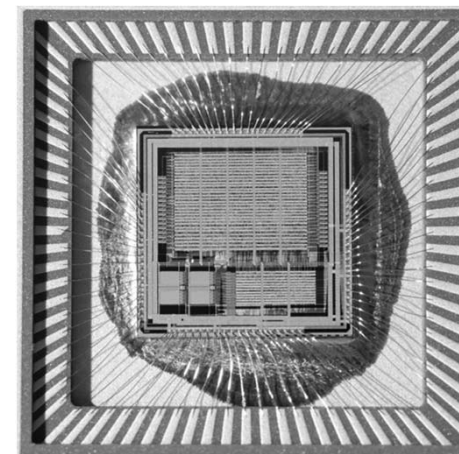
✓ 半导体微加工工艺



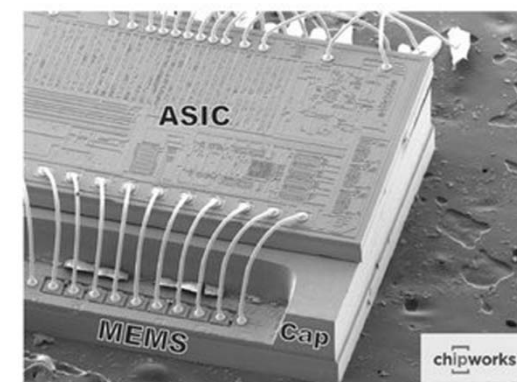
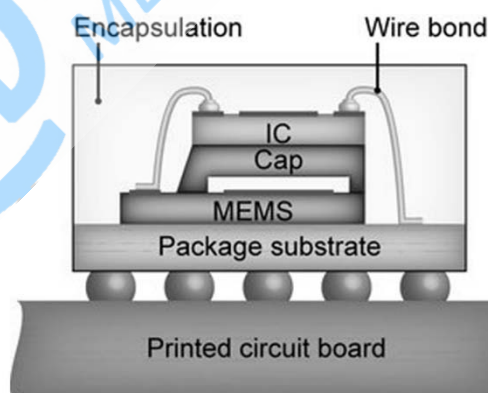
划片



引线



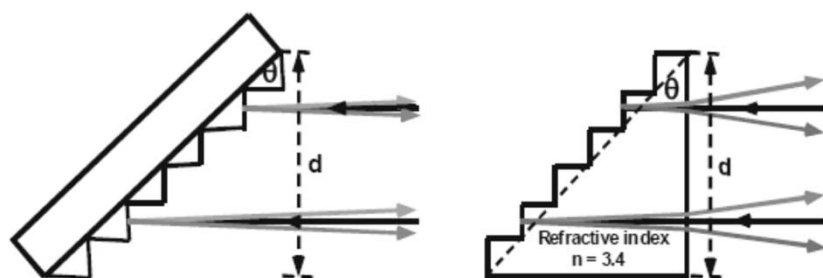
后道工艺：
减薄、划片、引线、键合、
封装、测试



键合

2.光栅色散型光谱仪

✓ 硅微浸没式光栅

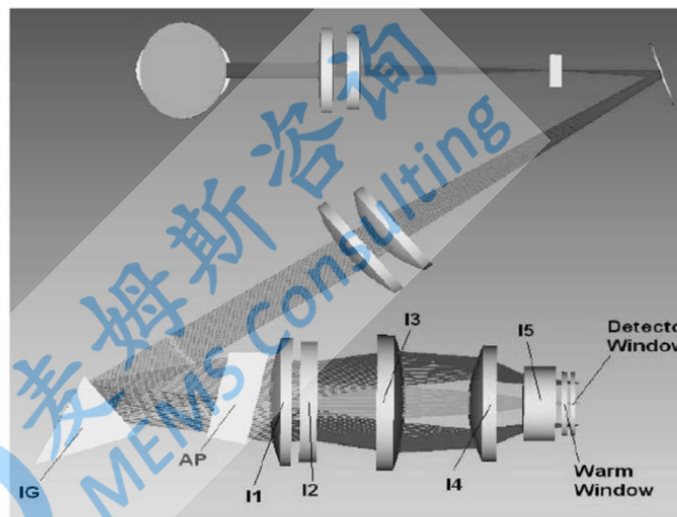


Reflection Grating

Silicon Immersion Grating

Fig. 1. Comparison between a conventional reflective grating and a silicon immersion grating. The spectral dispersion of the silicon immersion grating increases by a factor of 3.4 compared to the conventional grating.

J. Ge, et al. SPIE, 2002, doi.org/10.1117/12.454275.

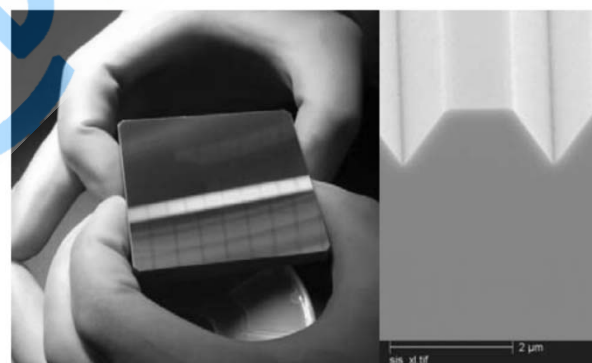


短波红外光谱仪:

1.6-2.4 μm

用于 CO_2 、 CH_4 、 H_2O
等探测

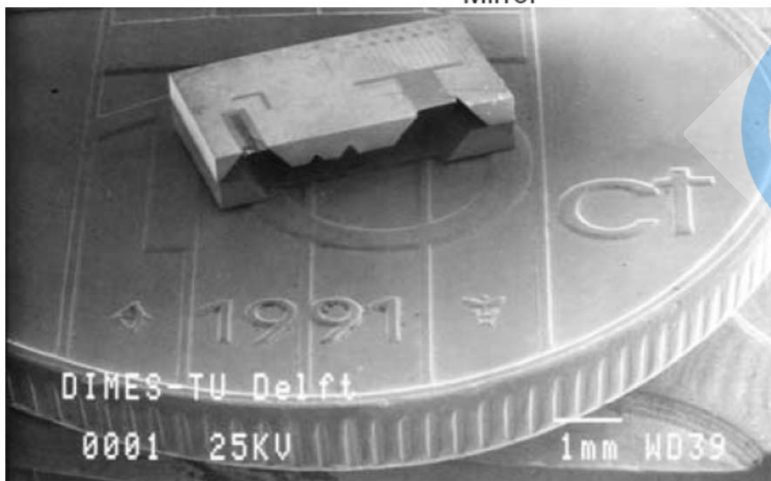
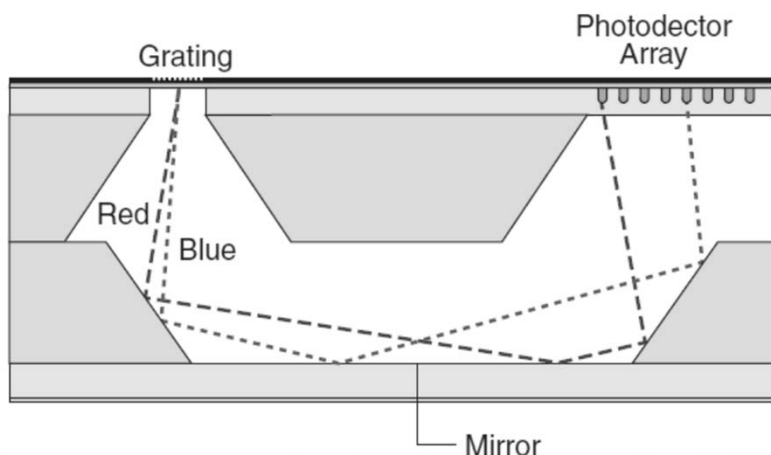
浸没, 意味着衍射发生在硅内部, 由于硅在该波段大的折射系数, 导致光谱仪体积大幅度减小, 这对于航天应用而言是非常有利的。



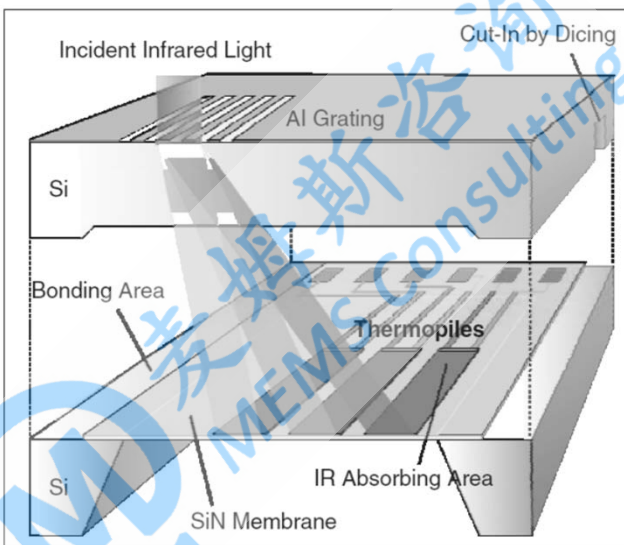
A. Amerongen, et al. Int'l
Conf. on Space Optics, 2012.

2.光栅色散型光谱仪

✓ 光栅式集成光谱仪

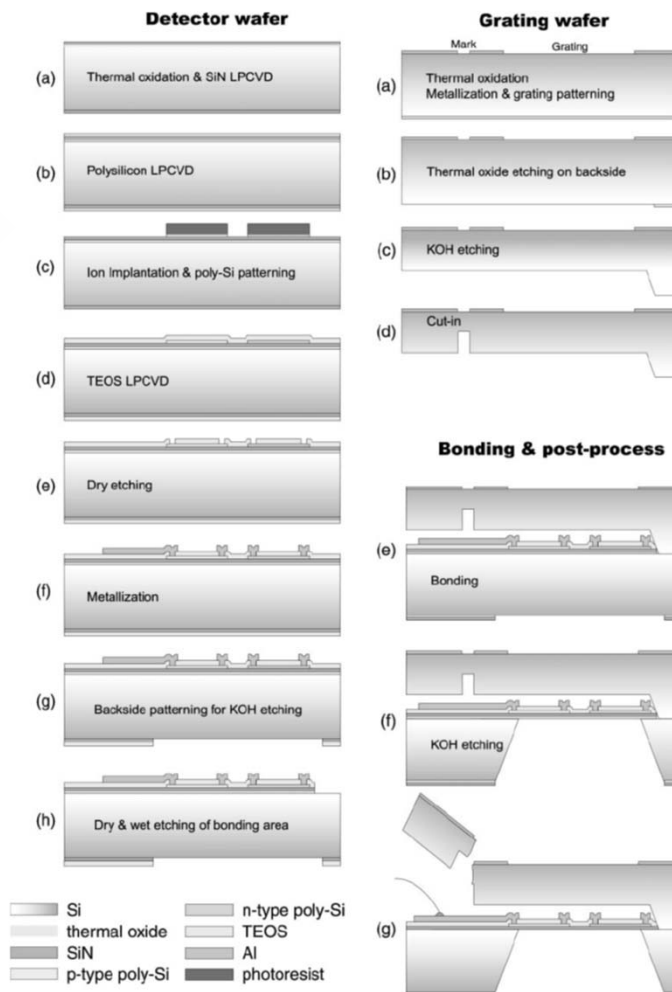


TU Delft
Department of Microelectronics



R.F. Wolffenbuttel, et al. *IEEE Instr. & Meas. Mag.*, 2001: 34.

R.F. Wolffenbuttel, et al. *Sensor. Actuat. A*, 2001, 92: 88.



2.光栅色散型光谱仪

✓ 光栅式集成光谱仪

Grating type compact spectrometer series

MS series

weight : 9g

C10988MA(340~750nm)

C11708MA(640~1050nm)

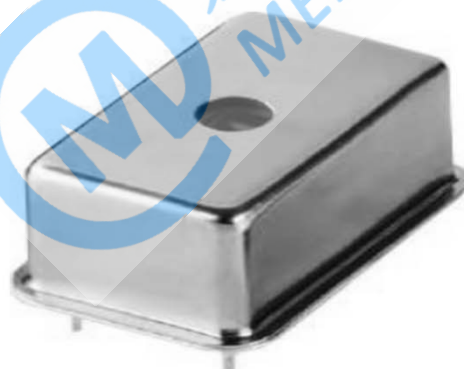


Micro series

weight : 5g

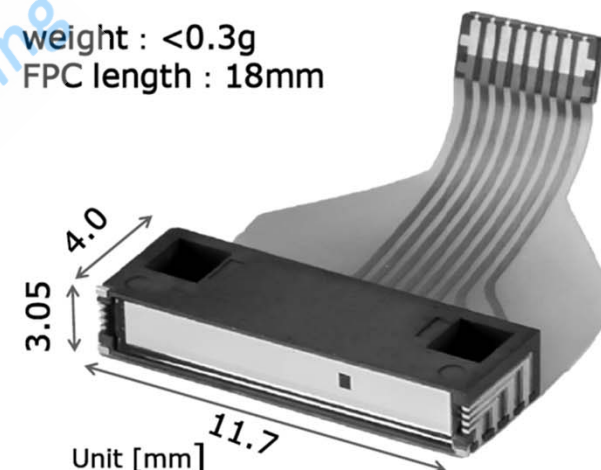
C12666MA(340~780nm)

C12880MA(340~850nm)



HAMAMATSU
PHOTON IS OUR BUSINESS

weight : <0.3g
FPC length : 18mm

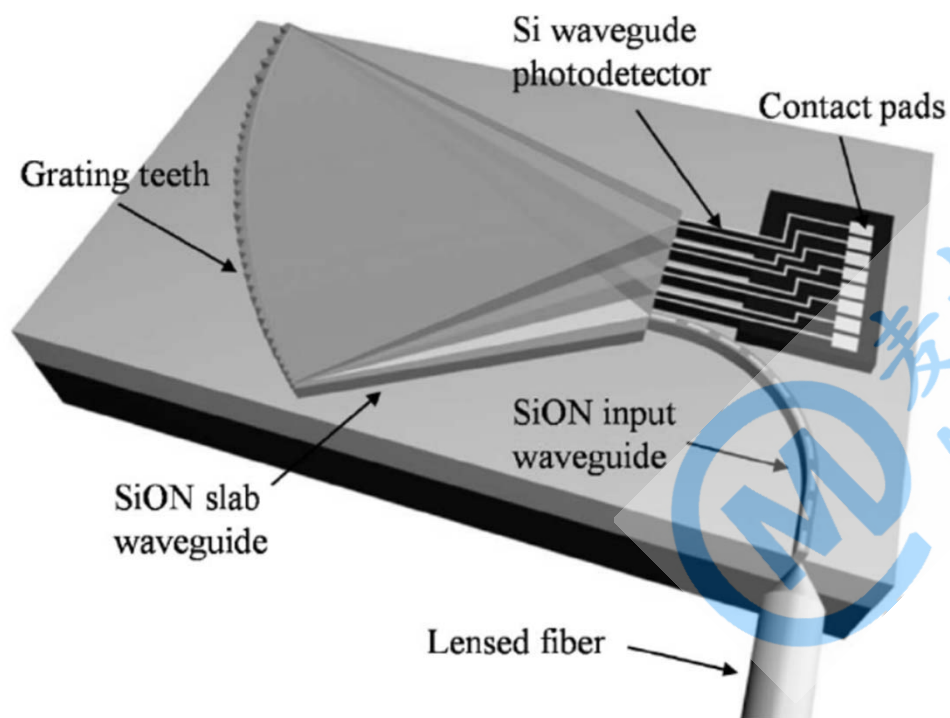


\1 Coin

volume:1/40

2.光栅色散型光谱仪

✓ 光栅式集成光谱仪



CMOS兼容集成光谱仪工作原理图

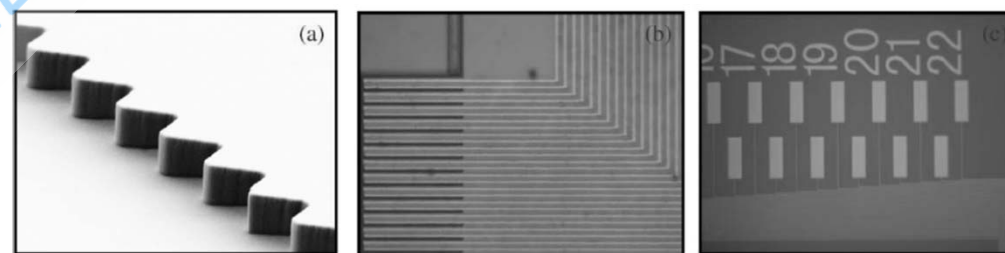
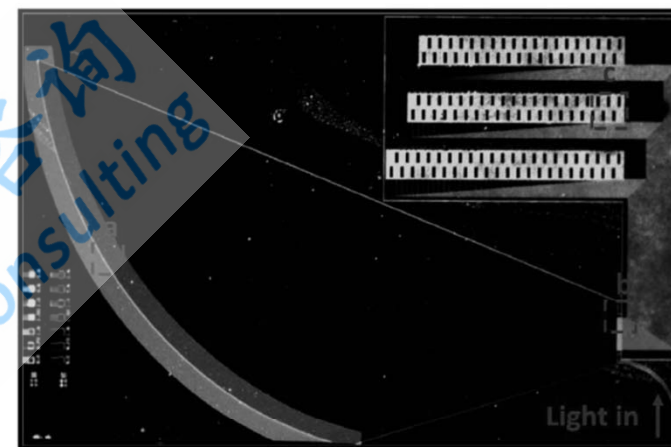
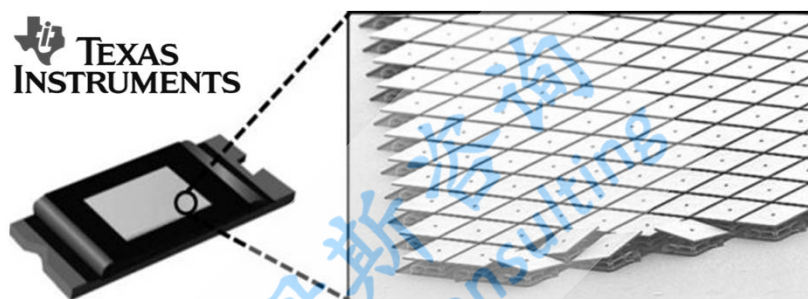
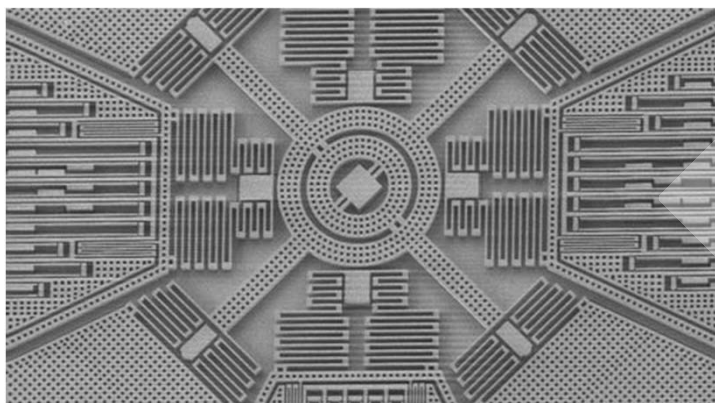
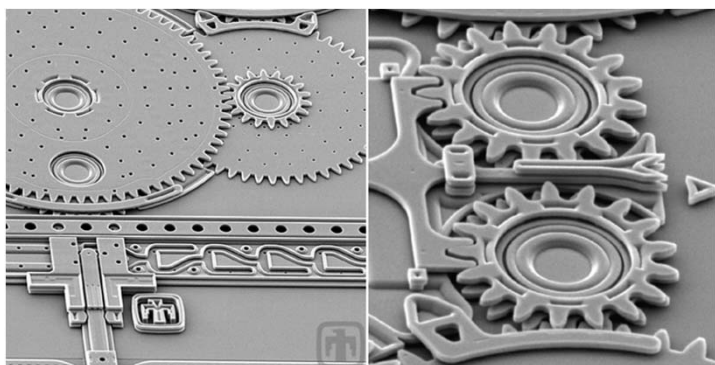


Fig. 4. (Top) image of the spectrometer chip and (bottom) the expanded views: (a) SEM image of the grating teeth; (b) microscope image of the electrode and electrical line; (c) microscope image of contact pads.

J.J. He, et al. *IEEE Photon. J.*, 2013, 5(2): 6600807.

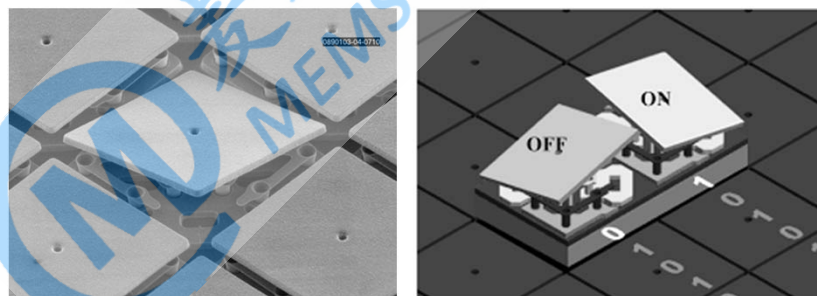
2.光栅色散型光谱仪

✓ MEMS光栅

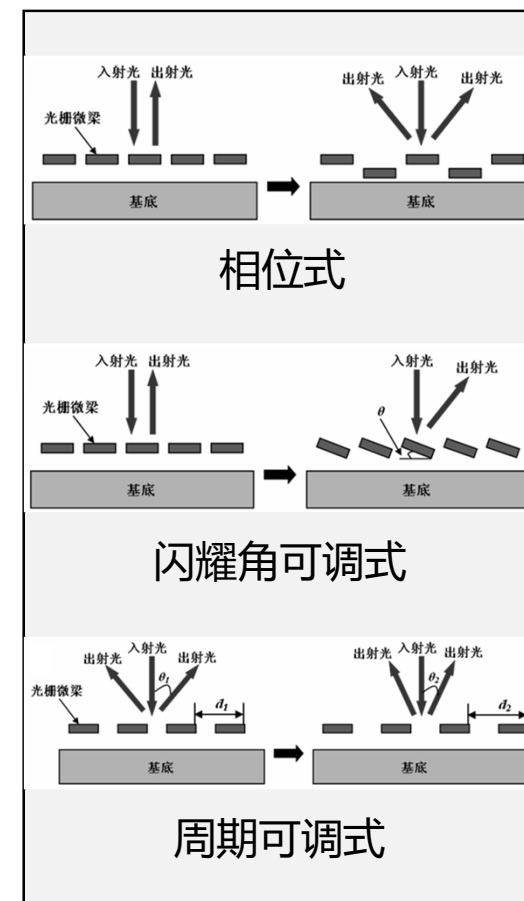


Digital Micromirror Device (DMD)

Array of micromirrors



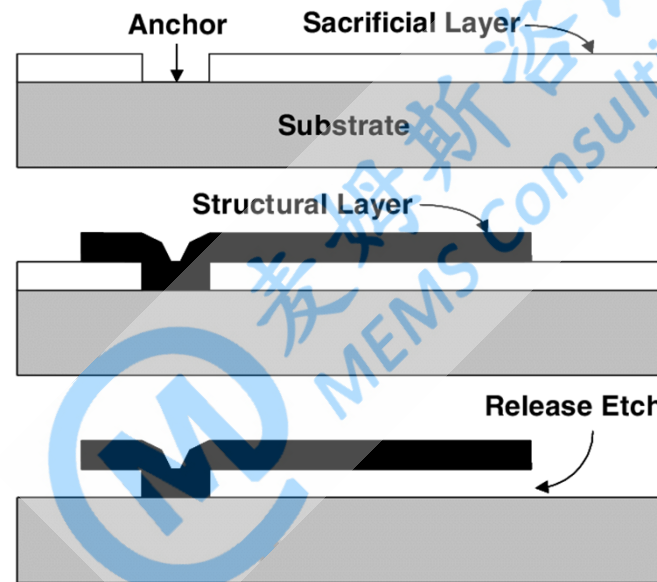
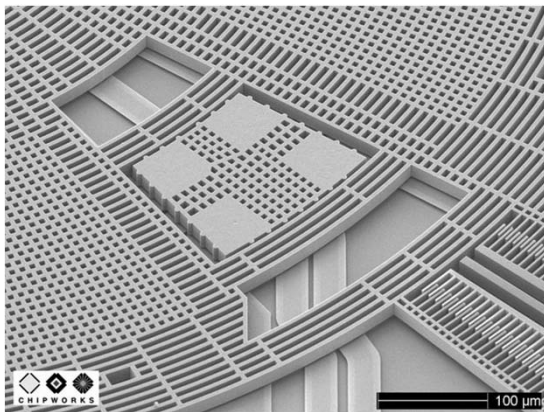
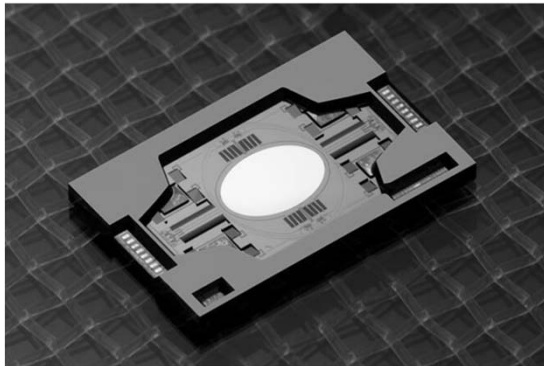
TI DMD (数字微镜器件)



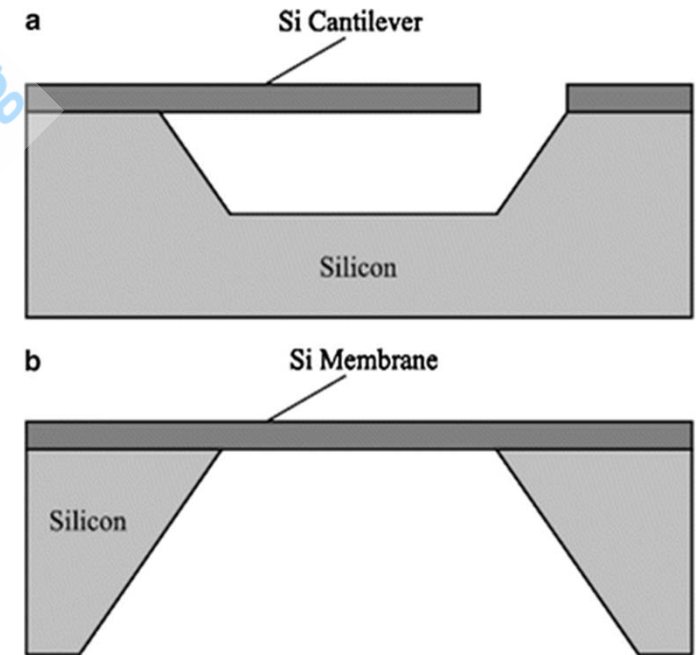
微机电系统MEMS：采用集成电路制造工艺，融合机械、电子、材料、力学、光学、热学、磁学等多个学科，属于前沿高技术领域，在智能革命浪潮中将扮演重要角色。

2.光栅色散型光谱仪

✓ MEMS工艺



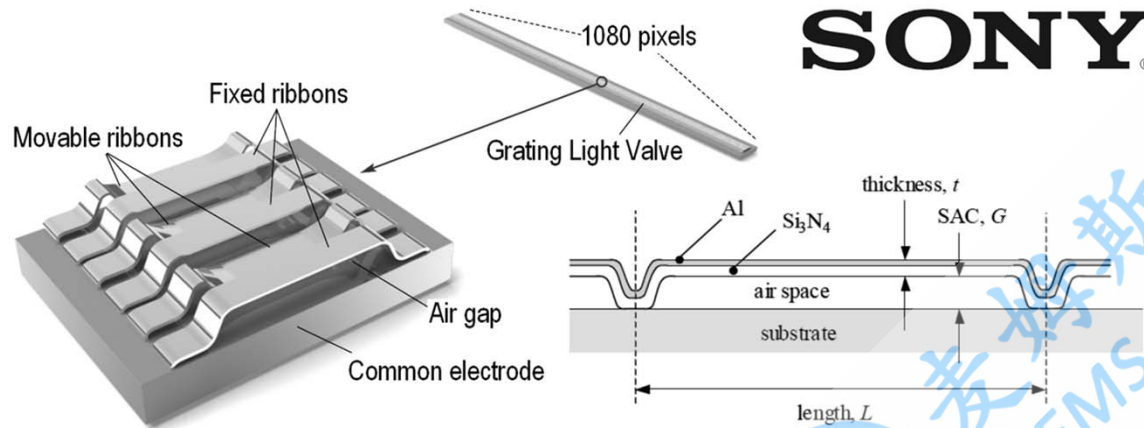
表面微加工工艺



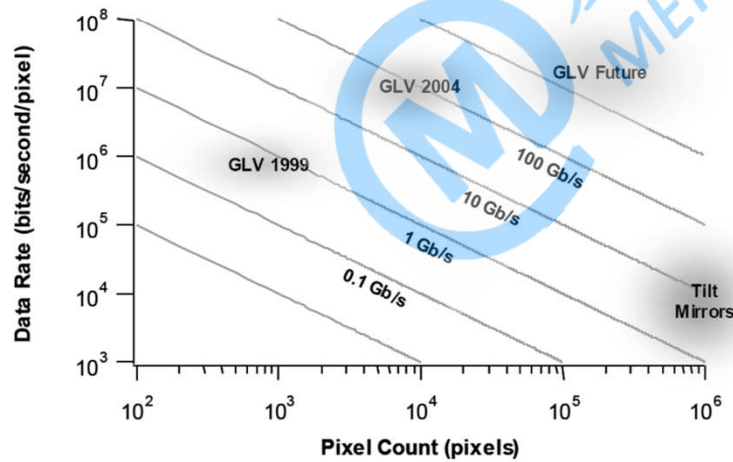
体微加工工艺

2.光栅色散型光谱仪

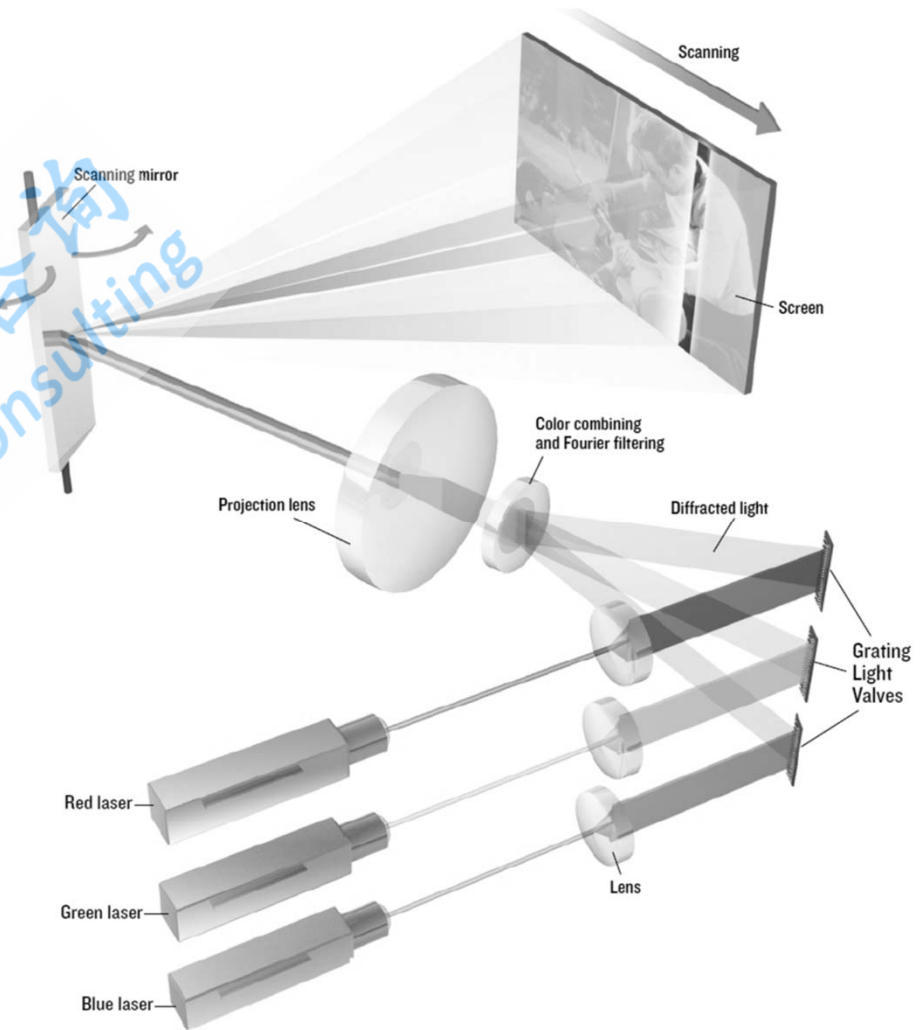
✓ MEMS光栅 相位式



GLV: 光栅光阀

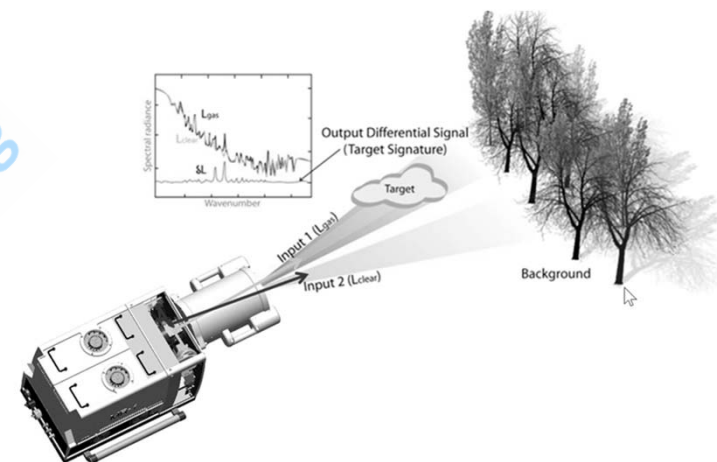
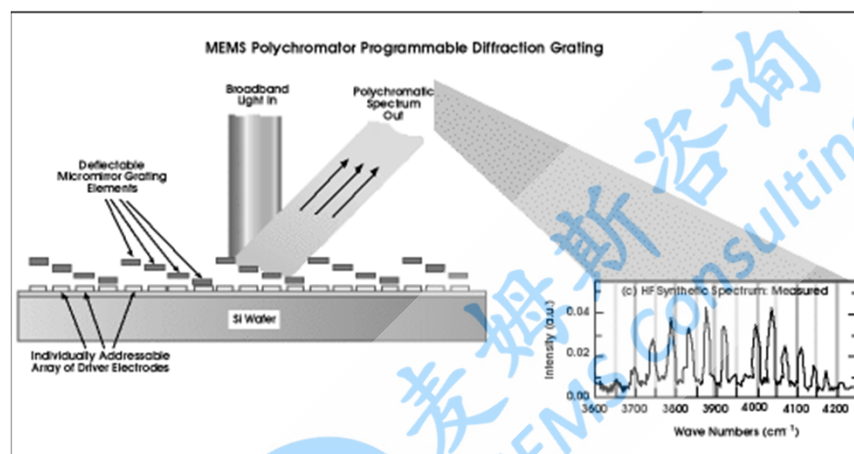
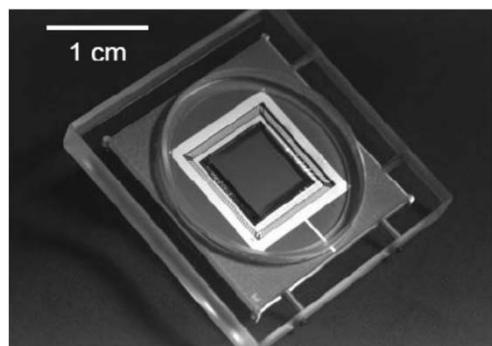


T.S. Perry. IEEE
Spectrum, 2004: 38.
A. Payne, et al.
Photonics West, 2004.



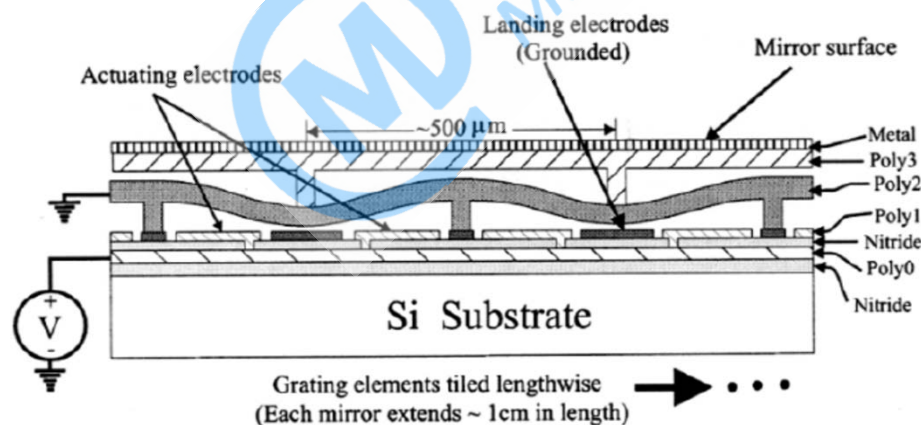
2.光栅色散型光谱仪

✓ MEMS光栅 相位式



Polychromator: 多色仪

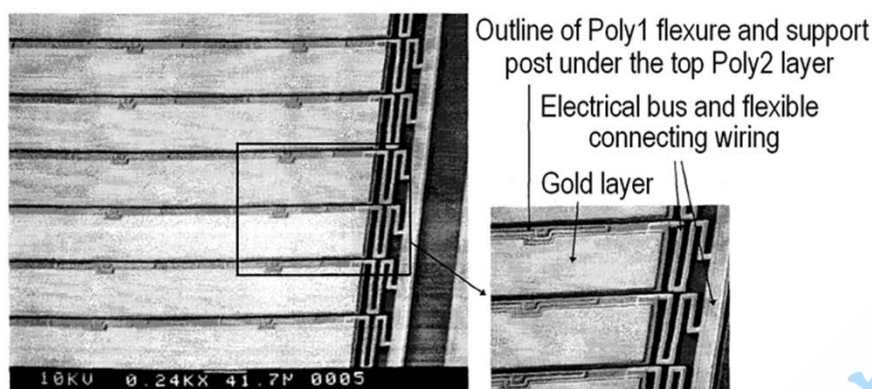
E.S. Hung, et al. JMEMS, 1999, 8(4): 497.



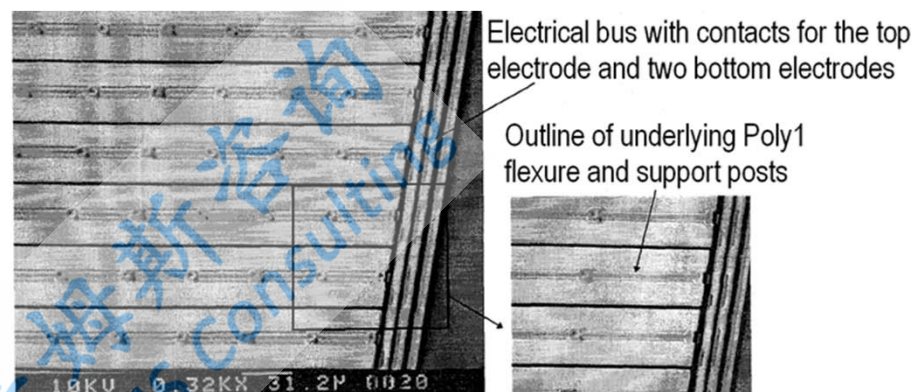
ThermoFisher
SCIENTIFIC

2.光栅色散型光谱仪

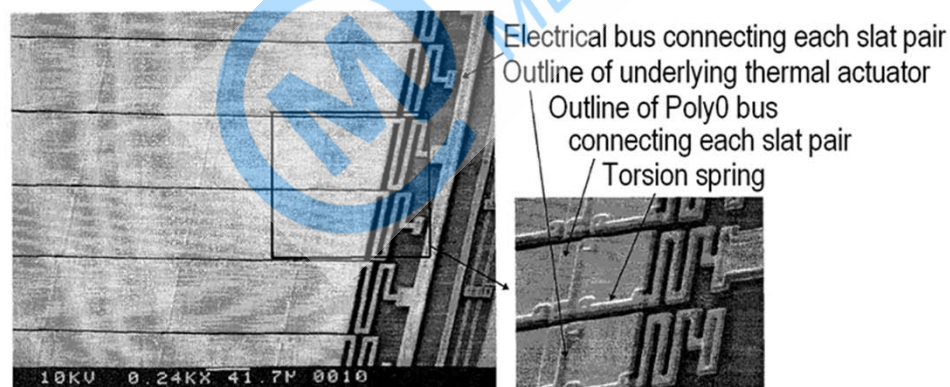
✓ MEMS光栅 闪耀角可调式



(a) 单向静电驱动



(b) 双向静电驱动



(c) 热驱动

D.M. Burns, et al. IEEE
MEMS, 1997.

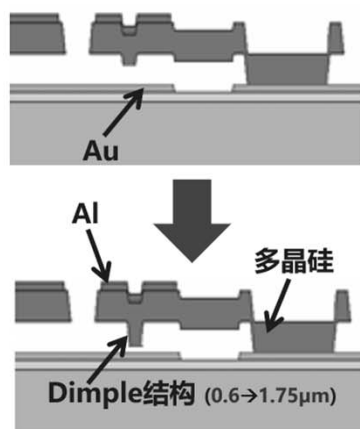


TABLE 1. Structural and Sacrificial Layers Used in MUMPs [2].

Layer Name	Nominal Thickness (μm)
Nitride (silicon nitride)	0.6
Poly-0 (bottom polysilicon layer)	0.5
1st Oxide (sacrificial layer - phosphosilicate glass)	2.0
Poly-1 (middle polysilicon layer)	2.0
2nd Oxide (sacrificial layer - phosphosilicate glass)	0.75
Poly-2 (top polysilicon layer)	1.5
Metal (gold)	0.5

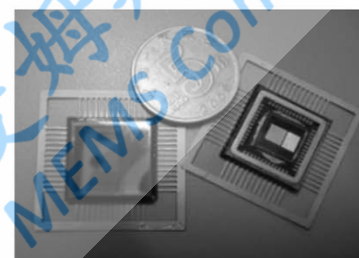
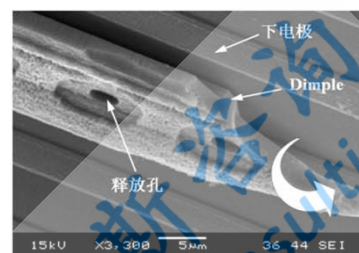
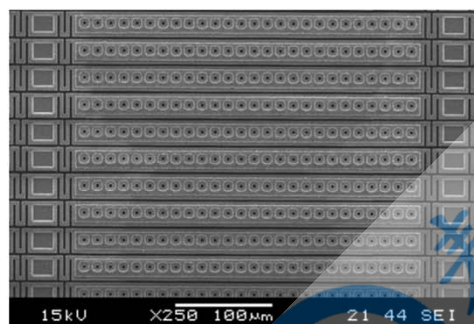
2.光栅色散型光谱仪

✓ MEMS光栅 闪耀角可调式

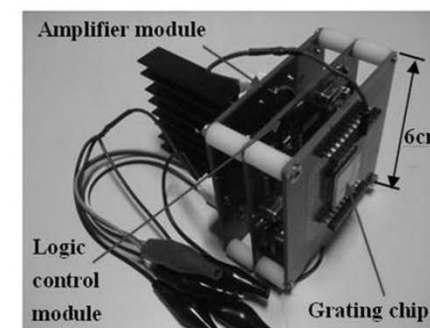
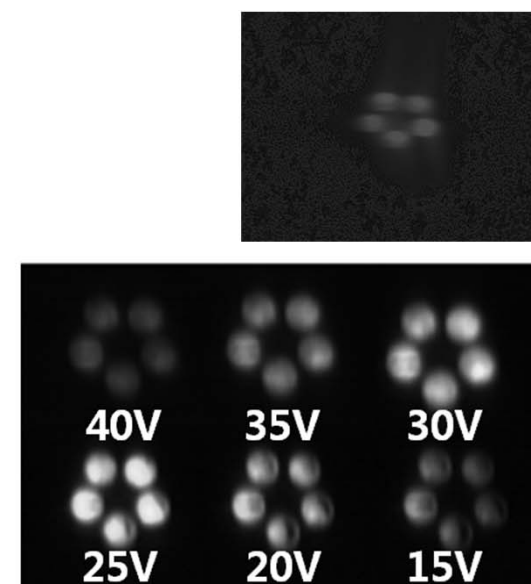
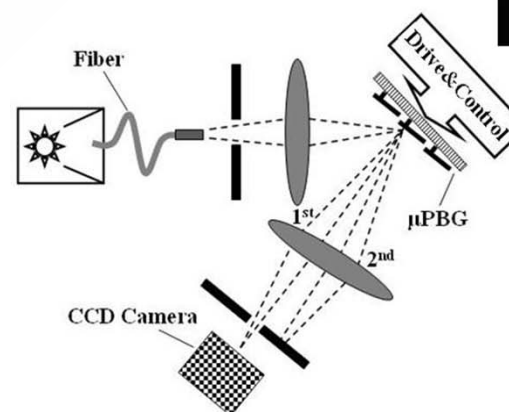
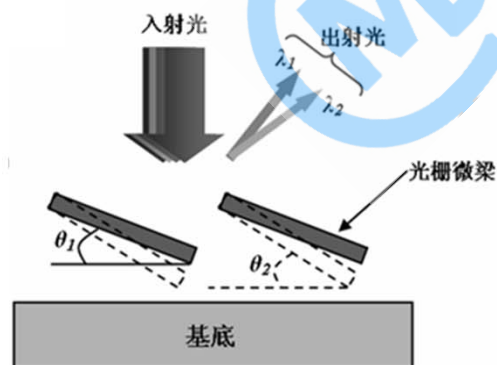


改进型Dimple结构

最大闪耀角: $1^\circ \rightarrow 5.1^\circ$
(单个芯片覆盖极紫外至长波红外)

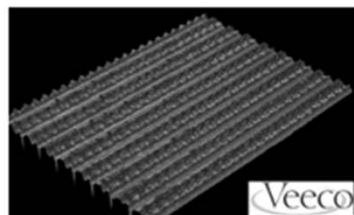
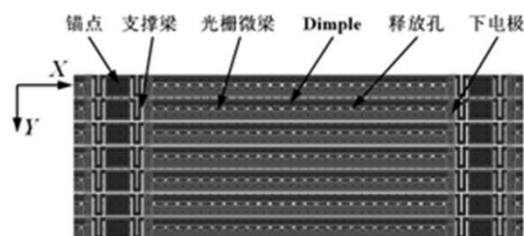


Appl. Opt., 2009, 48: 919.
Microsyst. Technol., 2009,
15: 853.
Chinese Sci. Bull., 2010,
55: 1112.



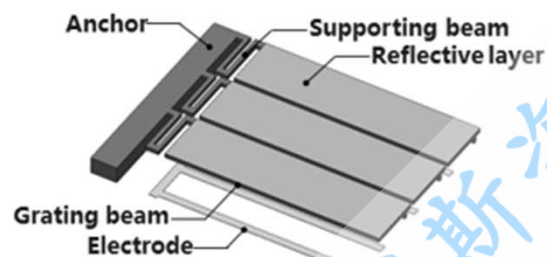
2.光栅色散型光谱仪

✓ MEMS光栅 闪耀角可调式

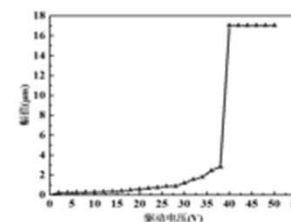
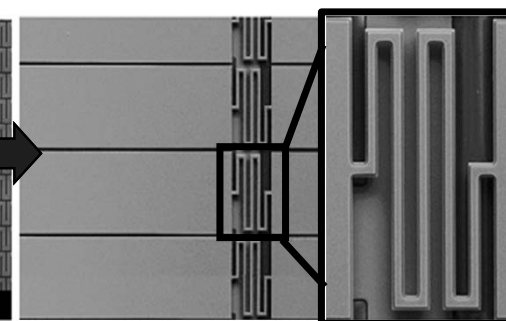
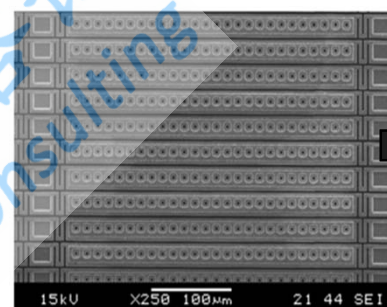


第I代: 表面微加工

Large surface roughness, release holes
→ Low diffraction efficiency (12-15%)



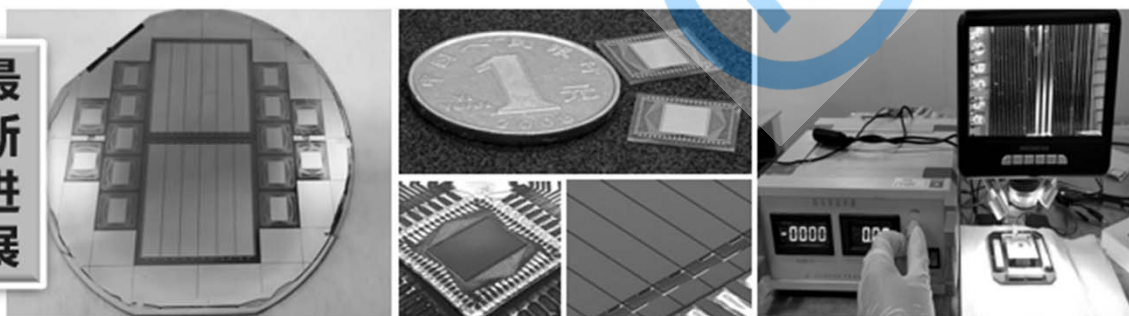
第II代: 体硅加工



下拉电压: 40V
回复电压: 30V
镜面粗糙度: 2-4nm
最大扭转角: 5°
有效工作面积: 3*3mm²

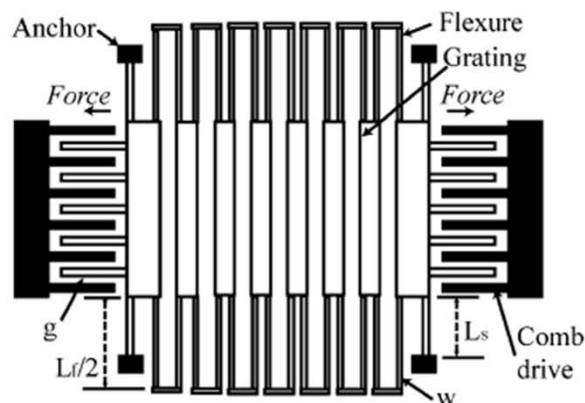
设计效率>90%
期望效率>60%

最新进展

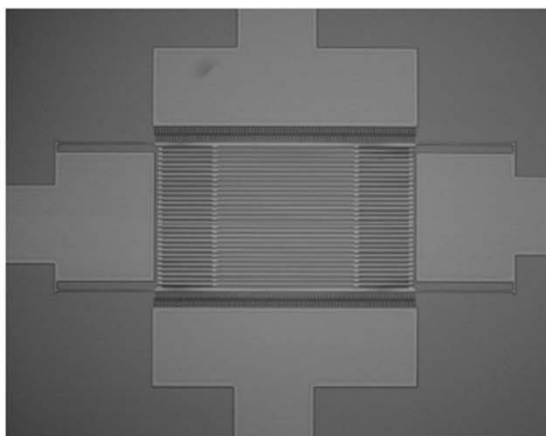


2.光栅色散型光谱仪

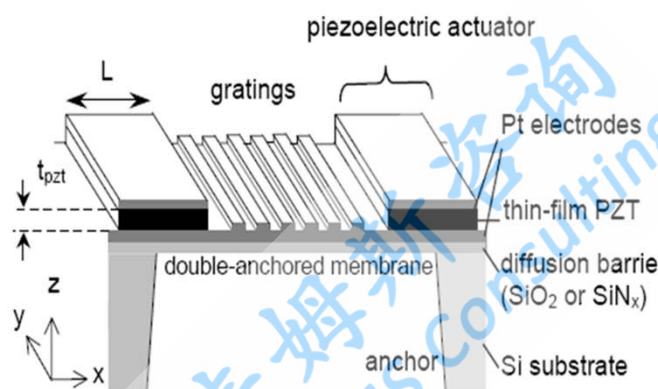
✓ MEMS光栅 周期可调式



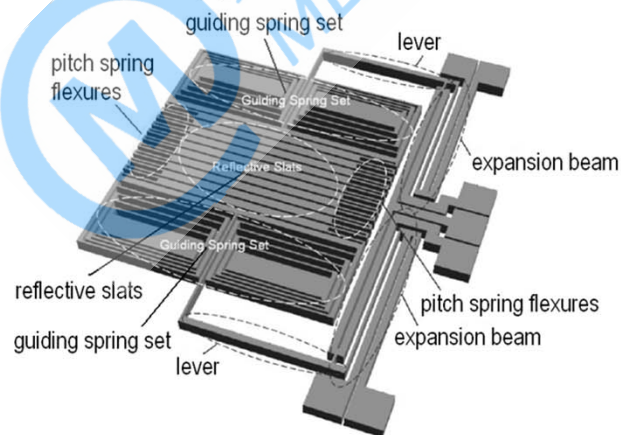
(a) 静电驱动



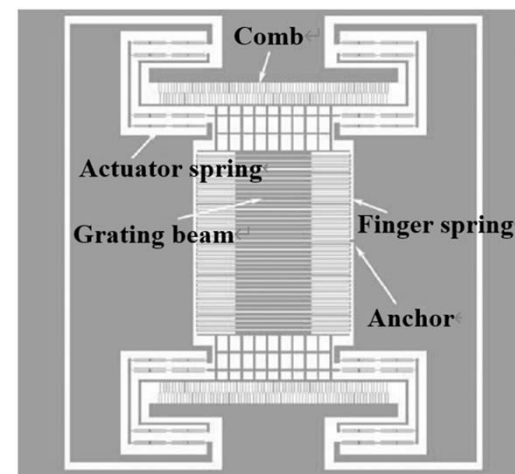
(c) CESM



(b) 压电驱动



(d) 台湾清华大学

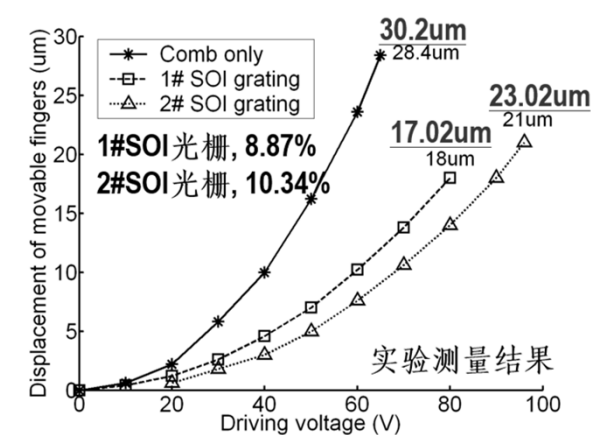
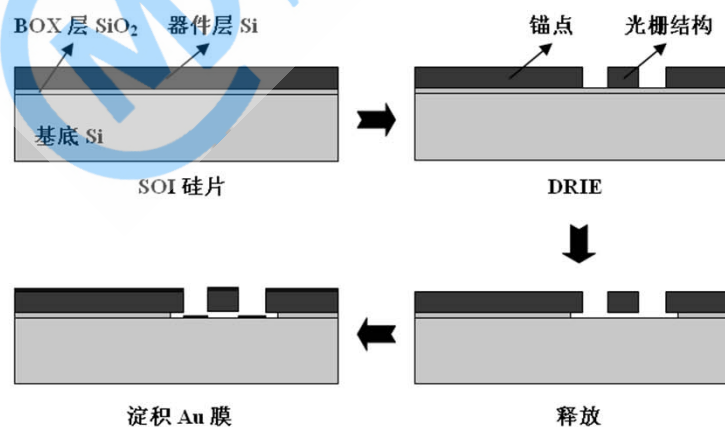
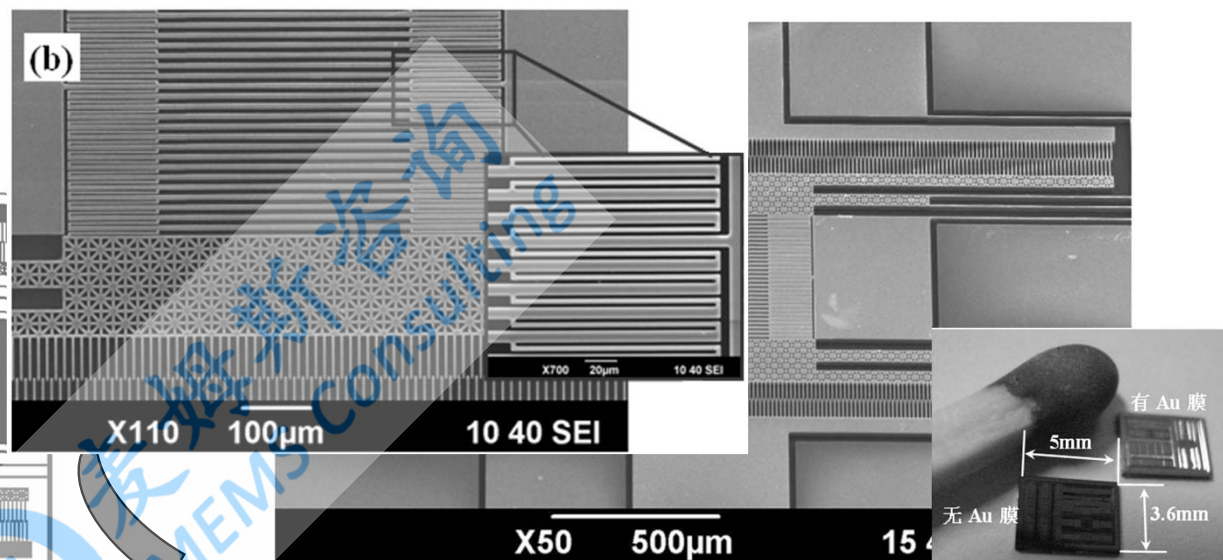
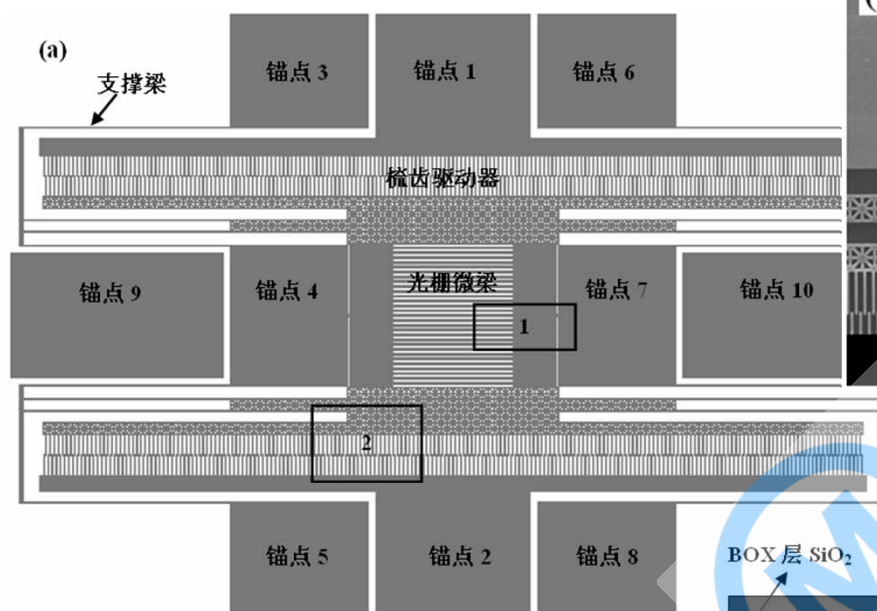


(e) Tohoku University

C.W. Wong, et al. *JMEMS*, 2004, 13: 998.
W.C. Shih, et al. *JMEMS*, 2006, 15: 763.
M. Tormen, et al. *SPIE*, 2006: 61140C.
Y.S. Yang, et al. *JMM*, 2009: 19: 015001.
Y. Wang, et al. *JMM*, 2009, 19: 025019.

2.光栅色散型光谱仪

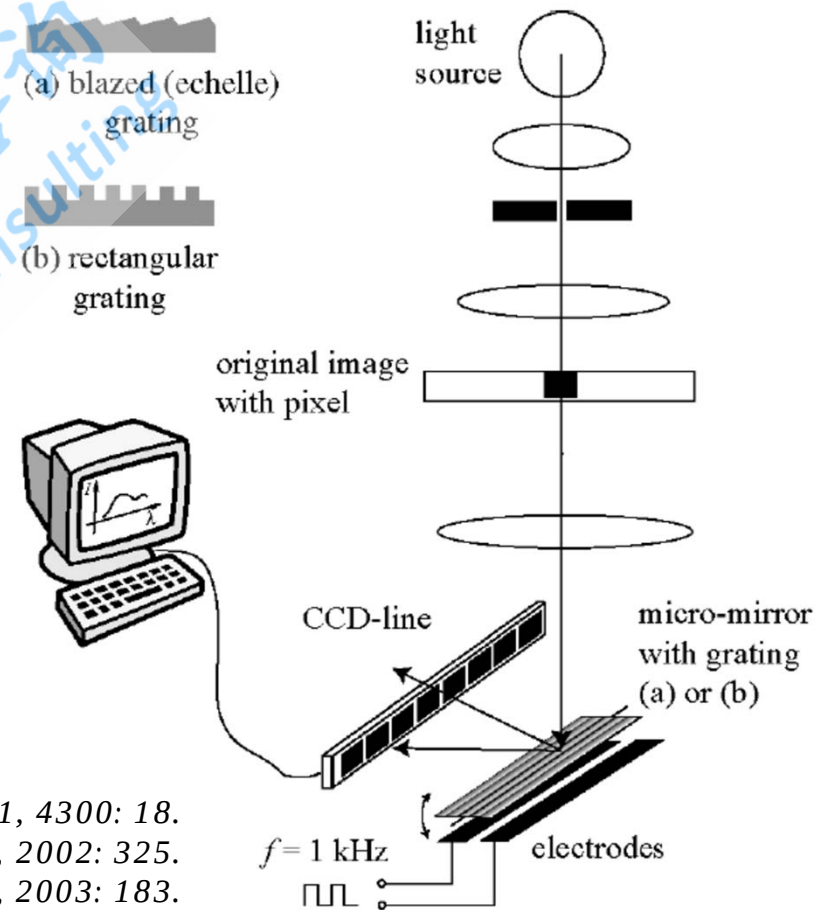
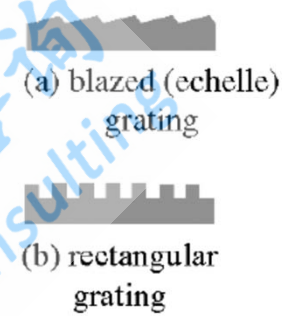
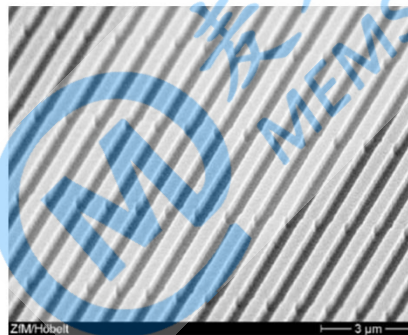
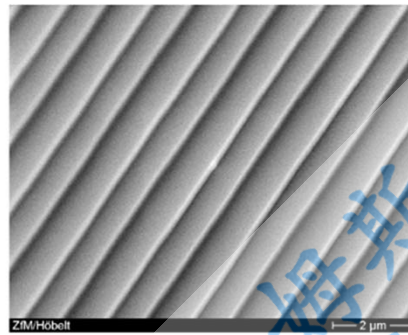
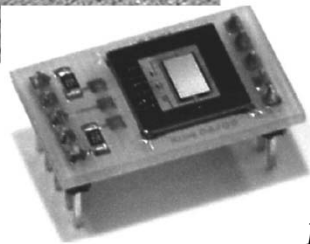
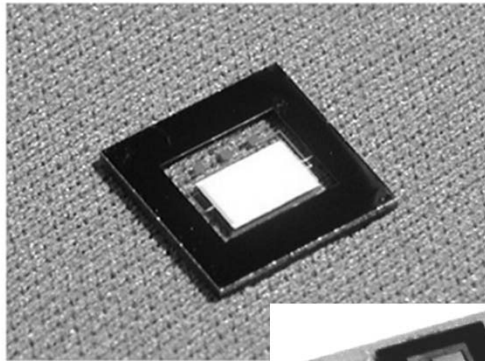
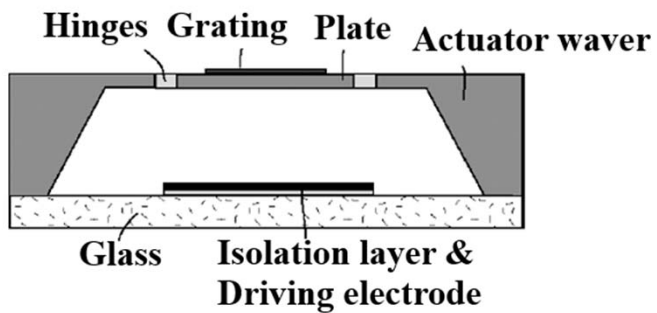
✓ MEMS光栅 周期可调式



JMEMS, 2010, 19: 1180.
JMM, 2010, 20: 065002.
JLT, 2009, 27: 5681.

2.光栅色散型光谱仪

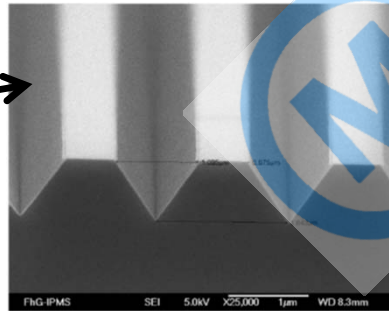
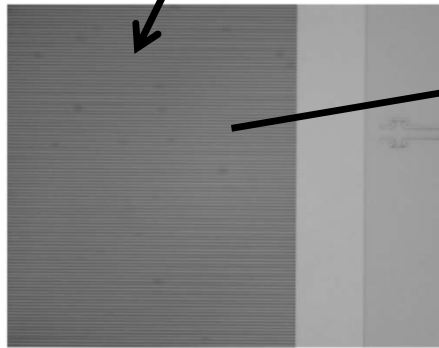
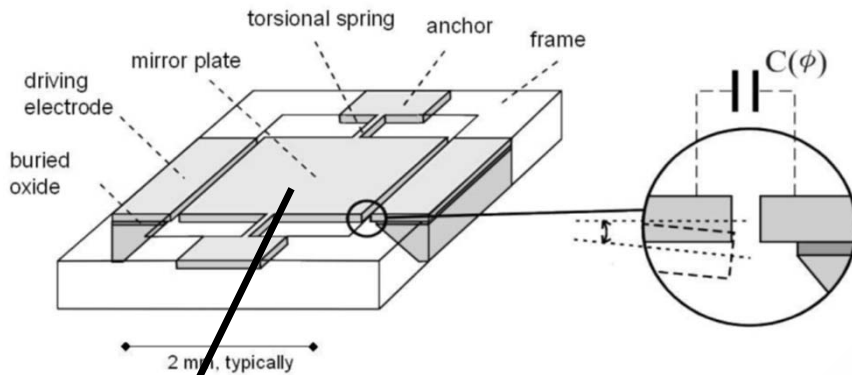
✓ MEMS光栅 光栅扫描式



A. Hübler, et al. *SPIE*, 2001, 4300: 18.
M. Flaspöhler, et al. *Proc. Actuators*, 2002: 325.
M. Flaspöhler, et al. *PICS Conf.*, 2003: 183.

2.光栅色散型光谱仪

✓ MEMS光栅 光栅扫描式



Fraunhofer
IPMS

Table 2.: Different scanning gratings and the mechanical angle required for selected wavelength ranges

Grating constant [nm]	Angle between incoming and diffracted beam (fixed) [°]	Wavelength range [nm]	Mechanical scan angle required for grating device [°]	Number of propagated orders
2000	15	900 - 2100	18.85	5
	15	900 - 3000	36.03	5
1600	15	900 - 2100	24.97	4
	15	900 - 3000	54.53	4
1400	15	900 - 2100	30.23	3
	15	900 - 2500*	45.32	3

(Grating area 3 x 3 mm², V-grooves, frequency 150 Hz or 500 Hz, *: diffraction limit 2500nm for this grating)

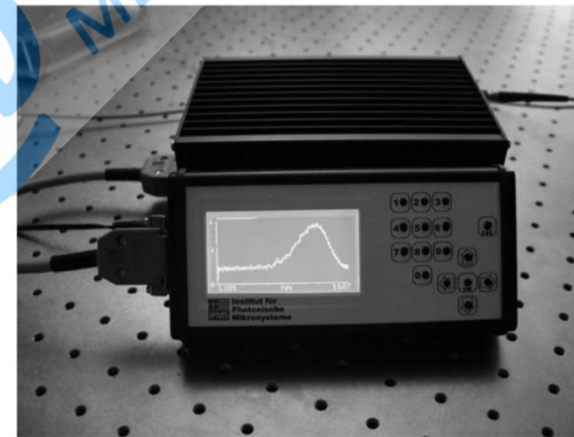


Fig. 7: Spectrometer set up with additional control and display device



H. Grueger, et al. SPIE, 2005, 58770J.
Spectrometers: MOEMS scanning grating chips reveal spectral images. www.laserfocusworld.com, 2008-7-1.

2.光栅色散型光谱仪

✓ MEMS光栅 光栅扫描式

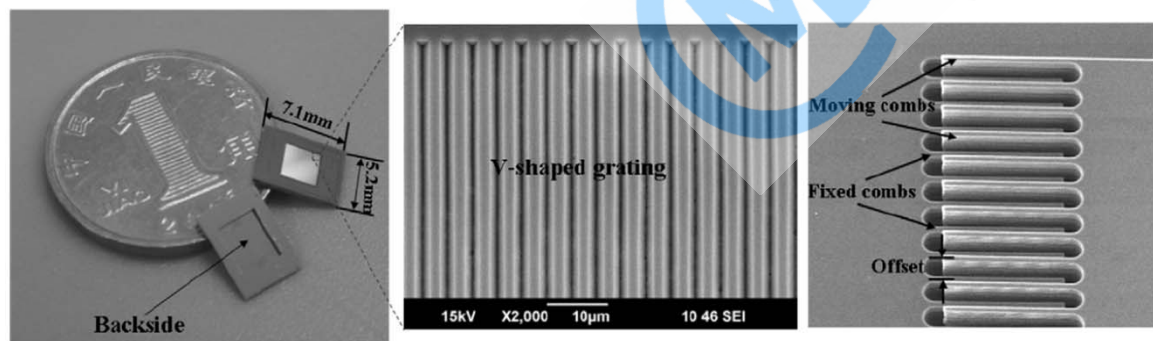
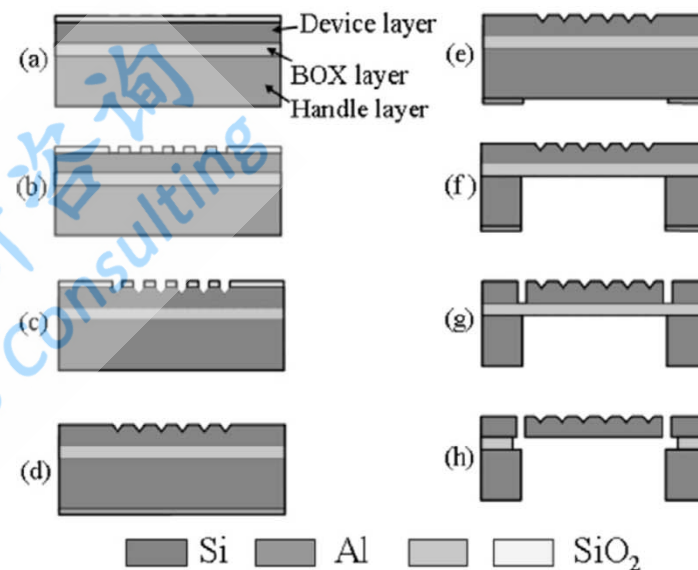
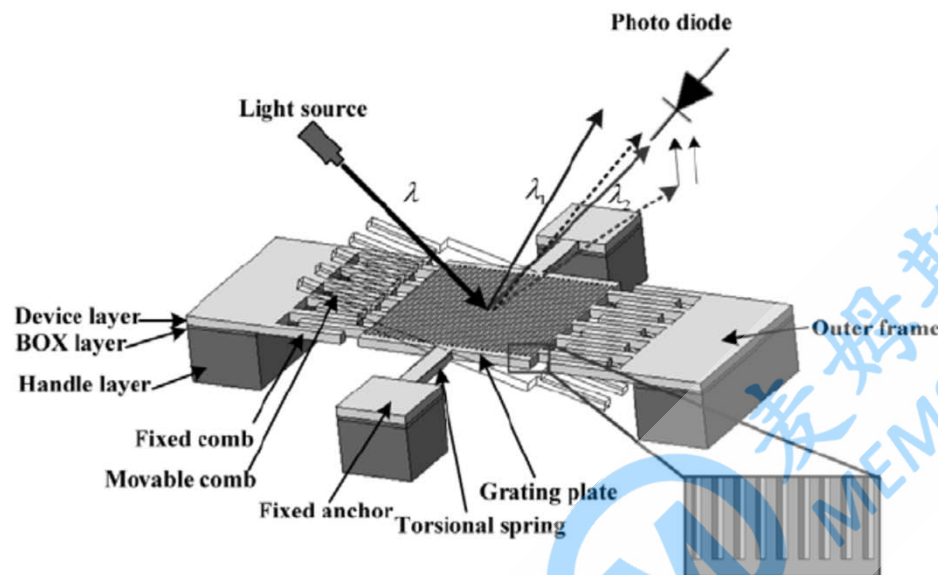


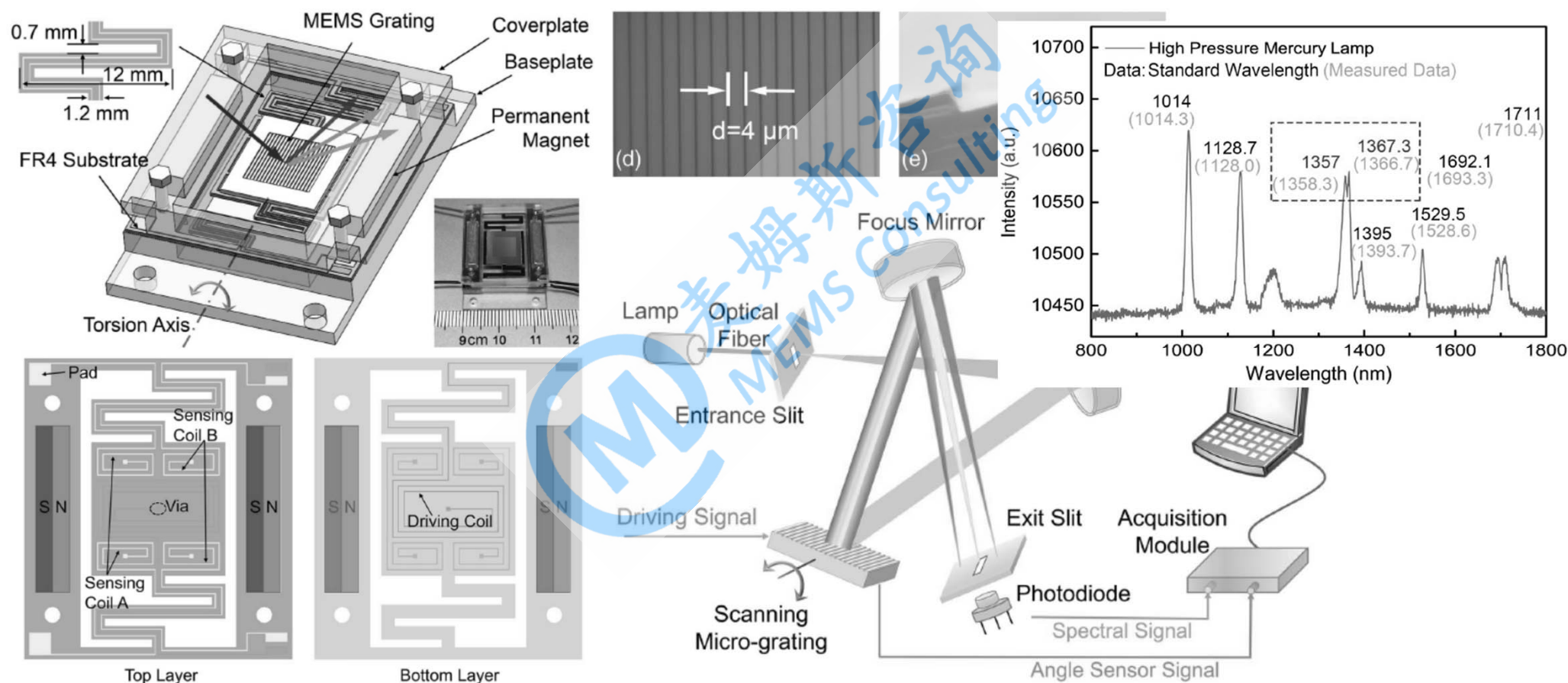
Table 2. The measured results for the diffraction efficiency.

Diffraction order	First order	Second order	Third order
Result	19%	1%	8%

D.Y. Qiao, et al. Laser Phys., 2013, 23: 035601.

2.光栅色散型光谱仪

✓ MEMS光栅 光栅扫描式

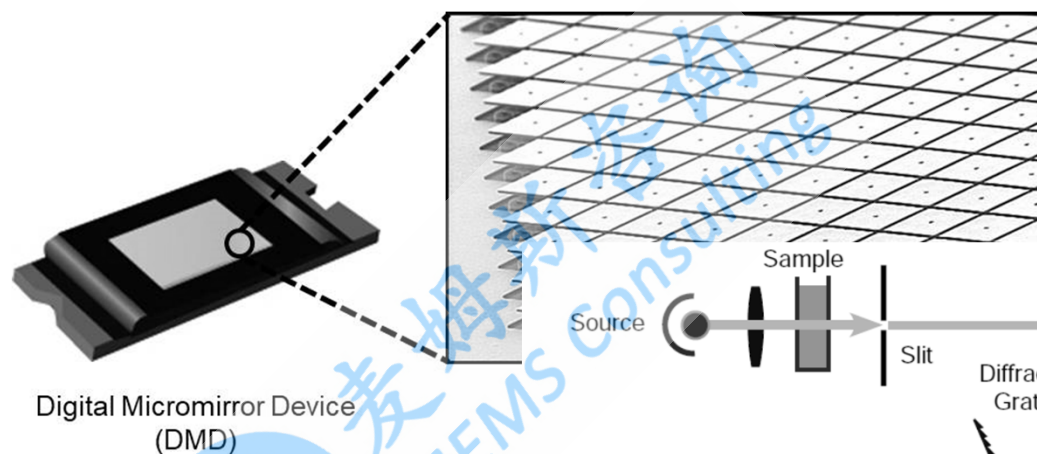


Q. Wen, et al. Appl. Opt., 2019, 58: 4642.

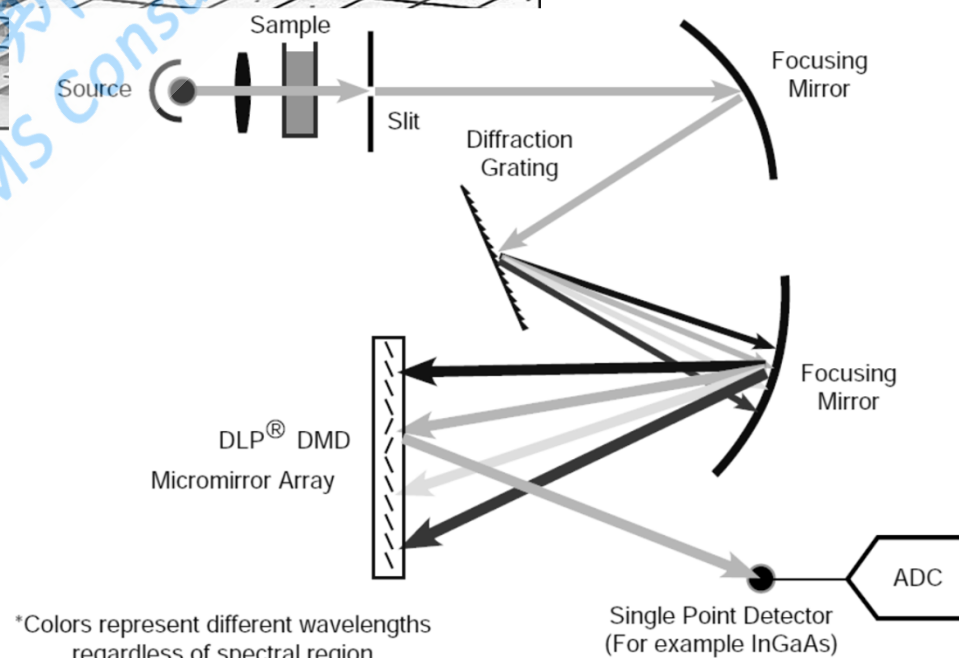
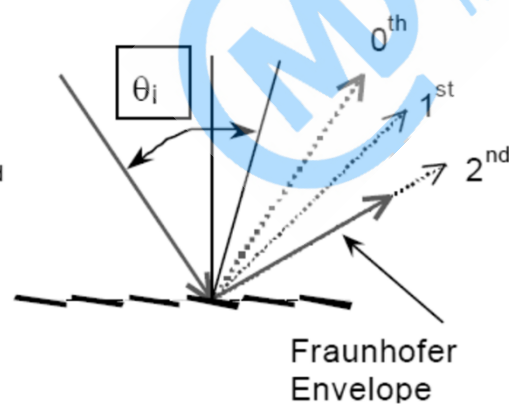
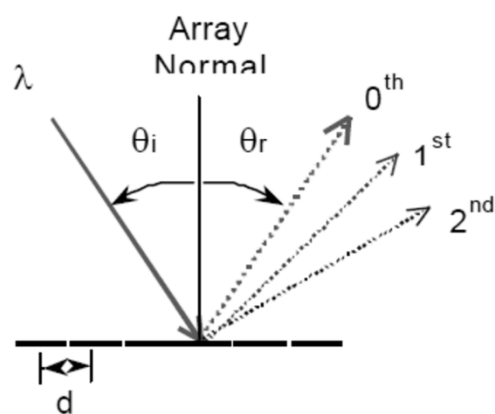
2.光栅色散型光谱仪

✓ MEMS光栅

DMD-based



TEXAS
INSTRUMENTS





提 纲

1、引言

2、光栅色散型光谱仪

- (1) 衍射光栅（反射/透射光栅）分光原理
- (2) 硅基微纳光栅：设计与加工
- (3) 基于硅基微纳光栅的微型光谱仪及其应用

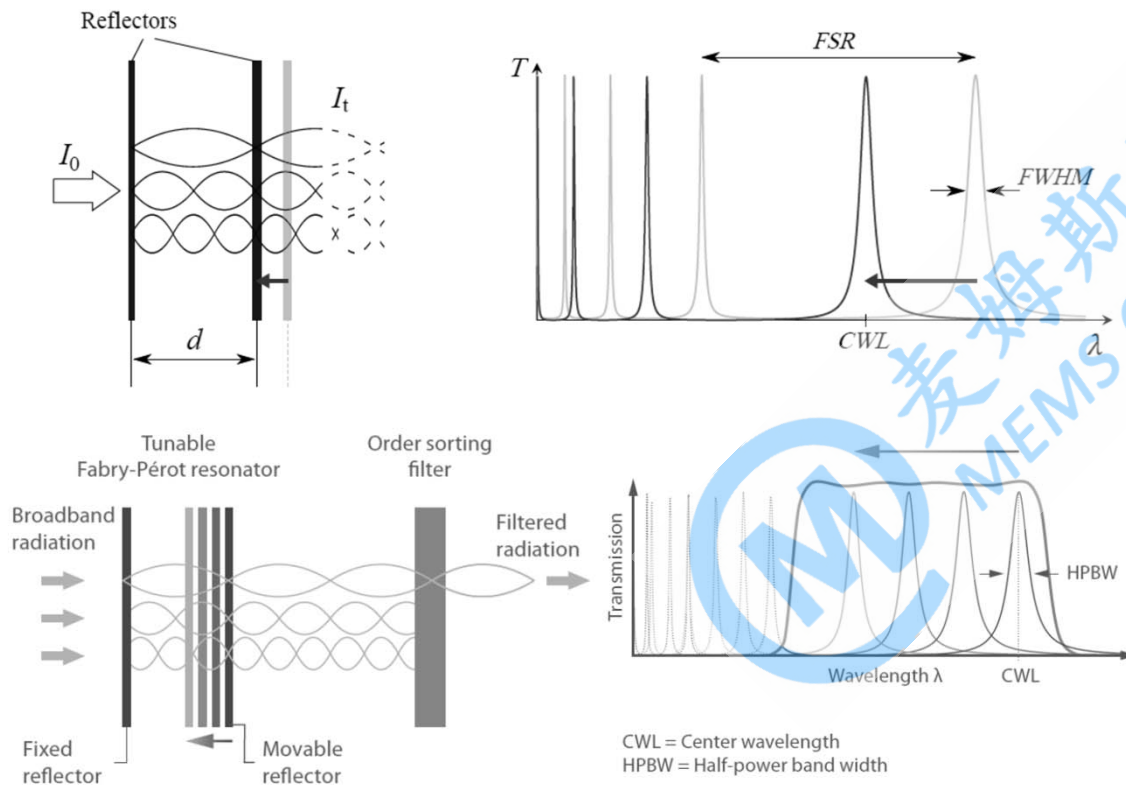
3、F-P干涉型光谱仪

- (1) F-P干涉仪分光原理
- (2) F-P全固态分光器件及其制造方法
- (3) 基于F-P全固态分光器件的微型光谱仪
- (4) F-P动态分光器件及其制造方法
- (5) 基于F-P动态分光器件的微型光谱仪
- (6) 基于F-P干涉技术的微型光谱仪典型应用

4、总结

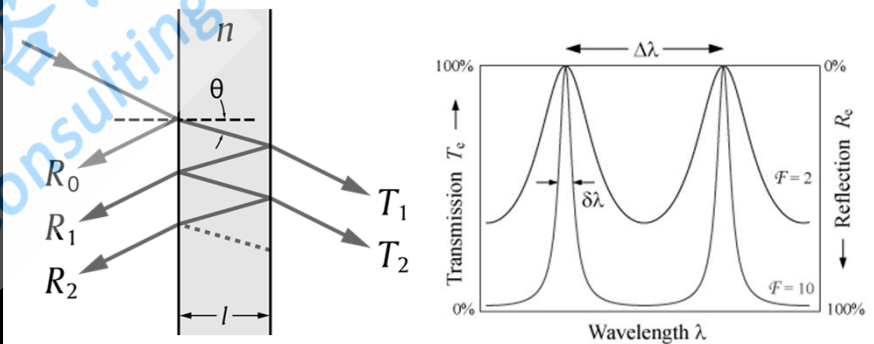
3、F-P干涉型光谱仪

✓ F-P干涉仪分光原理



基本结构与工作原理 (www.InfraTec.eu)

基本理论 (wikipedia.org)



$$\Delta\lambda = \frac{\lambda_0^2}{2n_g l \cos \theta + \lambda_0} \approx \frac{\lambda_0^2}{2n_g l \cos \theta}, \quad \text{自由谱范围FSR}$$

This is commonly approximated (for $R > 0.5$) by

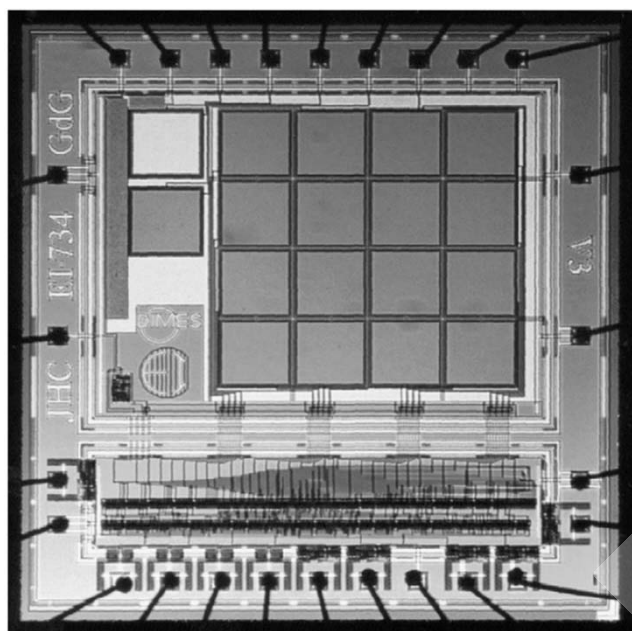
$$\mathcal{F} = \frac{\Delta\lambda}{\delta\lambda} = \frac{1}{2\varepsilon} \quad \mathcal{F} \approx \frac{\pi\sqrt{R}}{2} = \frac{\pi R^{\frac{1}{2}}}{1-R}$$

If the two mirrors are not equal, the finesse becomes

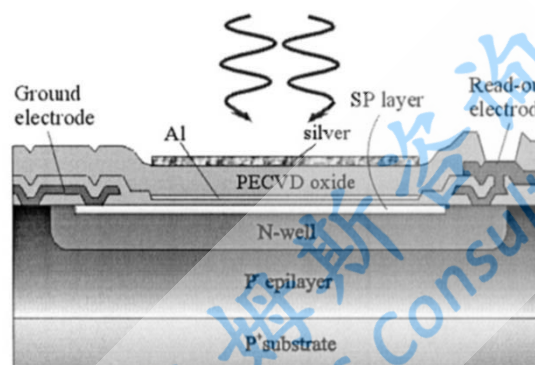
$$\mathcal{F} \approx \frac{\pi(R_1 R_2)^{\frac{1}{4}}}{1 - (R_1 R_2)^{\frac{1}{2}}}$$

3、F-P干涉型光谱仪

✓ F-P全固态分光器件



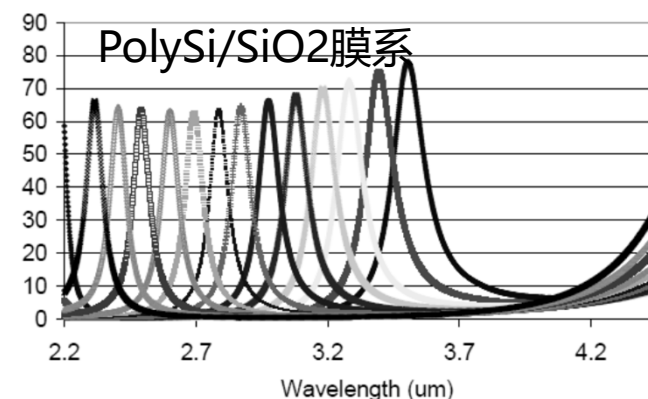
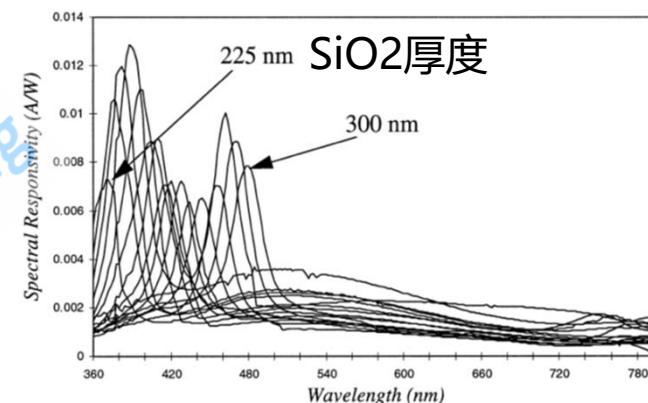
CMOS单芯片集成光谱仪



FPI: 45nm Ag/SiO₂/20nm Al

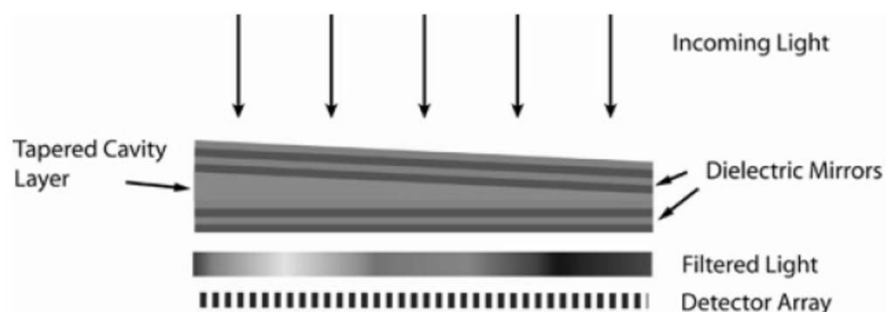
Single-chip optical microspectrometer test results

	Condition	Test result
Operating voltage	5 V	
Power dissipation at 1 MHz	Clk = 1 MHz	1250 μ W
Power dissipation at 100 kHz	Clk = 100 kHz	700 μ W
Max. clock frequency	-	6 MHz
Dark frequency	25°C	0.05 Hz
Free spectral range	-	380-500 nm
FWHM	-	18 nm
Diode responsivity (no Fabry-Perot etalon)	$\lambda = 480$ nm	0.18 A/W
Sensitivity (no Fabry-Perot etalon) [7] ref. TLS230 from Texas Instruments	$\lambda = 670$ nm	1.1 kHz/W m ⁻²
Sensitivity with Fabry-Perot etalon [7] ref. TLS230 from Texas Instruments	$\lambda = 670$ nm	511 Hz/W m ⁻²

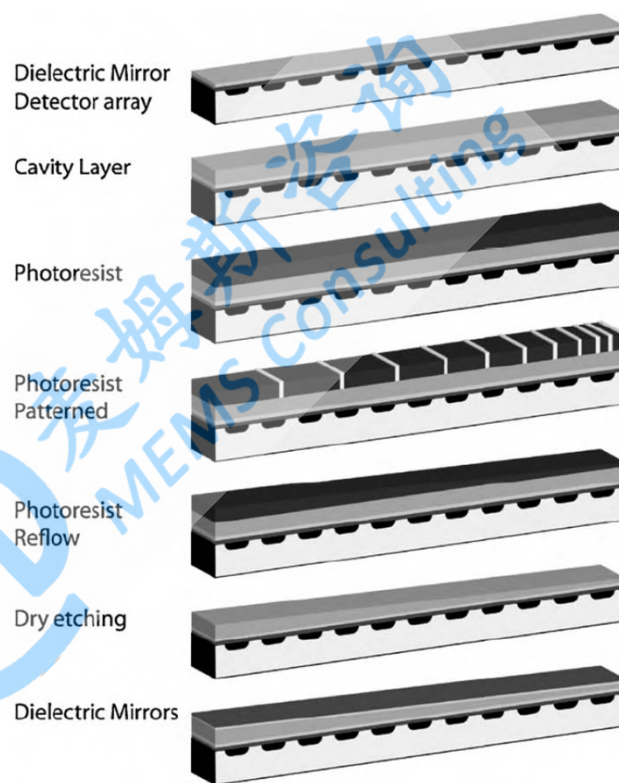


3、F-P干涉型光谱仪

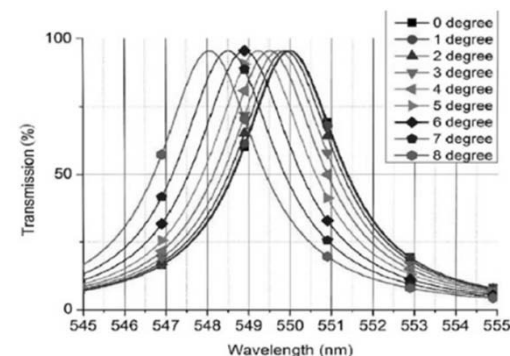
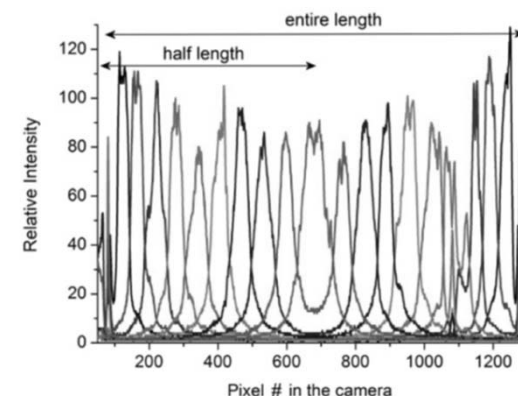
✓ F-P全固态分光器件



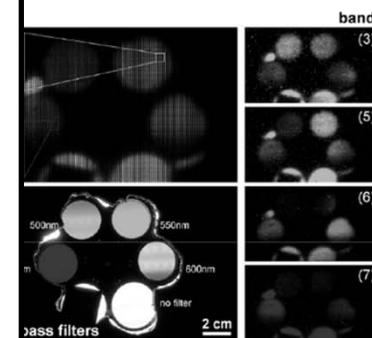
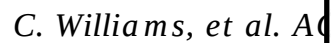
CMOS单芯片集成光谱仪 (LVOF)



光谱分辨率: 2.2nm / 0.7nm

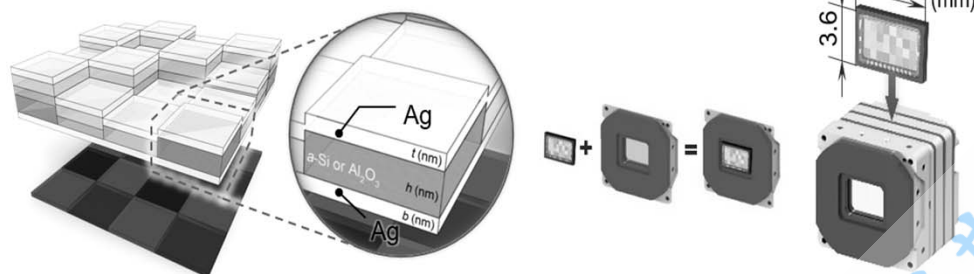


✓ F-P全固态



3、F-P干涉型光谱仪

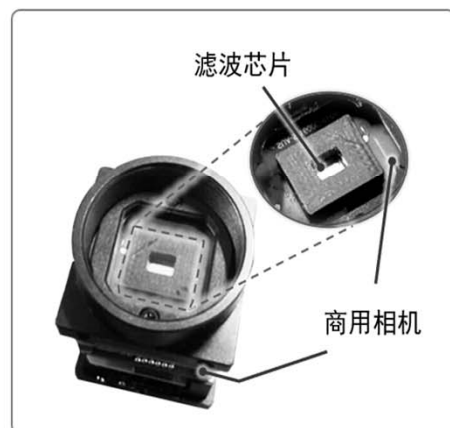
✓ F-P全固态分光器件



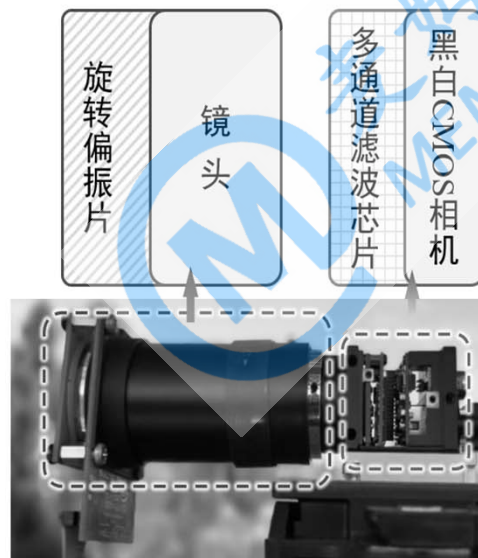
主要性能指标

工作波段：400~800nm，中心波长可任意定制
通道数：16通道，可任意定制
带宽：25~30nm
整机重量：100g

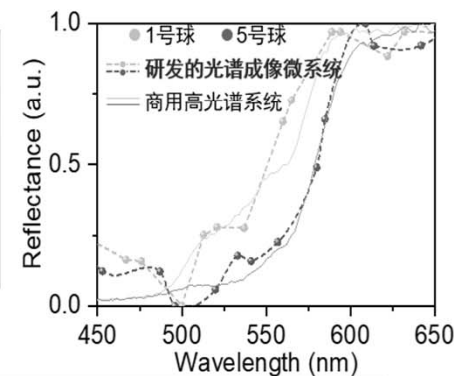
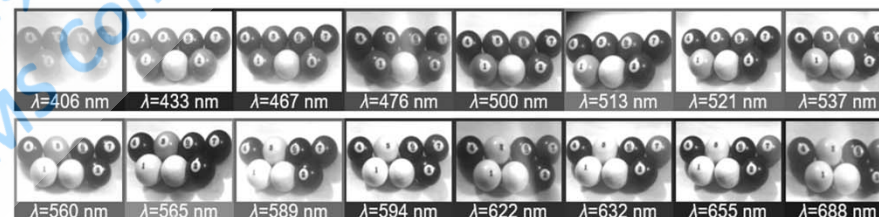
Opt. Express, 2021, 29: 30655.



即插即用的个性化集成模式



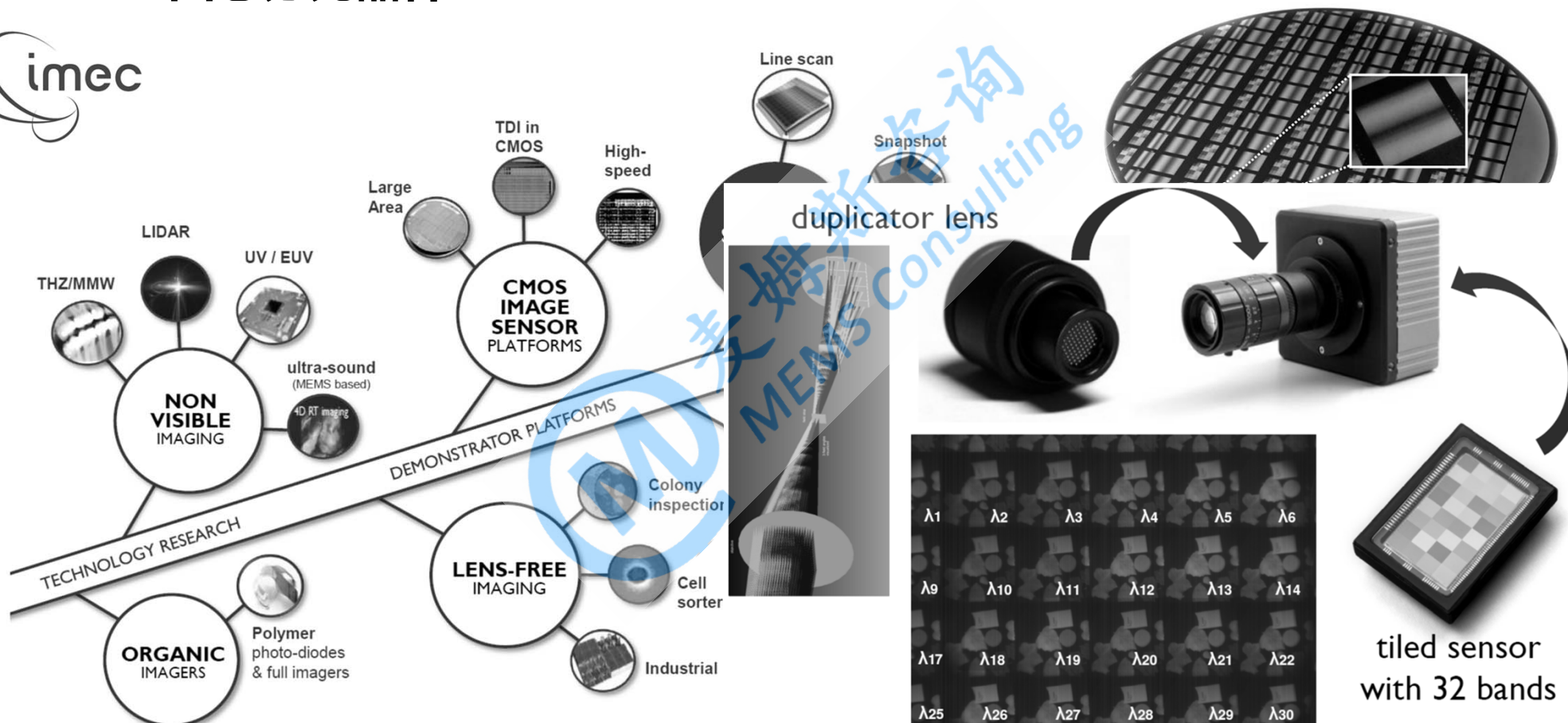
可扩展的模块化工作模式



准确的光谱测试与成像功能

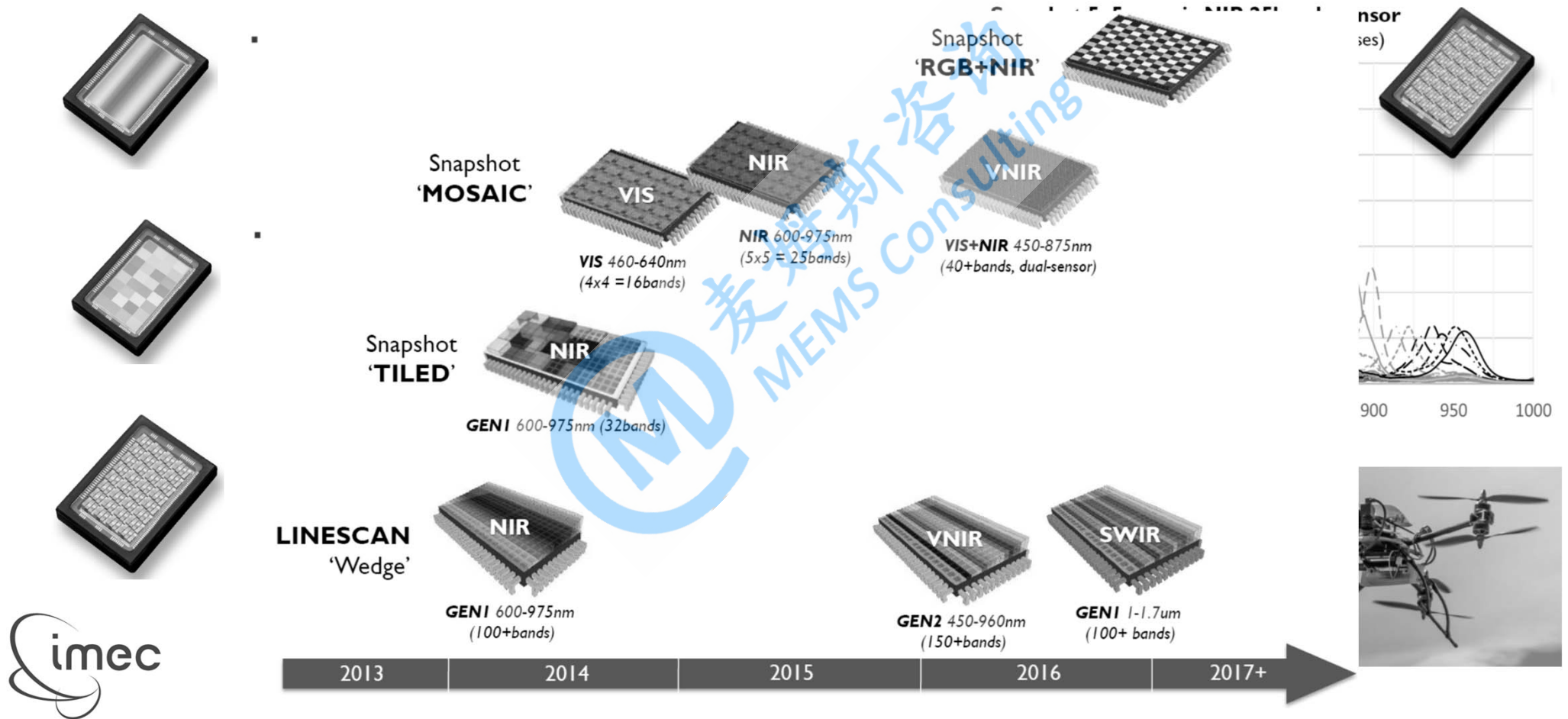
3、F-P干涉型光谱仪

✓ F-P全固态分光器件



3、F-P干涉型光谱仪

✓ F-P全固态分光器件



3、F-P干涉型光谱仪

Remote sensing



- UAVs & cube-sattelites for:
- Precision agriculture
 - Environment monitoring
 - Terrestrial / maritime earth observation

Machine vision / Optical sorting



- Food sorting / quality grading
- Pharmaceutical defect inspection
- Industrial process control (plastic, ceramic, glass, etc...)
- Robotic machine vision
- Bank-note inspection
- Print quality inspection

Medical imaging



- Surgery-guided imaging
- Fluorescence microscopy
- Endoscopy
- Ophthalmology / retina imaging
- Wounds imaging

Life-science / spectroscopy instrumentation



- Imaging spectroscopy analyzers
- DNA sequencers / flow cytometers
- Water monitoring analyzers
- Blood / urine analyzers

Security / Surveillance



- Industrial gas leaks monitoring
- Intrusion detection / authentication
- Forensics
- Anti-counterfeiting

Automotive & Transport

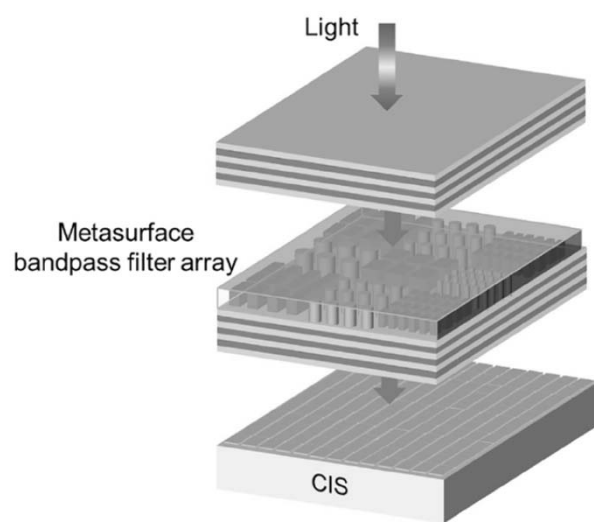


- Night vision systems
- Fuel monitoring systems
- Driver drowsiness

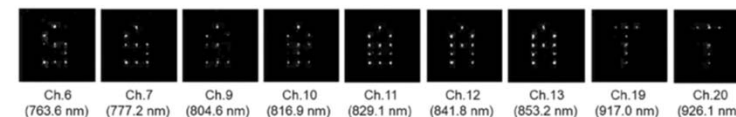
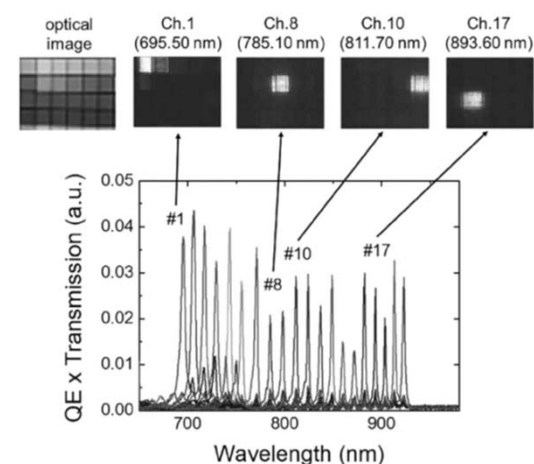
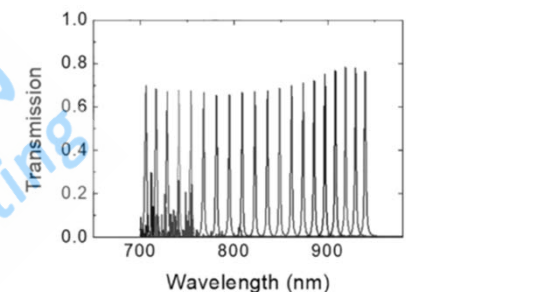
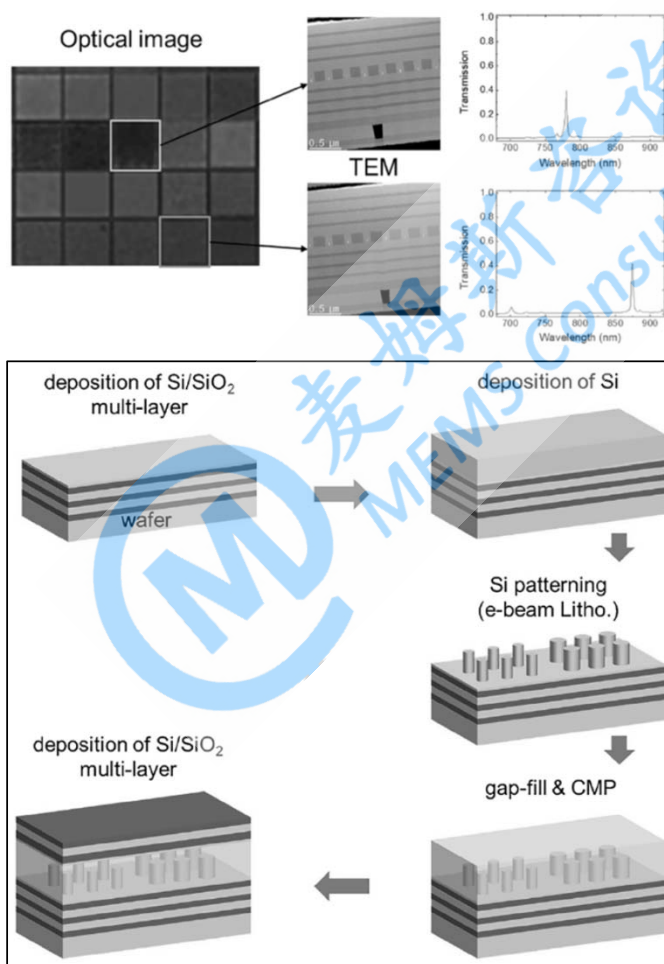
3、F-P干涉型光谱仪

✓ F-P全固态分光器件

J. Lee, et al. Nanophoton., 2022, doi: 10.1515/nanoph-2021-0706.



器件原理图



3、F-P干涉型光谱仪



Features

- Built-in Hamamatsu InGaAs PIN photodiode single device chip
- Ultra-compact: TO-5 package
- Ultra-lightweight: 1 g
- Hermetically sealed package: High reliability in high humidity environment
- Built-in thermistor
- Built-in band-pass filter for cutting off wavelengths outside the spectral response range

Applications

- Moisture detection
- Food inspection
- Farm product inspection
- Plastic screening
- Textile identification
- Installation into mobile measuring devices

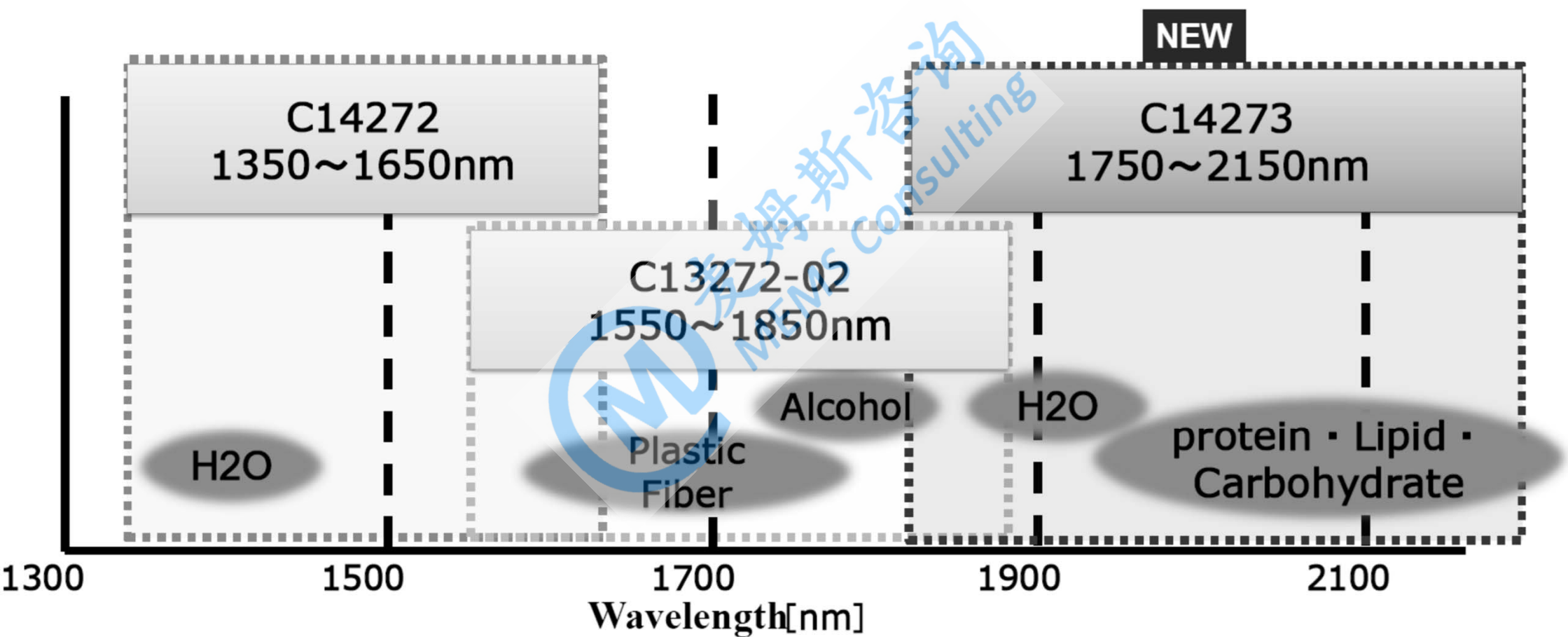
Type no.	Spectral response range (nm)									Spectral resolution (full width at half maximum) max. (nm)	Dark current max. (nA)	Photosensitive area (mm)
	Near infrared											
	800	1000	1200	1400	1600	1800	2000	2200				
<u>C14272</u>				1350 to 1650						18	10	φ0.3
NEW <u>C13272-03</u>					1550 to 1850					20	100	φ0.3
NEW <u>C14273</u>						1750 to 2150				22	150	φ0.3

48

3、F-P干涉型光谱仪

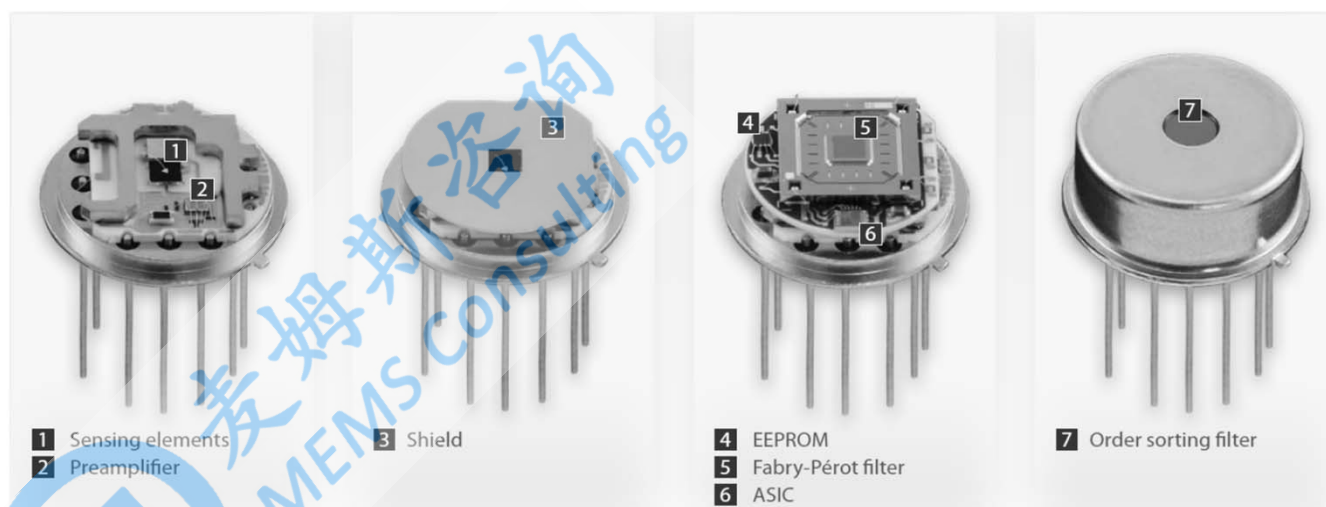
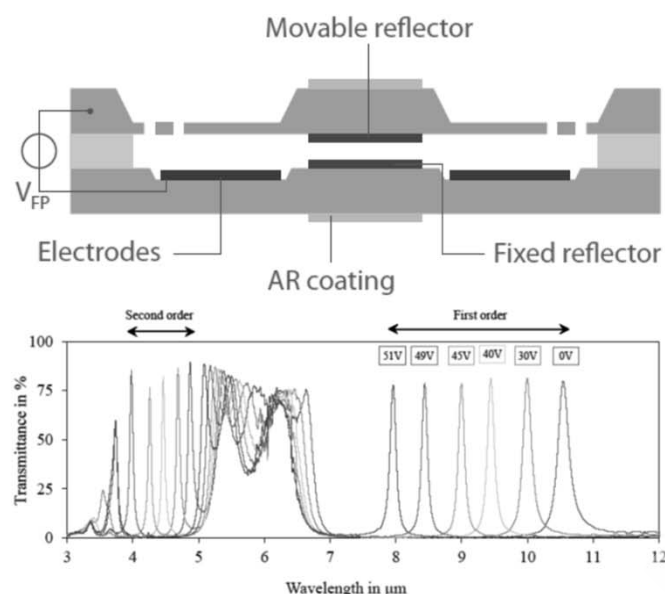
✓ F-P动态分光器件 变腔长

HAMAMATSU
PHOTON IS OUR BUSINESS



3、F-P干涉型光谱仪

✓ F-P动态分光器件 变腔长

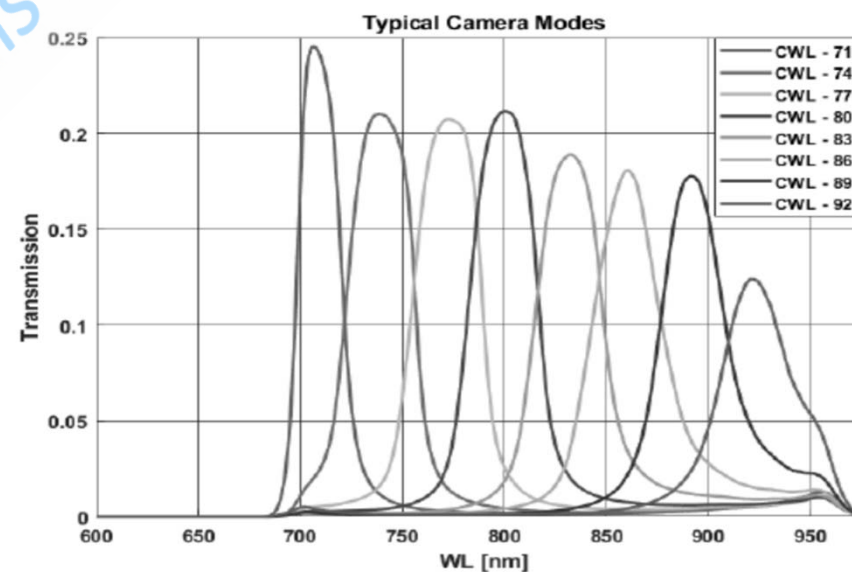
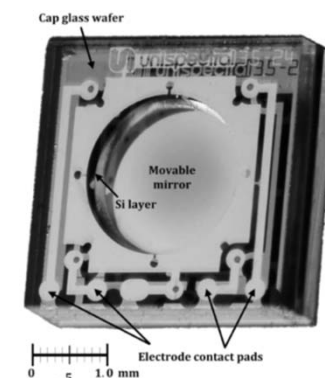
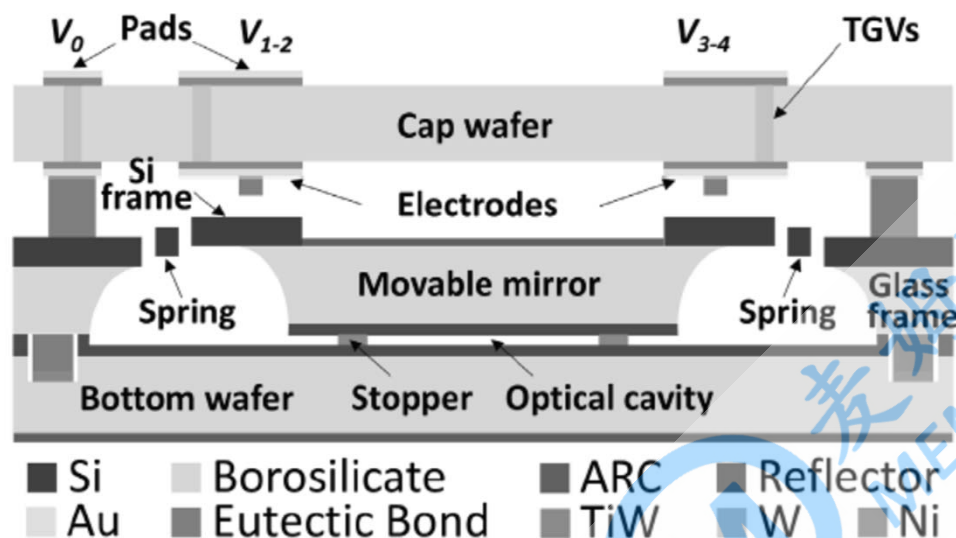


	LFP-3144C-337	LFP-3850C-337	LFP-5580C-337	LFP-80105C-337
Tuning CWL	(3.1 ... 4.4) μm	(3.8 ... 5.0) μm	(5.5 ... 8.0) μm	(8.0 ... 10.5) μm
Spectral resolution <i>HPBW</i>	(55 ... 70) nm	(60 ... 75) nm	(100 ... 130) nm	(130 ... 220) nm
Filter time constant	(3 ... 15) ms	(2 ... 12) ms	(1 ... 10) ms	(1 ... 8) ms
Type	LFP-3144C-337	LFP-3850C-337	LFP-5580C-337	LFP-80105C-337
Application examples	Environmental, climate, building and safety technology		Process measurement technology, liquid analysis	Medical technology, breath analysis

INFRAtec.

3、F-P干涉型光谱仪

✓ F-P动态分光器件 变腔长

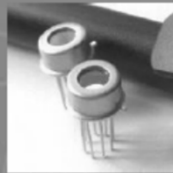


3、F-P干涉型光谱仪

✓ F-P动态分光器件 变腔长

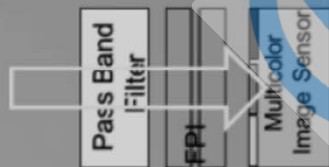
Single-point spectroscopy

- FPI combined with IR detector
- No array detector needed → reduced sensor cost
- Examples: NDIR sensors, portable NIR analyzers, selective gas sensors



Hyperspectral imaging

- FPI combined with imaging detector
- Both spectral and spatial data
- Example: Visualization of gas distribution in stand-off detection



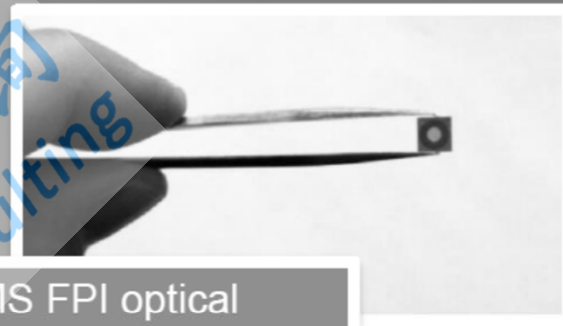
Comparison to pixel array-HSIs:

- ✓ **Smaller size** than pixel-based spectral imagers
- ✓ **Hundreds** of spectral bands
- ✓ **Does not require bulky and expensive telemetric lenses**
- ✓ Compatible with **low-cost mobile optics!**

3、F-P干涉型光谱仪

✓ F-P动态分光器件 变腔长

- **MEMS FPIs** for mass-producible microspectrometers in large-volume applications
 - Monolithic cleanroom fabrication processes
 - *Hyperspectral imaging, gas sensing, IR sensors*
 - *Automotive, mobile, IoT*
- **Piezo-actuated FPI** for small-to-medium volume applications
 - Separately assembled filter structure
 - Large optical apertures enable enhanced light throughput for high-performance applications
 - *CubeSats, drones, medical, defense*
- Wavelengths available from *UV to thermal IR*



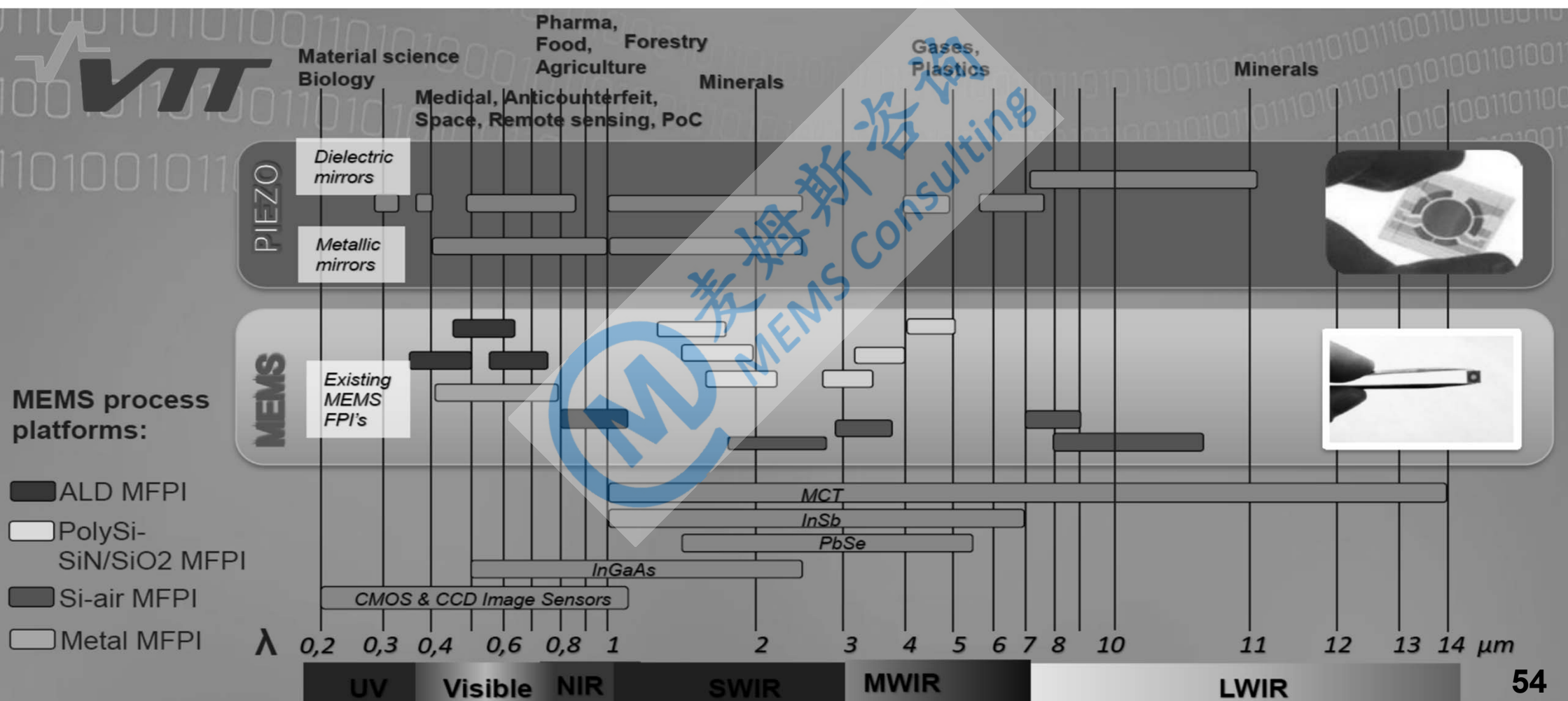
MEMS FPI optical apertures up to 4 mm



Piezo FPI optical apertures up to 24 mm

3、F-P干涉型光谱仪

✓ F-P动态分光器件 变腔长



FPI technology applications – examples and demonstrators

Mobile hyperspectral iPhone demo (2016)



Mobile and hand-held



Spectral Engines NIR sensor (2016)



MEMS-based hyperspectral imager demo (2012)



Mobile CO2 sensor demo (2014)

Space and environmental sensing



Hyperspectral imagers for space instruments

- NASA OMI (2006)
- Aalto-1 (2014-2017)*
- PICASSO Vision (2015-2018)*
- Hello World 2017-2018



Drone hyperspectral imagers for forestry, precision agriculture, gas sensing and UV-Raman

Visible-VNIR (2011), SWIR (2016), UV (2016)

SO₂/NO_x ship emissions imaging (2016)



Skin cancer hyperspectral imager (2014-2017)



Health and diagnostics



Fundus camera
Detection of glaucoma and diabetes, oxygen saturation (hypoxia, apnea)



Brain surgery spectral imaging integrated to the Zeiss Pentaro brain surgery microscope

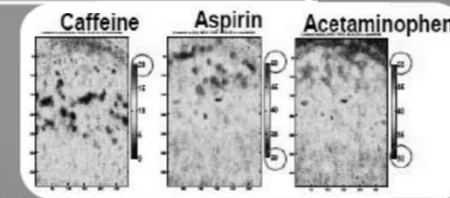
Stand-off - and chemical detection



Thermal IR hyperspectral imager (2014)



UV-FPI Raman stand-off trace detection (2014)

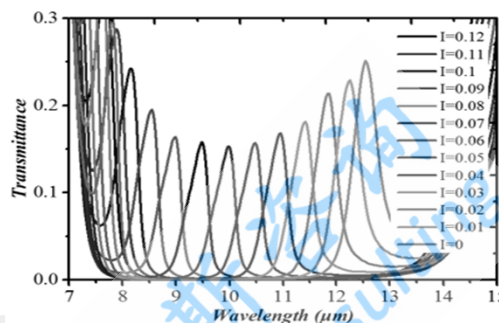
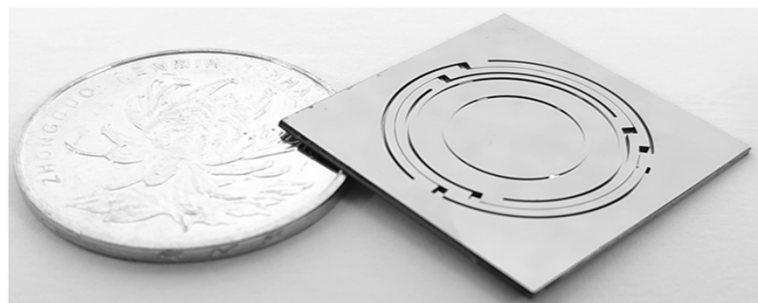


Chemical imager for 1-2.5 μm
Distribution of active ingredients

Hyperspectral microscope
Imaging of cells, micro well arrays & fluorescence imaging (2016)

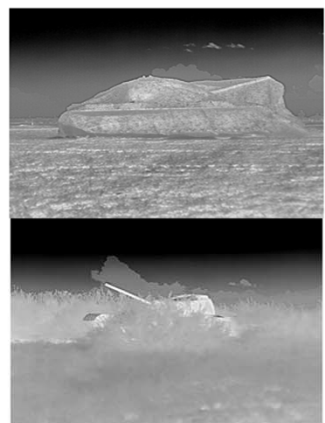


3、F-P干涉型光谱仪

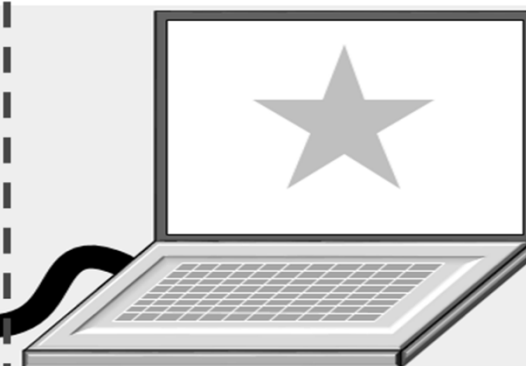
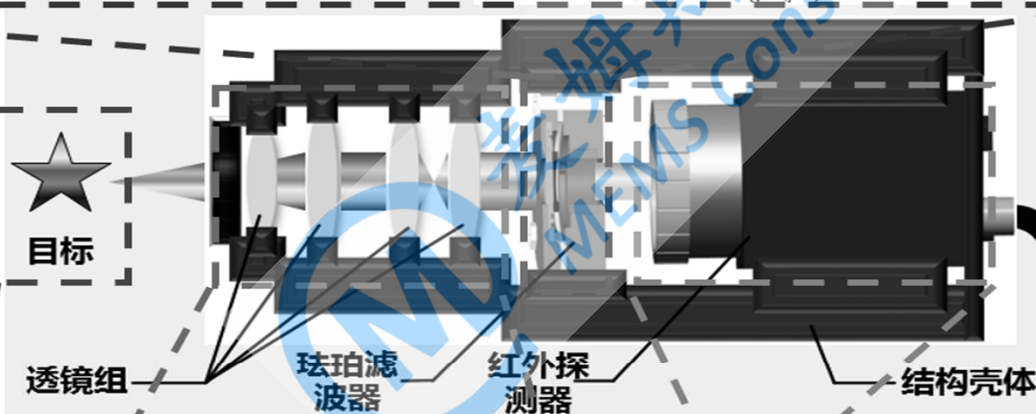


滤波芯片选择

- ◆ 噪声抑制
- ◆ 特征提取
- ◆ 图像融合
- ◆ 目标识别



典型目标场景



图像处理系统



19mm镜头

光学
镜头
定制

探测器参数:
型号: 高德 PLUG 617
工作波段: $8\sim 14\mu\text{m}$
分辨率: 640×512
帧频: 25Hz
接口: USB串口



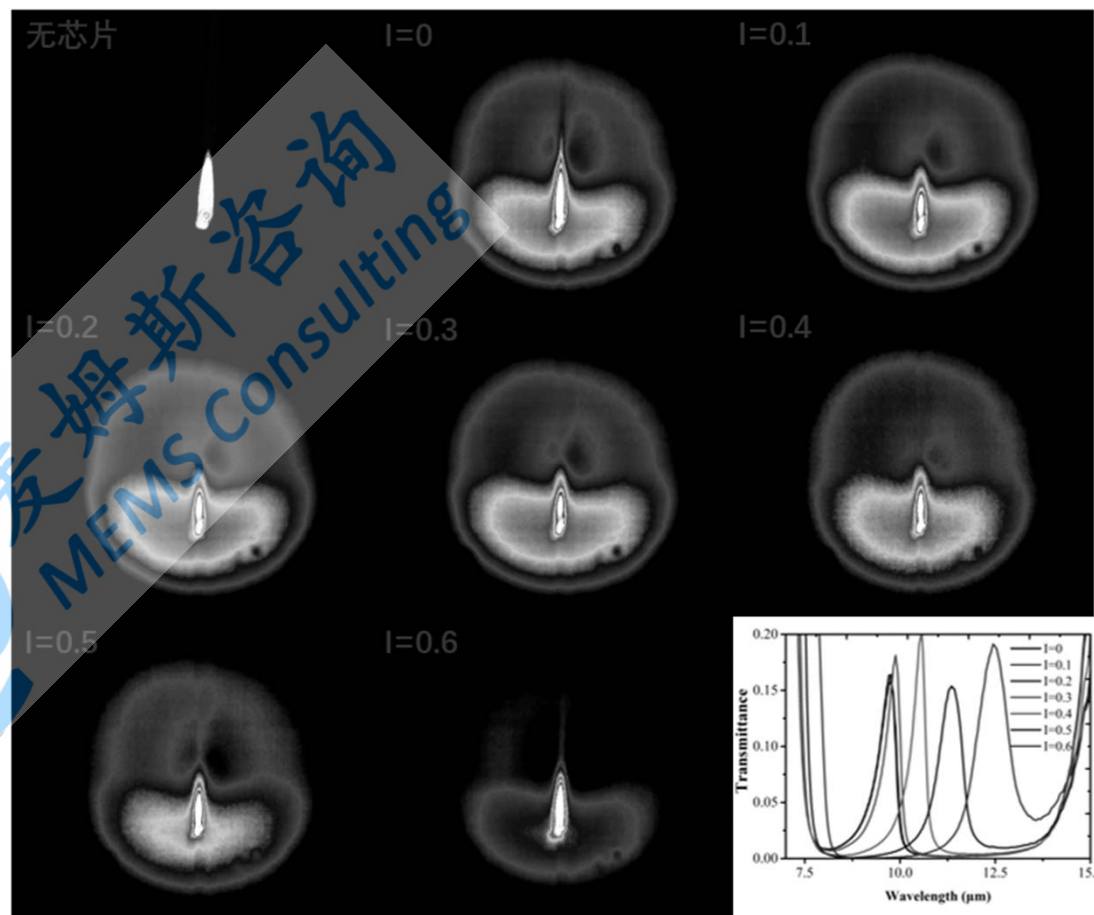
探测
器选
择

3、F-P干涉型光谱仪

✓ F-P动态分光器件 变腔长



小型化LWIR光谱成像功能样机

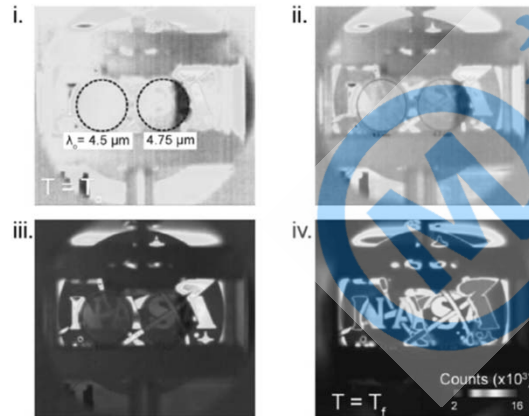
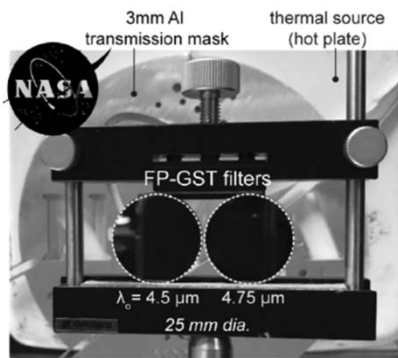


蜡烛火焰LWIR光谱成像试验

3、F-P干涉型光谱仪

✓ F-P动态分光器件 变折射率

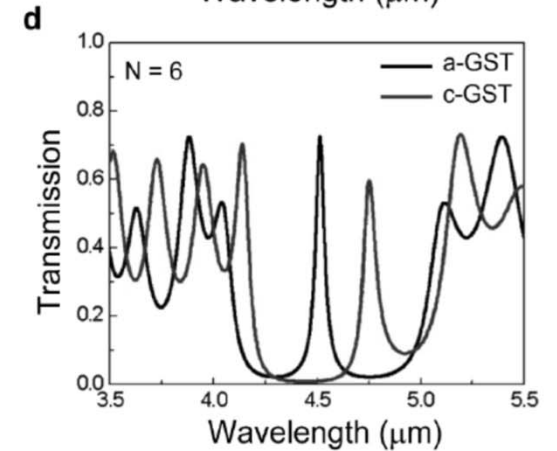
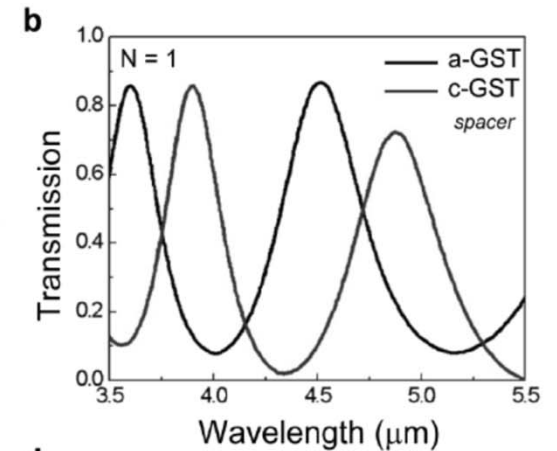
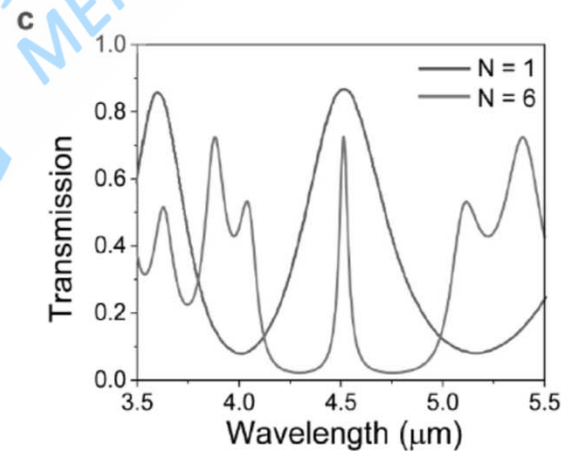
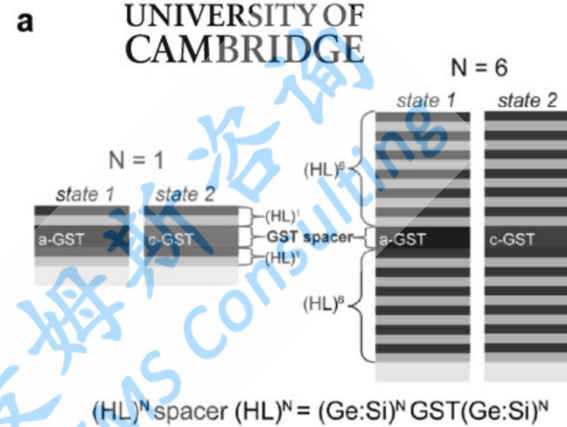
Material	$\lambda = 4.5\mu\text{m}$		$\lambda = 4.2\mu\text{m}$	
	n	k	n	k
Ge	4.100	~ 0	4.103	~ 0
GST-225 (amorphous)	2.948	0	2.952	~ 0
GST-225 (crystalline)	5.336	0.089	5.354	0.086
Si	3.42	~ 0	3.424	~ 0



C. Williams, et al. Opt. Express, 2020, 28: 10583.



UNIVERSITY OF
CAMBRIDGE





提 纲

1、引言

2、光栅色散型光谱仪

- (1) 衍射光栅（反射/透射光栅）分光原理
- (2) 硅基微纳光栅：设计与加工
- (3) 基于硅基微纳光栅的微型光谱仪及其应用

3、F-P干涉型光谱仪

- (1) F-P干涉仪分光原理
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- (5) 基于F-P动态分光器件的微型光谱仪
- (6) 基于F-P干涉技术的微型光谱仪典型应用

4、总结

4、总结

光谱作为“分子指纹”，对于物质成分检测具有显著优势，在即将到来的万物互联时代，将扮演重要角色。

无处不在的智能终端，在确保使用要求的前提下，亟需光谱感知设备的微小型化与低成本化，至于光谱仪还是光谱成像仪，看应用场景。

基于半导体制造工艺，芯片式光谱传感器将获得飞速发展，MEMS技术赋予核心分光元件动态可调能力，系统设计具有更大灵活性。

新兴材料的不断发展，有望实现全固态光谱可调滤波芯片。

人工智能技术将实现光谱滤波芯片的硬件优化，也将大幅度提高光谱信号、光谱图像的算法分析能力。

团队概况

- **教学专业：**微机电系统工程 (国家一流专业)
- **科研平台：**极端环境智能微系统国家重点实验室 (培育)
空天微纳系统教育部重点实验室
国防科工委先进工业技术(微系统)工程应用中心
- **学科分类：**机械工程 (国家一流学科)、仪器科学与技术
- **学科组成：**“机械电子工程” 国家重点学科
“微机电系统与纳米技术” 国防重点学科



2010



2011



2019

■ 在编28人，教授8人、副教授8人、工程技术5人、博士后4人，研究生~180人



苑伟政教授
(实验室主任
长江学者)



马炳和
教授
万人领军



常洪龙
教授
万人领军



乔大勇
教授
知微传感 创始人



虞益挺
教授
优青



西北工业大学
宁波研究院

智能传感芯片技术研究中心

SMART SENSING CHIPS INNOVATION CENTER OF NINGBO INSTITUTE, NPU

责任教授



虞益挺
教授

优青、洪堡学者



硅基8寸中试线

SPTS深硅刻蚀机
SUSS MA8/BA8光刻/键合机
全自动及手动涂胶/显影机
ALD
两台蒸镀（金属、介质）
两台激光隐切（Si、玻璃）
清洗/腐蚀/甩干机
打线机、CMP、热氧化炉
椭圆仪、台阶仪、SEM

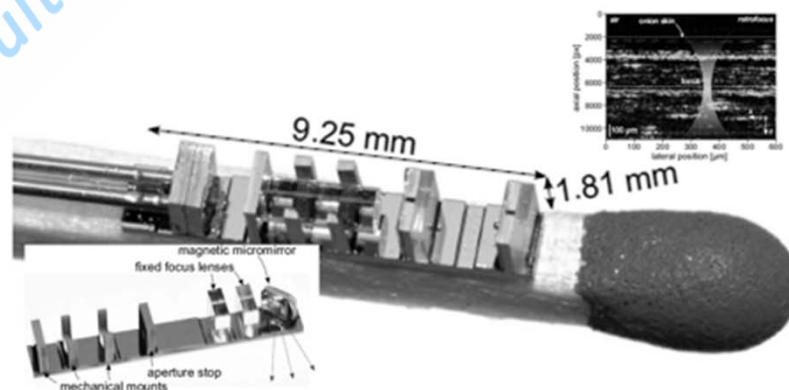
个人简介



虞益挺

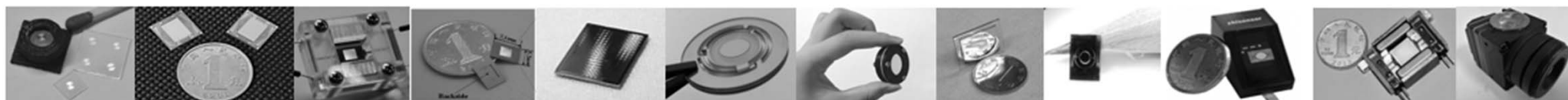
教授(2012)、博导(2014)、系主任(2018)、宁波中心负责人(2019)

- ✦ 2019: 国防173首席科学家
- ✦ 2016: 优青
- ✦ 2014: 省青年科技新星 (2019-优秀)
- ✦ 2013: 教育部新世纪优秀人才
- ✦ 2012: 教育部全国百篇优博论文奖
- ✦ 2010: 洪堡学者



硅基光学相干断层扫描系统 (OCT)

主要研究领域：光功能芯片、先进光学成像



研究领域



致谢

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Alexander von Humboldt
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谢 谢！期待合作！

